

Sharp Global-to-Local Theory for Fisher–Rao Gaussian Variational Inference

Abstract

We study full-covariance Gaussian variational inference for a target $\pi \propto e^{-V}$ through the Fisher–Rao natural gradient. For strongly log-concave targets, optimizer whitening separates three rate-limiting mechanisms. An underdispersed initialization first incurs a covariance bootstrap cost. Global localization then occurs on an intrinsic condition scale κ_* , and, in dimension at least two, a logarithmic-spiral construction shows that this dependence is unavoidable for an admissible anisotropic initialization. Near the optimizing Gaussian, the rate is instead controlled by a self-adjoint score operator. Its spectral gap satisfies

$$\max\left\{\alpha_*, \frac{1}{4+\Gamma}\right\} \leq \gamma_* \leq L_* \leq \min\left\{\beta_*, \frac{4+\Gamma}{2}\right\}, \quad \Gamma = O\left(1 + \sqrt{d} \log \kappa_*\right).$$

A smooth convex-ridge family with $d = \Theta(\kappa_*^2)$ has $\gamma_* = \Theta(\kappa_*^{-1/2})$, showing that the $\beta_* - 1$ branch of the universal spectral certificate is sharp up to constants while leaving the fixed-dimensional question open. For two discretizations—a Riemannian matrix-exponential retraction and a KL/Bregman resolvent—we prove global upper bounds and a fixed-step lower bound matching the κ/h dependence for constant-factor reduction. We also develop a joint Price/Hessian oracle with a sticking-the-landing score estimator, prove global geometric bounds with explicit constant-step residual levels, and, under Hessian-Lipschitzness, a floor-free decreasing-step rate and a finite-horizon local non-exit theory. Finally, for $d \geq 2$ we classify all affine-invariant Gaussian metrics, identify Fisher–Rao as the unique balanced Gaussian choice up to scale, and show by a smooth double-well example why covariance control can become essential beyond log-concavity.

Keywords. Gaussian variational inference; natural gradient; Fisher–Rao geometry; affine invariance; stochastic gradient; local convergence.

1 Introduction

Gaussian variational inference replaces an intractable target distribution $\pi(x) \propto e^{-V(x)}$ on $\mathbb{R}^d$ by the minimizer of

$$\mathcal{E}(m, C) := \text{KL}(\mathcal{N}(m, C) \parallel \pi), \quad (m, C) \in \mathbb{R}^d \times \mathbb{S}_{++}(d).$$

The approximation is computationally useful only if the optimization respects the strong coupling between the mean and covariance. Euclidean updates do not: their behavior changes under an invertible affine reparameterization of the target. The Fisher–Rao natural gradient is invariant under precisely these changes of coordinates and is therefore a canonical geometry for the full Gaussian family [1, 2, 7]. This paper develops a global-to-local rate theory and identifies several rates that this geometry can and cannot deliver.

The main conclusion is a global-to-local rate picture with three distinct bottlenecks. First, a small initial covariance suppresses the mean motion and creates a covariance burn-in. Second, after the covariance has reached the target curvature scale, global localization is controlled by an intrinsic condition number obtained by whitening with the optimizing covariance. Third, inside

a neighborhood of the optimizer, the rate is governed by a score- operator spectral gap that measures non-Gaussian coupling between the mean and covariance modes. These mechanisms are logically independent. A transport warm start removes the first, but neither the global lower construction nor the local spectral obstruction disappears.

1.1 The three bottlenecks

Assume initially that

$$\alpha I \preceq \nabla^2 V(x) \preceq \beta I, \quad \kappa := \beta/\alpha. \quad (1)$$

Let $a_\star = (m_\star, C_\star)$ be the optimizing Gaussian. In the optimizer-whitened coordinates

$$z = C_\star^{-1/2}(x - m_\star), \quad \hat{V}(z) = V(m_\star + C_\star^{1/2}z),$$

write $\alpha_\star, \beta_\star$ for the best curvature bounds of $\hat{V}$, and $\kappa_\star = \beta_\star/\alpha_\star$. These quantities are affine invariant, satisfy $1 \leq \kappa_\star \leq \kappa^2$, and equal one for every Gaussian target, even if its original precision is arbitrarily ill-conditioned.

The covariance bootstrap is visible directly in precision coordinates. If $\hat{C}_0 = C_\star^{-1/2}C_0C_\star^{-1/2}$ and $\lambda_{0,\star} = \lambda_{\min}(\hat{C}_0)$, then the Fisher–Rao flow satisfies

$$\mathcal{E}(a_t) - \mathcal{E}(a_\star) \leq \Delta_0 \left[1 + \beta_\star \lambda_{0,\star} (e^t - 1)\right]^{-1/\kappa_\star}. \quad (2)$$

The first logarithm in the resulting hitting time depends on $\lambda_{0,\star}$; the subsequent localization time is proportional to $\kappa_\star$. A two-dimensional logarithmic spiral forces $\Omega(\kappa_\star)$ continuous time under an admissible anisotropic initialization. For fixed-step Riemannian and KL schemes, a one-dimensional bump train, run with $h = \gamma/\kappa$ for $0 < \gamma \leq 1/2$, forces $\Omega(\kappa/h)$ iterations for a constant-factor reduction even after covariance burn-in. The two constructions therefore isolate distinct continuous-time and fixed-step obstructions.

At the optimizer, strong log-concavity alone does not reduce the problem to a Gaussian quadratic. The first- and second-Hermite components of the whitened Hessian form a self-adjoint linearized generator. We prove the spectral sandwich

$$\max\left\{\alpha_\star, \frac{1}{4+\Gamma}\right\} \leq \gamma_\star \leq L_\star \leq \min\left\{\beta_\star, \frac{4+\Gamma}{2}\right\}, \quad \Gamma = O(1 + \sqrt{d} \log \kappa_\star). \quad (3)$$

The lower bound is dimension dependent, and this is not merely a proof artifact in growing dimension. A smooth convex-ridge family with $d = \Theta(\kappa_\star^2)$ satisfies $\gamma_\star = \Theta(\kappa_\star^{-1/2})$ and a squared first-Hermite coupling of order $\kappa_\star^{1/2}$. It saturates the $\beta_\star - 1$ branch of Γ . It does not settle whether a stronger bound holds in fixed dimension or whether the $\sqrt{d} \log \kappa_\star$ branch is sharp.

1.2 Contributions

The deterministic theory gives both upper and lower bounds. The continuous estimate (2) is affine invariant and separates initial covariance from intrinsic conditioning. The two deterministic discretizations admit two-sided covariance bootstraps and have the common complexity

$$O\left(h^{-1} \log_+ \frac{1}{\beta \lambda_{\min}(C_0)} + \frac{\kappa}{h} \log_+ \frac{\Delta_0}{\delta}\right),$$

with a lower bound matching the κ/h factor for constant-factor reduction. A Bures–Wasserstein warm start or one pointwise gradient–Hessian quadratic rescue can remove the covariance term, but not the other two stages.

The local theory keeps the nonlinear Gaussian covariance dynamics exactly and perturbs only the non-Gaussian part. This Gaussian-core decomposition gives an explicit energy-sublevel

region for the flow, constructive local regions for both maps, exact continuous and KL invariant regions on Gaussian targets, and an online residual gate for entry. For general discrete targets, the radius depends on a supremum of second derivatives of the update map over a compact hull. The result is constructive but is not advertised as a closed-form practical radius.

The stochastic theory uses a joint Price/Hessian oracle and a sticking-the-landing mean estimator [8]. It supplies pathwise covariance safety after the quadratic rescue under the stated stepsizes and global constant-step bounds with explicit residual levels. Under Hessian-Lipschitzness it also gives a floor-free $O(1/N)$ decreasing-step guarantee. Under the same assumption, the local result is deliberately separated into a killed-process mean-square contraction and a finite-horizon containment probability. Under the universal spectral certificate its iteration dependence is quadratic, rather than linear, in $4 + \Gamma$. Infinite-horizon containment is not claimed.

Two extensions mark the scope of the theory. First, for $d \geq 2$ we classify the complete three-parameter family of affine-invariant Riemannian metrics on Gaussians. On a Gaussian target the parameters precondition the mean, trace, and traceless covariance modes separately; Fisher–Rao is the unique choice, up to scale, that balances all three. This modal statement does not extend every sharp rate theorem to every metric. Second, a smooth non-log-concave double well produces a Riemannian covariance cascade and a pole in the KL resolvent. A covariance-constrained KL step nevertheless has an $O(1/N)$ stationarity guarantee in a corrected split displacement. General Wasserstein–Fisher–Rao dynamics are left to a separate paper.

1.3 Relation to prior work

Natural gradients and information geometry provide the classical background for the Fisher–Rao metric [1, 2, 7]. Gaussian variational inference and stochastic variational inference are reviewed in Blei et al. [3] and Hoffman et al. [5]. The present analysis differs from a generic natural-gradient convergence argument by exploiting the exact Gaussian covariance equation and by retaining affine-invariant, optimizer-whitened quantities throughout.

Wasserstein Gaussian variational methods offer a complementary geometry. Lambert et al. [6] study variational inference through Wasserstein gradient flows, while Diao et al. [4] analyze a Bures–Wasserstein forward–backward scheme. We use Bures–Wasserstein only as a warm start and as a nonconvex comparison; it is not blended with Fisher–Rao into a general WFR algorithm here. Affine-invariant metrics on positive-definite matrices were classified in several forms by Thanwerdas and Pennec [10, 11]. Our classification includes the mean block of the Gaussian family and, for $d \geq 2$, uses it to expose the exact trace and traceless modal rates.

The Price/Hessian stochastic construction is related to the sticking-the-landing principle of Roeder et al. [8]. Recent natural-gradient VI work also studies nonconjugate models and constrained covariance schemes [9]. Our stochastic results focus on the joint geometry of the mean and full covariance, distinguish constant-step residual bounds from decreasing-step convergence, and keep local containment separate from local contraction.

1.4 Organization and notation

Section 2 introduces the objective, the affine metric family, and the algorithms. Section 3 develops the deterministic global theory and its lower bounds. Section 4 gives the local spectral and nonlinear theory. Section 5 treats Price/Hessian algorithms. Section 6 identifies the non-log-concave boundary, and Section 7 presents focused numerical illustrations.

We write d for the ambient dimension, $\|\cdot\|_{\text{op}}$ and $\|\cdot\|_{\text{F}}$ for the operator and Frobenius norms, and $\Delta(a) = \mathcal{E}(a) - \mathcal{E}(a_*)$. The symbol κ always denotes the original-coordinate curvature ratio in (1); κ_* denotes its optimizer-whitened counterpart. Constants hidden by $O(\cdot)$ are numerical unless their dependence is stated, and $\log_+ x := \max\{\log x, 0\}$.

2 Gaussian variational inference and affine-invariant geometry

Let $d \geq 1$ and

$$\mathcal{A} := \mathbb{R}^d \times \mathbb{S}_{++}(d), \quad a = (m, C) \in \mathcal{A}, \quad \rho_a := \mathcal{N}(m, C).$$

We identify the tangent space $T_a \mathcal{A}$ with $\mathbb{R}^d \times \mathbb{S}(d)$. The target distribution has density $\pi(x) = Z^{-1} e^{-V(x)}$, and Gaussian variational inference minimizes the reverse relative entropy over $\mathcal{A}$.

2.1 The objective and Gaussian differential identities

Up to a constant independent of a , the variational objective is

$$\mathcal{E}(a) := \text{KL}(\rho_a \| \pi) = \mathbb{E}_{\rho_a}[V(X)] - \frac{1}{2} \log \det C + \text{const.} \quad (4)$$

The three averaged quantities used throughout the paper are

$$g(a) := -\mathbb{E}_{\rho_a}[\nabla V(X)], \quad A(a) := \mathbb{E}_{\rho_a}[\nabla^2 V(X)], \quad R(a) := C^{-1} - A(a). \quad (5)$$

Thus g is the averaged target score, A is the Price curvature, and R is the precision residual.

We use the following regularity convention. Whenever a Bonnet or Price identity is invoked, differentiation under the Gaussian expectation and the corresponding Gaussian integration by parts are assumed to be valid. It is enough, for example, that $V \in C^2$ and $\nabla^2 V$ is bounded. In particular, all identities below hold under the strong log-concavity and smoothness assumptions imposed in the convergence sections.

Lemma 2.1 (Bonnet–Price identities). *For $(u, X) \in \mathbb{R}^d \times \mathbb{S}(d)$,*

$$D_m \mathbb{E}_{\rho_a}[V][u] = \mathbb{E}_{\rho_a}[\nabla V]^\top u, \quad D_C \mathbb{E}_{\rho_a}[V][X] = \frac{1}{2} \text{Tr}(A(a)X). \quad (6)$$

Consequently, the differential of the reverse-KL objective is

$$D\mathcal{E}(a)[(u, X)] = -g(a)^\top u - \frac{1}{2} \text{Tr}(R(a)X). \quad (7)$$

If $a_\star = (m_\star, C_\star)$ is an interior minimizer, then

$$g(a_\star) = 0, \quad R(a_\star) = 0, \quad \text{or equivalently} \quad A(a_\star) = C_\star^{-1}. \quad (8)$$

Proof. For the mean derivative, write $X = m + C^{1/2}\xi$ with $\xi \sim \mathcal{N}(0, I)$, differentiate the translation $m \mapsto m + \epsilon u$, and use dominated convergence. For the covariance derivative, if p_C is the density of $\mathcal{N}(m, C)$, then the Gaussian heat identity gives

$$D_C p_C[X] = \frac{1}{2} \text{Tr}(X \nabla_x^2 p_C).$$

Twice integrating by parts therefore yields $D_C \mathbb{E}_{\rho_a}[V][X] = \frac{1}{2} \mathbb{E}_{\rho_a}[\text{Tr}(X \nabla^2 V)] = \frac{1}{2} \text{Tr}(A(a)X)$. Jacobi's formula gives $D_C(-\frac{1}{2} \log \det C)[X] = -\frac{1}{2} \text{Tr}(C^{-1}X)$, proving (7). Since the trace pairing is nondegenerate on $\mathbb{S}(d)$, vanishing of the differential in every tangent direction is equivalent to (8). $\square$

2.2 Classification of affine-invariant metrics

For $B \in \text{GL}(d)$ and $b \in \mathbb{R}^d$, the change of variables $x \mapsto Bx + b$ acts on Gaussian parameters and tangent vectors by

$$\Phi_{B,b}(m, C) = (Bm + b, BCB^\top), \quad D\Phi_{B,b}(u, X) = (Bu, BXB^\top). \quad (9)$$

A Riemannian metric is affine invariant if every map $\Phi_{B,b}$ is an isometry.

Theorem 2.2 (Classification of affine-invariant Gaussian metrics). *Suppose $d \geq 2$. Every affine-invariant Riemannian metric on $\mathcal{A}$ is uniquely of the form*

$$\langle (u, X), (v, Y) \rangle_a^{\eta, \omega, \tau} = \eta u^\top C^{-1} v + \omega \text{Tr}(C^{-1} X C^{-1} Y) \quad (10)$$

$$+ \tau \text{Tr}(C^{-1} X) \text{Tr}(C^{-1} Y), \quad (11)$$

where

$$\eta > 0, \quad \omega > 0, \quad \omega + d\tau > 0 \quad \left(\text{i.e. } \tau > -\frac{\omega}{d} \right). \quad (12)$$

Conversely, every choice satisfying (12) defines an affine-invariant Riemannian metric. Multiplication of (η, ω, τ) by a common positive constant only changes the overall metric scale.

Proof. The affine group acts transitively on $\mathcal{A}$, so the metric is determined at $(0, I)$ by invariance under its isotropy group $O(d)$. At that point the tangent representation splits as

$$\mathbb{R}^d \oplus \mathbb{S}_0(d) \oplus \mathbb{R}I, \quad \mathbb{S}_0(d) := \{X \in \mathbb{S}(d) : \text{Tr} X = 0\}.$$

If $L(u, X)$ is a possible mean–covariance cross term, invariance under the sign element $Q = -I$ gives $L(u, X) = L(-u, X) = -L(u, X)$; hence $L = 0$. The only $O(d)$ -invariant bilinear form on the mean block is $\eta u^\top v$. The trace and traceless decomposition gives the two-parameter covariance family

$$\omega \text{Tr}(XY) + \tau \text{Tr}(X) \text{Tr}(Y),$$

the standard classification of $O(d)$ -invariant metrics on symmetric matrices [10, 11]. Positivity on the mean, traceless, and scalar directions respectively gives $\eta > 0$, $\omega > 0$, and $d(\omega + d\tau)s^2 > 0$ for $X = sI$.

Finally, transport from $(0, I)$ to (m, C) by $B = C^{1/2}$ and $b = m$ pulls a tangent vector back to

$$(C^{-1/2}u, C^{-1/2}XC^{-1/2}).$$

Substitution gives (10). The displayed expression is directly invariant under (9), proving the converse. $\square$

Remark 2.3 (The one-dimensional family). When $d = 1$ there is no traceless covariance block. Formula (10) still describes every affine-invariant metric, but only the sum $\omega + \tau$ is identifiable; positivity requires $\eta > 0$ and $\omega + \tau > 0$. The uniqueness statements below concern the full three-mode geometry and therefore assume $d \geq 2$.

Because an overall scaling only reparametrizes gradient-flow time, one may normalize $\eta = 1$. We retain η temporarily so that the relative speeds of the mean, trace, and traceless modes remain visible.

2.3 The general affine-invariant gradient flow

Introduce the covariance-whitened residual and its trace–traceless split,

$$S(a) := C^{1/2}R(a)C^{1/2}, \quad s(a) := \text{Tr } S(a), \quad S_0(a) := S(a) - \frac{s(a)}{d}I. \quad (13)$$

Theorem 2.4 (Affine-invariant gradient and exact modal preconditioning). *For the metric (10), the Riemannian gradient of $\mathcal{E}$ is*

$$\text{grad}_{\eta, \omega, \tau} \mathcal{E}(a) = \left(-\frac{1}{\eta} Cg(a), -\frac{1}{2\omega} CR(a)C + \frac{\tau}{2\omega(\omega + d\tau)} \text{Tr}(CR(a)) C \right). \quad (14)$$

Hence its negative gradient flow is

$$\dot{m} = \frac{1}{\eta} Cg(a), \quad \dot{C} = \frac{1}{2\omega} CR(a)C - \frac{\tau}{2\omega(\omega + d\tau)} \text{Tr}(CR(a)) C. \quad (15)$$

Equivalently, the covariance equation has the exact trace–traceless decomposition

$$C^{-1/2} \dot{C} C^{-1/2} = \frac{1}{2\omega} S_0(a) + \frac{s(a)}{2d(\omega + d\tau)} I. \quad (16)$$

Along this flow,

$$\frac{d}{dt} \mathcal{E}(a_t) = -\frac{1}{\eta} g(a_t)^\top C_t g(a_t) - \frac{1}{4\omega} \|S_0(a_t)\|_F^2 - \frac{s(a_t)^2}{4d(\omega + d\tau)}. \quad (17)$$

Proof. The mean block follows by matching $\eta u^\top C^{-1}v$ with the first term of (7). For the covariance block, write $W = C^{-1/2}(\text{grad } \mathcal{E})_C C^{-1/2}$ and $Z = C^{-1/2}XC^{-1/2}$. Gradient duality requires

$$\omega \text{Tr}(WZ) + \tau \text{Tr}(W) \text{Tr}(Z) = -\frac{1}{2} \text{Tr}(SZ) \quad \text{for every } Z \in \mathbb{S}(d).$$

Testing separately against traceless matrices and against I gives

$$W_0 = -\frac{1}{2\omega} S_0, \quad \text{Tr } W = -\frac{s}{2(\omega + d\tau)}.$$

Recombining the two components yields the covariance block in (14). Negating the gradient gives (15) and (16). Finally, $\frac{d}{dt} \mathcal{E} = -\|\text{grad } \mathcal{E}\|_a^2$; evaluating the metric norm on the three orthogonal blocks gives (17). $\square$

2.4 Gaussian modal rates and the balanced metric

The metric parameters have a particularly transparent interpretation on a Gaussian target. By an affine change of variables, any nondegenerate Gaussian target can be placed in optimizer-whitened coordinates, where $\pi = \mathcal{N}(0, I)$.

Theorem 2.5 (Exact Gaussian modal rates). *Let $\pi = \mathcal{N}(0, I)$ and consider the flow (15). Its exact nonlinear equations are*

$$\dot{m} = -\frac{1}{\eta} Cm, \quad \dot{C} = \frac{1}{2\omega} (C - C^2) - \frac{\tau}{2\omega(\omega + d\tau)} \text{Tr}(I - C) C. \quad (18)$$

Write $C = I + X$, $\sigma = \text{Tr}(X)/d$, and $X_0 = X - \sigma I$. Then the covariance equation is exactly

$$\dot{X} = -\frac{1}{2\omega} X + \frac{\tau}{2\omega(\omega + d\tau)} \text{Tr}(X)I + \mathcal{Q}(X), \quad (19)$$

$$\mathcal{Q}(X) := -\frac{1}{2\omega}X^2 + \frac{\tau}{2\omega(\omega + d\tau)}\text{Tr}(X)X. \quad (20)$$

Consequently, the Jacobian of the flow at $(0, I)$ has the three exact decay rates

$$\gamma_{\text{mean}} = \frac{1}{\eta}, \quad \gamma_{\text{traceless}} = \frac{1}{2\omega}, \quad \gamma_{\text{trace}} = \frac{1}{2(\omega + d\tau)}. \quad (21)$$

More explicitly, $\dot{m} = -\eta^{-1}m + O(\|(m, X)\|^2)$, $\dot{X}_0 = -(2\omega)^{-1}X_0 + O(\|X\|^2)$, and $\dot{\sigma} = -[2(\omega + d\tau)]^{-1}\sigma + O(\|X\|^2)$.

Proof. For the standard Gaussian potential, $g(a) = -m$, $A(a) = I$, and $R(a) = C^{-1} - I$. Substitution in (15) gives (18). Expanding at $C = I + X$ gives (19). Projection onto $\mathbb{S}_0(d)$ and $\mathbb{R}I$, together with the linearization of the mean equation, yields (21). $\square$

Theorem 2.6 (Fisher–Rao is the unique balanced choice). *Assume $d \geq 2$. The three Gaussian modal rates in (21) are equal if and only if*

$$(\eta, \omega, \tau) = c \left(1, \frac{1}{2}, 0\right) \quad \text{for some } c > 0. \quad (22)$$

Thus, up to an overall metric scale, the Fisher–Rao metric is the unique affine-invariant Gaussian metric that balances the mean, trace-covariance, and traceless-covariance modes. Under the normalization $\eta = 1$, its parameters are $\omega = 1/2$ and $\tau = 0$.

Proof. Equality of the mean and traceless rates gives $\eta = 2\omega$. Equality of the two covariance rates gives $\omega = \omega + d\tau$, hence $\tau = 0$. These conditions are also sufficient, and removing the common scale gives (22). $\square$

Remark 2.7 (Scope of the balancing statement). The rates in (21) are exact eigenvalues of the Gaussian flow linearization. They are not asserted to be global rates for the nonlinear covariance equation, nor rates for a general non-Gaussian target. A different affine-invariant metric may accelerate a particular direction or target; Fisher–Rao is canonical here because it is the unique choice that does not privilege any of the three Gaussian modes.

2.5 Canonical Fisher–Rao dynamics and deterministic algorithms

With the balanced normalization, the Gaussian Fisher–Rao metric is

$$\langle (u, X), (v, Y) \rangle_{a, \text{FR}} := u^\top C^{-1}v + \frac{1}{2} \text{Tr}(C^{-1}XC^{-1}Y). \quad (23)$$

The factor $1/2$ is the covariance block of the Gaussian Fisher information.

Proposition 2.8 (Fisher–Rao gradient, flow, and dissipation). *For the metric (23),*

$$\text{grad}_{\text{FR}} \mathcal{E}(a) = (-Cg(a), -CR(a)C), \quad \dot{m} = Cg(a), \quad \dot{C} = CR(a)C = C - CA(a)C. \quad (24)$$

Moreover,

$$\begin{aligned} G(a)^2 &:= \|\text{grad}_{\text{FR}} \mathcal{E}(a)\|_{a, \text{FR}}^2 = g(a)^\top Cg(a) + \frac{1}{2} \|C^{1/2}R(a)C^{1/2}\|_{\text{F}}^2, \\ \frac{d}{dt} \mathcal{E}(a_t) &= -\|\text{grad}_{\text{FR}} \mathcal{E}(a_t)\|_{a_t, \text{FR}}^2. \end{aligned} \quad (25)$$

Proof. This is Theorem 2.4 with $(\eta, \omega, \tau) = (1, 1/2, 0)$. $\square$

We use two first-order discretizations of (24). The first is obtained from the map

$$\text{Retr}_a(u, U) := \left(m + u, C^{1/2} \exp(C^{-1/2} U C^{-1/2}) C^{1/2}\right). \quad (26)$$

Applying (26) to $-h \text{grad}_{\text{FR}} \mathcal{E}(a)$ gives the Riemannian matrix-exponential scheme

$$m^+ = m + h C g(a), \quad C^+ = C^{1/2} \exp\left(h[I - C^{1/2} A(a) C^{1/2}]\right) C^{1/2}. \quad (27)$$

The covariance block in (26) is the exponential map for the affine-invariant metric on $\mathbb{S}_{++}(d)$. The combined mean-covariance map is, however, only a retraction for the full Gaussian Fisher–Rao geometry; it is not the full Gaussian-manifold exponential map.

For the second scheme, split $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$, where $\mathcal{E}_1(m, C) = -\frac{1}{2} \log \det C + \text{const}$ and $\mathcal{E}_2(m, C) = \mathbb{E}_{\rho_a}[V]$. The KL/Bregman forward-backward step is

$$a^+ := \underset{a'=(m', C') \in \mathcal{A}}{\text{argmin}} \left\{ h D \mathcal{E}_2(a)[a' - a] + h \mathcal{E}_1(a') + \text{KL}(\rho_{a'} \parallel \rho_a) \right\}. \quad (28)$$

Whenever $C^{-1} + h A(a) \succ 0$, its unique minimizer is the rational resolvent

$$m^+ = m + h C g(a), \quad C^+ = (1 + h)(C^{-1} + h A(a))^{-1}, \quad (29)$$

or equivalently $(C^+)^{-1} = (C^{-1} + h A(a))/(1 + h)$. Indeed, the first-order conditions in m' and C' give these two identities. This scheme treats the potential term explicitly and the Gaussian entropy implicitly; it is not a full-gradient mirror step. Under log-concavity $A(a) \succeq 0$, so (29) is well defined for every $h > 0$.

Exact and stochastic oracle conventions. The deterministic theory treats $g(a)$ and $A(a)$ in (5) as exact Gaussian expectations. The stochastic theory instead draws $\xi_s \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I)$, $X_s = m + C^{1/2} \xi_s$, and uses the Price/Hessian pair

$$\widehat{g}_{\text{STL}}(a) := \frac{1}{B} \sum_{s=1}^B \left[-\nabla V(X_s) + C^{-1/2} \xi_s \right], \quad \widehat{A}(a) := \frac{1}{B} \sum_{s=1}^B \nabla^2 V(X_s). \quad (30)$$

Both estimators are conditionally unbiased for $g(a)$ and $A(a)$, respectively. The first is the sticking-the-landing score estimator, while the second implements the Price identity through sampled Hessians. The same samples may be used for the two components; no independence between the score and Hessian estimators is assumed. Replacing (g, A) by $(\widehat{g}_{\text{STL}}, \widehat{A})$ in (27) or (29) gives the two Price/Hessian stochastic algorithms analyzed later.

3 Sharp global deterministic theory under strong log-concavity

The global behavior separates into two mechanisms. An underdispersed covariance must first reach the curvature scale; after that, the objective localizes at a rate set by target conditioning. For the Fisher–Rao flow these mechanisms are visible in a single exact estimate, while for the two discretizations they appear as a covariance bootstrap followed by geometric descent. Throughout this section we assume

$$\alpha I \preceq \nabla^2 V(x) \preceq \beta I, \quad x \in \mathbb{R}^d, \quad \kappa := \frac{\beta}{\alpha}, \quad (31)$$

and all Gaussian expectations are evaluated exactly. We write

$$\begin{aligned} g(a) &= -\mathbb{E}_{\mathcal{N}(m, C)}[\nabla V(X)], \quad A(a) = \mathbb{E}_{\mathcal{N}(m, C)}[\nabla^2 V(X)], \\ R(a) &= C^{-1} - A(a). \end{aligned} \quad (32)$$

The energy gap is $\Delta(a) = \mathcal{E}(a) - \mathcal{E}(a_\star)$, with $\Delta_0 = \Delta(a_0)$, and $\lambda_0 := \lambda_{\min}(C_0)$.

3.1 Continuous covariance bootstrap and the intrinsic upper bound

The natural conditioning of the continuous flow is obtained by whitening with the covariance of the optimizing Gaussian. Set

$$\hat{m} = C_\star^{-1/2}(m - m_\star), \quad \hat{C} = C_\star^{-1/2} C C_\star^{-1/2}, \quad \hat{V}(z) = V(m_\star + C_\star^{1/2} z). \quad (33)$$

Then $a_\star$ becomes $(0, I)$, and we define the optimal pointwise curvature constants

$$\alpha_\star := \inf_z \lambda_{\min}(\nabla^2 \hat{V}(z)), \quad \beta_\star := \sup_z \lambda_{\max}(\nabla^2 \hat{V}(z)), \quad \kappa_\star := \frac{\beta_\star}{\alpha_\star}, \quad (34)$$

and the initial whitened covariance floor

$$\lambda_{0,\star} := \lambda_{\min}(C_\star^{-1/2} C_0 C_\star^{-1/2}). \quad (35)$$

The first-order condition $C_\star^{-1} = A(a_\star)$ gives $\alpha_\star \leq 1 \leq \beta_\star$. Moreover, $1 \leq \kappa_\star \leq \kappa^2$, although the upper comparison can be arbitrarily loose: an anisotropic Gaussian target has $\kappa_\star = 1$. Unlike κ , all quantities in (34)–(35) are unchanged by invertible affine reparametrization.

The covariance floor follows most transparently from the precision. If $P_t = C_t^{-1}$, then the Fisher–Rao covariance equation gives the exact Duhamel formula

$$\dot{P}_t = A(a_t) - P_t, \quad P_t = e^{-t} P_0 + \int_0^t e^{-(t-s)} A(a_s) ds. \quad (36)$$

The same identity holds after optimizer whitening.

Proposition 3.1 (Continuous covariance bootstrap). *Along the Fisher–Rao flow,*

$$C_t \succeq \frac{\lambda_0 e^t}{1 + \beta \lambda_0 (e^t - 1)} I, \quad \hat{C}_t \succeq \ell_\star(t) I, \quad \ell_\star(t) := \frac{\lambda_{0,\star} e^t}{1 + \beta_\star \lambda_{0,\star} (e^t - 1)}. \quad (37)$$

Consequently, the two covariance floors are reached after the respective times

$$\begin{aligned} C_t &\succeq (2\beta)^{-1} I \quad \text{for } t \geq \log_+ \left(\frac{1}{\beta \lambda_0} \right), \\ \hat{C}_t &\succeq (2\beta_\star)^{-1} I \quad \text{for } t \geq \log_+ \left(\frac{1}{\beta_\star \lambda_{0,\star}} \right). \end{aligned} \quad (38)$$

Proof. Since $A(a_t) \preceq \beta I$, (36) implies $P_t \preceq [e^{-t} \lambda_0^{-1} + \beta(1 - e^{-t})] I$; inversion gives the first bound. Applying the same argument to the whitened flow gives the second. The two hitting times follow by making the decaying initial-precision term no larger than the corresponding curvature term. $\square$

Theorem 3.2 (Affine-invariant continuous global rate). *For every $t \geq 0$,*

$$\boxed{\Delta(a_t) \leq \Delta_0 [1 + \beta_\star \lambda_{0,\star} (e^t - 1)]^{-1/\kappa_\star}}. \quad (39)$$

Thus, for $0 < \delta < \Delta_0$, the hitting time $T_c^{\text{glob}}(\delta) := \inf\{t \geq 0 : \Delta(a_t) \leq \delta\}$ satisfies

$$T_c^{\text{glob}}(\delta) \leq \log \left(1 + \frac{(\Delta_0/\delta)^{\kappa_\star} - 1}{\beta_\star \lambda_{0,\star}} \right), \quad (40)$$

$$T_c^{\text{glob}}(\delta) \leq \boxed{\log_+ \frac{1}{\beta_\star \lambda_{0,\star}} + \kappa_\star \log \frac{\Delta_0}{\delta}}. \quad (41)$$

The same argument in the original coordinates gives

$$T_c^{\text{glob}}(\delta) \leq \log \left(1 + \frac{(\Delta_0/\delta)^\kappa - 1}{\beta \lambda_0} \right); \quad (42)$$

one may take the smaller of the intrinsic and original-coordinate bounds.

Proof. Optimizer whitening preserves both the energy and Fisher–Rao dissipation. In whitened coordinates the exact dissipation identity, the Gaussian Bures–Wasserstein PL inequality, and $\hat{C}_t \succeq \ell_\star(t)I$ give

$$\dot{\Delta}(a_t) = -\mathbf{G}(a_t)^2 \leq -\alpha_\star \ell_\star(t) \Delta(a_t). \quad (43)$$

The scalar integral is explicit:

$$\int_0^t \ell_\star(s) \, ds = \frac{1}{\beta_\star} \log[1 + \beta_\star \lambda_{0,\star}(e^t - 1)]. \quad (44)$$

Grönwall’s inequality and $\alpha_\star/\beta_\star = 1/\kappa_\star$ prove (39). Solving that estimate for t gives (40). Finally, for $R \geq 1$ and $b > 0$, $1 + (R - 1)/b \leq R/b$ when $b \leq 1$, and it is at most R when $b \geq 1$; this yields (41). $\square$

The two terms in (41) have different meanings. The first is the multiplicative covariance burn-in caused by underdispersion. The second is the global localization cost after the covariance reaches the intrinsic curvature scale. The condition number alone cannot encode arbitrary input size: even for a quadratic target, reaching a fixed energy level from an unrestricted initial mean takes arbitrarily long.

3.2 A matching continuous lower bound

The intrinsic dependence in Theorem 3.2 cannot be replaced by a dimension-free logarithmic dependence. The obstruction already occurs in two dimensions and is geometric: an anisotropic covariance can follow a slowly descending spiral valley while remaining inside the globally admissible spectral band.

Theorem 3.3 (Logarithmic-spiral lower bound). *For every $K \geq 16$, there is a globally C^3 potential $V_K : \mathbb{R}^2 \rightarrow \mathbb{R}$ and an initialization $a_0 = (m_0, C_0)$ such that*

$$\alpha I \preceq \nabla^2 V_K \preceq \beta I, \quad \kappa = \frac{\beta}{\alpha} = 2K + 1, \quad \frac{1}{\beta} I \preceq C_0 \preceq \frac{1}{\alpha} I. \quad (45)$$

Its optimizer-whitened condition number and initial covariance obey

$$K \leq \kappa_\star(V_K) \leq 1.03K, \quad \beta_\star \lambda_{0,\star} \geq \frac{1 - \eta_K}{1 + \eta_K}, \quad 0 \leq \eta_K \leq 10^{-3}. \quad (46)$$

For every level

$$\delta_{\text{loc}} < \frac{\alpha}{2} e^{-0.08} \|m_0\|^2, \quad (47)$$

the exact Fisher–Rao flow satisfies

$$T_{\text{loc}} := \inf\{t \geq 0 : \Delta(a_t) \leq \delta_{\text{loc}}\} \geq \frac{K}{100} \geq \frac{\kappa_\star(V_K)}{103}. \quad (48)$$

The construction extends to every dimension $d \geq 2$ by adjoining independent quadratic coordinates initialized at equilibrium.

Proof outline. The potential has a high-curvature logarithmic-spiral valley outside an isotropic quadratic core. Its Hessian is certified globally between αI and βI . A rescaled four-dimensional system for radial displacement, angular displacement, and the two independent covariance errors has a uniformly stable linear part; a semigroup estimate and a stopping-time bootstrap control the nonlinear remainder and the Gaussian tails. The resulting exact trajectory obeys $\|m_t\| \geq e^{-0.04} \|m_0\|$ for $0 \leq t \leq K/100$.

The optimizing Gaussian is concentrated inside the quadratic core. Hence $C_\star^{-1}$ is within relative error η_K of a scalar matrix, which gives (46). Strong convexity in the mean then yields $\Delta(a_t) \geq (\alpha/2) \|m_t\|^2$, proving (48). The Hessian certificate, shadow system, semigroup bounds, Gaussian-smoothing estimates, and dimension extension are given in Section B.1. $\square$

Remark 3.4 (Scope of continuous sharpness). The lower bound uses an admissible but anisotropic initial covariance. It does not prove the same bound under the stricter initialization $C_0 = \beta^{-1}I$. The initial radius grows with K ; if a prescribed positive Hessian-Lipschitz constant is imposed, it also grows with the inverse of that constant. These input-size factors belong to the logarithmic terms suppressed in the shorthand statement $T_{\text{loc}} = \tilde{\Theta}(\kappa_*)$. No analogous linear lower bound is asserted for the endpoint of zero Hessian Lipschitz constant, where the target is quadratic.

3.3 Covariance bootstraps and descent for the two discretizations

Assume

$$\lambda_{0,\min}I \preceq C_0 \preceq \lambda_{0,\max}I, \quad \lambda_0 := \lambda_{0,\min}, \quad \lambda_{\max} := \max \left\{ \lambda_{0,\max}, \frac{1}{\alpha} \right\}. \quad (49)$$

The upper barrier $\lambda_{\max}$ controls the globally safe explicit stepsize. The lower barrier evolves much more favorably than a frozen estimate suggests.

Proposition 3.5 (Two-sided covariance bootstrap). *Let $L_0 = \min\{\lambda_0, (2\beta)^{-1}\}$.*

Riemannian retraction. If $0 < h \leq \min\{1, (\beta\lambda_{\max})^{-1}\}$, then $C_n \preceq \lambda_{\max}I$ and $C_n \succeq L_n^R I$, where

$$L_{n+1}^R = \frac{e^h L_n^R}{1 + (e-1)h\beta L_n^R}. \quad (50)$$

With

$$y_0 = \frac{1}{\beta L_0}, \quad y_\infty = \frac{(e-1)h}{e^h - 1} \in [1, e-1), \quad (51)$$

one has $C_n \succeq (2\beta)^{-1}I$ for every

$$n \geq N_{\text{cov},-}^R := \left\lceil \frac{1}{h} \log_+ \frac{y_0 - y_\infty}{2 - y_\infty} \right\rceil = O\left(\frac{1}{h} \log_+ \frac{1}{\beta\lambda_0}\right). \quad (52)$$

For the upper curvature band, put $z_0 = (\alpha\lambda_{0,\max})^{-1}$ and $z_\infty = h/(e^h - 1) > 1/2$. Then $C_n \preceq 2\alpha^{-1}I$ for all

$$n \geq N_{\text{cov},+}^R := \begin{cases} 0, & z_0 \geq \frac{1}{2}, \\ \left\lceil \frac{1}{h} \log \frac{z_\infty - z_0}{z_\infty - 1/2} \right\rceil, & z_0 < \frac{1}{2}. \end{cases} \quad (53)$$

KL/Bregman resolvent. For every $h > 0$, $C_n \preceq \lambda_{\max}I$ and $C_n \succeq L_n^{\text{KL}}I$, where

$$L_{n+1}^{\text{KL}} = \frac{(1+h)L_n^{\text{KL}}}{1 + h\beta L_n^{\text{KL}}}, \quad L_n^{\text{KL}} = \frac{1}{\beta[1 + (1/(\beta L_0) - 1)(1+h)^{-n}]}. \quad (54)$$

Consequently, $C_n \succeq (2\beta)^{-1}I$ for

$$n \geq N_{\text{cov},-}^{\text{KL}} := \left\lceil \frac{\log_+(1/(\beta L_0) - 1)}{\log(1+h)} \right\rceil. \quad (55)$$

When $h \leq 1$, this is $O(h^{-1} \log_+(1/(\beta\lambda_0)))$. In addition, $C_n \preceq 2\alpha^{-1}I$ for

$$n \geq N_{\text{cov},+}^{\text{KL}} := \left\lceil \frac{\log 2}{\log(1+h)} \right\rceil. \quad (56)$$

Proof. For the Riemannian update, let $B_n = C_n^{1/2} A(a_n) C_n^{1/2}$. The precision update is $C_{n+1}^{-1} = e^{-h} C_n^{-1/2} e^{hB_n} C_n^{-1/2}$. The upper barrier follows from $e^X \succeq I + X$. Under $0 \preceq hB_n \preceq I$, the chord bound $e^{hB_n} \preceq I + (e-1)hB_n$ gives (50). Inverting the scalar recurrence produces $(\beta L_n^R)^{-1} = y_\infty + (y_0 - y_\infty)e^{-nh}$, which yields (52). A lower affine recursion for $\lambda_{\min}(C_n^{-1})/\alpha$ gives (53).

For the KL update the precision recursion is exactly affine: $C_{n+1}^{-1} = (C_n^{-1} + hA(a_n))/(1+h)$. Comparing it with αI and βI gives (54) and the upper barrier. Solving the resulting scalar recurrences proves (55)–(56). $\square$

The descent mechanisms differ even though both methods share the mean update. Let $K_{\text{glob}} := \beta\lambda_{\text{max}} \geq 1$, and suppose at the current iterate that $LI \preceq C \preceq \lambda_{\text{max}}I$. We use the Gaussian Bures–Wasserstein gradient norm

$$\left\| \text{grad}^{W_2} \mathcal{E}(a) \right\|_{W_2}^2 := \|g(a)\|^2 + \text{Tr}(R(a)CR(a)). \quad (57)$$

Proposition 3.6 (One-step deterministic descent). *For the Riemannian retraction, if $0 < h \leq K_{\text{glob}}^{-1}$, then*

$$\mathcal{E}(a^+) \leq \mathcal{E}(a) - h(1 - K_{\text{glob}}h)\mathbf{G}(a)^2. \quad (58)$$

In particular, $h \leq (2K_{\text{glob}})^{-1}$ gives decrease by at least $(h/2)\mathbf{G}(a)^2$.

For the KL/Bregman resolvent, if $0 < h \leq K_{\text{glob}}^{-1}$, then the useful orientation of the Gaussian divergence is forward:

$$\mathcal{E}(a^+) \leq \mathcal{E}(a) - \frac{1}{h} \text{KL}(\rho_{a^+} \parallel \rho_a), \quad (59)$$

$$\text{KL}(\rho_{a^+} \parallel \rho_a) \geq \frac{Lh^2}{4(1 + hK_{\text{glob}})^2} \left\| \text{grad}^{W_2} \mathcal{E}(a) \right\|_{W_2}^2. \quad (60)$$

The reverse orientation $\text{KL}(\rho_a \parallel \rho_{a^+})$ does not provide the corresponding bound for arbitrary mean displacement.

Proof idea. For the Riemannian map, couple the old and new Gaussians through a shared standard normal variable. Smoothness of V , Gaussian integration by parts, and the scalar inequality $e^w \leq 1 + w + w^2$ for $|w| \leq 1$ control the exponential covariance retraction and yield (58).

For the KL map, diagonalize $C^{1/2}A(a)C^{1/2}$. Its covariance update is a rational function of this matrix, so the covariance part reduces to a scalar inequality on eigenvalues; the mean part is controlled by $hK_{\text{glob}} \leq 1$. This gives (59). The elementary inequality $q^{-1} - 1 + \log q \geq (q - 1)^2/[2\max\{q, 1\}^2]$ then proves (60). $\square$

Theorem 3.7 (Global complexity of the deterministic discretizations). *Let $N_{\bullet}^{\text{glob}}(\delta) := \inf\{n \geq 0 : \Delta(a_n) \leq \delta\}$ for $0 < \delta < \Delta_0$.*

If the Riemannian retraction is run with

$$0 < h \leq \frac{1}{2\beta\lambda_{\text{max}}}, \quad (61)$$

then

$$N_{\text{R}}^{\text{glob}}(\delta) = O\left(\frac{1}{h} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{h} \log_+ \frac{\Delta_0}{\delta}\right). \quad (62)$$

If the KL/Bregman resolvent is run with

$$0 < h \leq \frac{1}{\beta\lambda_{\text{max}}}, \quad (63)$$

then

$$N_{\text{KL}}^{\text{glob}}(\delta) = O\left(\frac{1}{h} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{h} \log_+ \frac{\Delta_0}{\delta}\right). \quad (64)$$

The constants hidden by $O(\cdot)$ are independent of the dimension and the conditioning parameters.

Proof. After the covariance burn-in established in Theorem 3.5, one has $C_n \succeq (2\beta)^{-1}I$. The Fisher–Rao/Bures comparison and the Gaussian Bures–Wasserstein PL inequality imply

$$\mathbf{G}(a_n)^2 \geq \frac{1}{2\kappa} \Delta(a_n). \quad (65)$$

For the Riemannian scheme, (58) therefore gives $\Delta_{n+1} \leq (1 - h/(4\kappa))\Delta_n$. For the KL scheme, (59)–(60) and $1 + hK_{\text{glob}} \leq 2$ give $\Delta_{n+1} \leq (1 - h/(16\kappa))\Delta_n$. Adding the covariance burn-in and summing these geometric contractions proves (62) and (64). $\square$

The KL covariance recursion is an affine precision resolvent and preserves the global upper barrier for every $h > 0$; the global descent theorem certifies $h \leq (\beta\lambda_{\text{max}})^{-1}$. The Riemannian method uses a matrix exponential and its stated descent rate uses the smaller range $h \leq (2\beta\lambda_{\text{max}})^{-1}$. These are certified theoretical ranges, rather than a claim that either method is stable beyond them on every target.

3.4 Fixed-step sharpness in one dimension

At the globally safe scale $h = \Theta(\kappa^{-1})$, the localization term in Theorem 3.7 is $\Theta(\kappa^2)$ for a constant-factor energy reduction. This polynomial dependence is attained even when the covariance starts at the curvature scale and the Hessian is Lipschitz with a constant independent of κ .

Theorem 3.8 (Bump-train lower bound for both fixed-step maps). *Fix $\gamma \in (0, 1/2]$ and $L_0 > 0$. There exist constants $T, q, Y > 0$ and $\kappa_0 \geq 1$, depending only on γ, L_0 , and a fixed smooth bump profile, such that for every $\kappa \geq \kappa_0$ there is an even potential $V_\kappa \in C^\infty(\mathbb{R})$ satisfying*

$$1 \leq V_\kappa'' \leq \kappa, \quad \|V_\kappa'''\|_\infty \leq L_0, \quad \kappa_\star(V_\kappa) = \kappa. \quad (66)$$

Initialize

$$m_0 = \frac{Y\kappa^4}{\gamma}, \quad C_0 = c_0 = \frac{1}{\kappa}, \quad (67)$$

and run either deterministic discretization with the fixed step

$$h = \frac{\gamma}{\kappa}. \quad (68)$$

Then

$$\Delta_n \geq q\Delta_0 \quad \text{for every } 0 \leq n \leq \left\lfloor \frac{T\kappa^2}{\gamma} \right\rfloor. \quad (69)$$

Consequently, for a constant-factor objective reduction,

$$N_{\text{R}} = \Omega\left(\frac{\kappa^2}{\gamma}\right) = \Omega\left(\frac{\kappa}{h}\right), \quad N_{\text{KL}} = \Omega\left(\frac{\kappa^2}{\gamma}\right) = \Omega\left(\frac{\kappa}{h}\right). \quad (70)$$

Proof outline. The construction places flat-top curvature bumps along a prescribed scalar reference orbit. Each bump has height $\kappa - 1$ and width $\Theta(\kappa)$, which keeps $\|V_\kappa'''\|_\infty$ bounded. At the matched covariance $c_\kappa = 1/\kappa$, the Gaussian-averaged curvature satisfies $c_\kappa A(m, c_\kappa) = 1 + o(1)$, so both covariance maps are almost stationary. Meanwhile $b(m, c_\kappa)/m = \Theta(1)$, and hence the relative mean displacement per step is only $hc_\kappa b(m, c_\kappa)/m = \Theta(\gamma/\kappa^2)$.

Gaussian leakage between neighboring bumps is small enough to shadow the reference orbit for $\Theta(\kappa^2/\gamma)$ steps. Strong convexity in the mean and control of the shadowing error then show that the objective gap retains a fixed fraction of its initial value. The bump placement, uniform regularity certificate, simultaneous shadowing of the two covariance maps, and objective comparison are proved in Section B.2. $\square$

Remark 3.9 (Scope of discrete sharpness). The target in Theorem 3.8 may depend on the fixed value of γ . The theorem proves sharpness of the factor κ/h for a constant-factor reduction by an explicit method committed to one globally fixed step. It does not cover adaptive relative-curvature steps, backtracking or Armijo line searches, or implicit and Rosenbrock-stabilized updates, and it does not claim that the entire factor $\log(\Delta_0/\delta)$ is necessary at every target accuracy.

3.5 Removing the covariance burn-in

The burn-in can be removed without changing the subsequent Fisher–Rao dynamics. Two optional initializers are useful in different oracle models: a short Bures–Wasserstein transport stage uses the averaged quantities $g(a)$ and $A(a)$, whereas a quadratic rescue uses one pointwise gradient–Hessian query.

Proposition 3.10 (Bures–Wasserstein transport warm start). *Consider first the Gaussian Bures–Wasserstein flow*

$$\dot{m}_t = g(a_t), \quad \dot{C}_t = 2I - C_t A(a_t) - A(a_t) C_t. \quad (71)$$

If $C_0 \succeq \lambda_0 I$, then

$$C_t \succeq \left[\frac{1}{\beta} + \left(\lambda_0 - \frac{1}{\beta} \right) e^{-2\beta t} \right] I. \quad (72)$$

For $c \in (0, 1)$, define

$$T_W(c) := \begin{cases} 0, & \lambda_0 \geq c/\beta, \\ \frac{1}{2\beta} \log \frac{1 - \beta\lambda_0}{1 - c}, & \lambda_0 < c/\beta. \end{cases} \quad (73)$$

Then $C_{T_W(c)} \succeq (c/\beta)I$, the energy has not increased, and a permanent switch to the Fisher–Rao flow reaches $\Delta \leq \delta$ within the additional time

$$\min \left\{ \frac{\kappa}{c} \log_+ \frac{\Delta_0}{\delta}, \log_+ \frac{\kappa}{c\beta_\star} + \kappa_\star \log_+ \frac{\Delta_0}{\delta} \right\}. \quad (74)$$

For fixed c , the warm-up (73) is $O(1/\beta)$, independently of how small λ_0 is.

There is also a one-step discrete version. For any $c \in (0, 1]$ and $a = (m, C)$, let

$$\eta = \frac{c}{\beta}, \quad M = I - \eta A(a), \quad m^+ = m + \eta g(a), \quad \tilde{C} = MCM, \quad (75)$$

and set

$$C^+ = \frac{1}{2} \left(\tilde{C} + 2\eta I + [\tilde{C}(\tilde{C} + 4\eta I)]^{1/2} \right). \quad (76)$$

If $C \preceq \lambda_{\max} I$, then $\eta I \preceq C^+ \preceq \lambda_{\max} I$ and $\mathcal{E}(a^+) \leq \mathcal{E}(a)$. Thus, if $\lambda_0 < c/\beta$, one such step followed by either Fisher–Rao discretization replaces the burn-in term by one iteration and gives

$$N_{W \rightarrow \bullet}^{\text{glob}}(\delta; c) = 1 + O\left(\frac{\kappa}{ch} \log_+ \frac{\Delta_0}{\delta} \right), \quad \bullet \in \{\text{R}, \text{KL}\}, \quad (77)$$

under the corresponding stepsize restriction in (61) or (63). If the initial floor already exceeds c/β , the transport step is skipped and the additive one is absent.

Proof idea. An eigenvalue $\mu_i(t)$ of C_t along (71) satisfies $\dot{\mu}_i = 2(1 - \mu_i h_i)$ for some $h_i \in [\alpha, \beta]$. Scalar comparison with $\dot{\ell} = 2(1 - \beta\ell)$ proves (72). The flow is the Gaussian Bures–Wasserstein gradient flow of $\mathcal{E}$, so its energy decreases [6]. After the switch, the floor c/β persists under Fisher–Rao evolution; the original-coordinate PL estimate gives the first term in (74), while Theorem 3.2 applied to $\hat{C} \succeq (c/\kappa)I$ gives the second.

For the discrete step, the scalar map $\psi_\eta(x) = (x + 2\eta + \sqrt{x(x + 4\eta)})/2$ is increasing and satisfies $\psi_\eta(0) = \eta$, proving the lower floor. Its identity $\psi_\eta((1 - \eta/L)^2 L) = L$ proves the upper barrier. Energy decrease is the standard Bures–Wasserstein forward–backward inequality for $\eta \leq 1/\beta$ [4]. The bounds in Theorem 3.7, restarted from the new floor, yield (77). $\square$

The additive covariance growth in (72) is coordinate-dependent: the Bures–Wasserstein flow is not affine invariant. We use it only as a finite warm start followed by a permanent switch; no general Wasserstein–Fisher–Rao dynamics is needed.

Proposition 3.11 (One-query quadratic rescue). *Suppose a pointwise Hessian oracle is available. Given m_{in} , query*

$$G_{\text{in}} = \nabla V(m_{\text{in}}), \quad H_{\text{in}} = \nabla^2 V(m_{\text{in}}), \quad (78)$$

and initialize

$$\boxed{m_0 = m_{\text{in}} - H_{\text{in}}^{-1} G_{\text{in}}, \quad C_0 = H_{\text{in}}^{-1}.} \quad (79)$$

Then

$$\frac{1}{\beta} I \preceq C_0 \preceq \frac{1}{\alpha} I, \quad (80)$$

so both deterministic Fisher–Rao schemes start with zero covariance burn-in. In particular, with Δ_0 denoting the post-rescue gap,

$$N_{\text{rescue} \rightarrow \bullet}^{\text{glob}}(\delta) = 1 + O\left(\frac{\kappa}{h} \log_+ \frac{\Delta_0}{\delta}\right), \quad \begin{cases} 0 < h \leq (2\kappa)^{-1}, & \bullet = \text{R}, \\ 0 < h \leq \kappa^{-1}, & \bullet = \text{KL}, \end{cases} \quad (81)$$

where the additive one counts the gradient–Hessian query.

If $V(x) = \frac{1}{2}(x - \mu)^\top H(x - \mu) + c$ with $H \succ 0$, then (79) gives $(m_0, C_0) = (\mu, H^{-1}) = a_\star$ and $\Delta_0 = 0$. Over the class of unknown Gaussian targets, the pointwise gradient–Hessian oracle complexity of exact recovery is therefore exactly one.

Proof. The inverse map is operator antitone, so (31) immediately gives (80). Hence $\lambda_0 \geq 1/\beta$, $\lambda_{\text{max}} = 1/\alpha$, and Theorem 3.7 starts directly in its localization phase, proving (81). For a quadratic potential, $G_{\text{in}} = H(m_{\text{in}} - \mu)$ and $H_{\text{in}} = H$, which proves exact recovery. Zero queries cannot recover every unknown pair (μ, H) , while one query returns both H and $H(m_{\text{in}} - \mu)$; thus one query is also necessary. $\square$

The transport warm start and quadratic rescue modify only the initialization stage. They do not remove the global-conditioning obstruction: the bump-train lower bound begins at $C_0 = \kappa^{-1}$, after covariance burn-in has already ended. Nor do they change the local non-Gaussian stiffness studied in the next section. Covariance warm-up, global localization, and local contraction are therefore distinct parts of the deterministic rate landscape.

4 Local spectral theory and global-to-local convergence

Global energy decay eventually places the iterates near the optimizing Gaussian, but it does not determine the local rate. After optimizer whitening, the derivative of the Fisher–Rao field contains first- and second-Hermite correlations of the target Hessian. This section identifies the corresponding self-adjoint generator, bounds its spectrum, and then controls the nonlinear dynamics by retaining the Gaussian part of the field exactly.

4.1 Optimizer whitening and the score generator

Work in the optimizer-whitened coordinates introduced in Section 1.1. Thus $a_\star = (0, I)$,

$$\mathbb{E}_{\mathcal{N}(0, I)} \nabla \widehat{V}(Z) = 0, \quad \mathbb{E}_{\mathcal{N}(0, I)} \nabla^2 \widehat{V}(Z) = I,$$

and

$$\alpha_\star I \preceq \nabla^2 \widehat{V}(z) \preceq \beta_\star I.$$

For $\xi = (u, X) \in \mathbb{R}^d \times \mathbb{S}(d)$, use the equilibrium Fisher–Rao norm

$$\|\xi\|_\star^2 := \|u\|^2 + \frac{1}{2} \|X\|_{\text{F}}^2. \quad (82)$$

Let $B(Z) := \nabla^2 \hat{V}(Z)$. The first- and second-Hermite score operators are

$$\mathcal{T}[X]_k := \mathbb{E}[\text{Tr}(XB(Z))Z_k], \quad \mathcal{T}^*[u]_{ij} := \mathbb{E}[(u^\top Z)B_{ij}(Z)], \quad (83)$$

$$\mathcal{H}[X]_{ij} := \mathbb{E}[B_{ij}(Z)(Z^\top XZ - \text{Tr } X)]. \quad (84)$$

We use the operator norm

$$\|\mathcal{T}\| := \sup_{X \neq 0} \frac{\|\mathcal{T}[X]\|}{\|X\|_F}. \quad (85)$$

Gaussian integration by parts gives

$$u^\top \mathcal{T}[X] = \text{Tr}(\mathcal{T}^*[u]X), \quad \text{Tr}(\mathcal{H}[X]Y) = \text{Tr}(X\mathcal{H}[Y]).$$

Consequently, the negative Jacobian of the Fisher–Rao field at equilibrium is the self-adjoint operator

$$\mathcal{L}_\star(u, X) = \left(u + \frac{1}{2}\mathcal{T}[X], X + \mathcal{T}^*[u] + \frac{1}{2}\mathcal{H}[X] \right). \quad (86)$$

Write

$$\gamma_\star := \lambda_{\min}(\mathcal{L}_\star), \quad L_\star := \lambda_{\max}(\mathcal{L}_\star).$$

The dimension dependence enters through a centered diagonal block. Set

$$t_0 := \max\{1, \log(4\beta_\star)\}, \quad \Gamma := \min\left\{\beta_\star - 1, \sqrt{d}(4\sqrt{2t_0} + 4t_0 + 4)\right\}. \quad (87)$$

In particular, $\Gamma = O(1 + \sqrt{d} \log \kappa_\star)$.

Theorem 4.1 (Spectral sandwich for the local generator). *Under the whitened curvature bounds,*

$$\boxed{\max\left\{\alpha_\star, \frac{1}{4 + \Gamma}\right\} \leq \gamma_\star \leq L_\star \leq \min\left\{\beta_\star, \frac{4 + \Gamma}{2}\right\}}. \quad (88)$$

If the target is Gaussian, then $\mathcal{T} = \mathcal{H} = 0$ and $\gamma_\star = L_\star = 1$.

Proof. Diagonalize $X = \text{diag}(\lambda)$ and define

$$\tilde{T}_{ki} := \mathbb{E}[B_{ii}(Z)Z_k], \quad D_{ij} := \mathbb{E}[B_{ii}(Z)(Z_j^2 - 1)].$$

The restriction of $\mathcal{H}$ to diagonal matrices is self-adjoint, so D is symmetric. For $\eta(Z) = u + \text{diag}(\lambda)Z$, two Gaussian integrations by parts give

$$\mathbb{E}[\eta^\top B \eta] = \|u\|^2 + 2u^\top \tilde{T} \lambda + \|\lambda\|^2 + \lambda^\top D \lambda. \quad (89)$$

Positivity of B implies $\tilde{T}^\top \tilde{T} \preceq I + D$. The pointwise upper curvature bound gives $\lambda_{\max}(D) \leq \beta_\star - 1$.

For the second branch of Γ , normalize $\|\lambda\| = 1$ and put $Q_X(Z) = Z^\top XZ - \text{Tr } X$. The exact representation

$$\lambda^\top D \lambda = \mathbb{E}[\text{Tr}(XB(Z))Q_X(Z)]$$

is combined with $\mathbb{E} \text{Tr}(|X|B) = \text{Tr } |X| \leq \sqrt{d}$ and a two-sided Gaussian quadratic-form tail bound. Splitting at t_0 yields

$$|\lambda^\top D \lambda| \leq \sqrt{d}(4\sqrt{2t_0} + 4t_0 + 4).$$

The quadratic form of $\mathcal{L}_\star$ is

$$Q(u, \text{diag } \lambda) = \|u\|^2 + u^\top \tilde{T} \lambda + \frac{1}{2} \|\lambda\|^2 + \frac{1}{4} \lambda^\top D \lambda.$$

Completing the square in u , using the Schur bound, and taking $\nu = (4 + \Gamma)^{-1}$ gives $Q \geq \nu \|(u, X)\|_\star^2$. The same estimates give $Q \leq (4 + \Gamma) \|(u, X)\|_\star^2 / 2$. Finally, applying (89) with u replaced by $2u$ gives the exact representation

$$Q(u, X) = \frac{1}{4} \mathbb{E} \left[(2u + XZ)^\top B(Z) (2u + XZ) \right] + \frac{1}{4} \|X\|_F^2.$$

Since $\mathbb{E} \|2u + XZ\|^2 = 4\|u\|^2 + \|X\|_F^2$ and $\alpha_\star \leq 1 \leq \beta_\star$, the pointwise curvature sandwich gives the direct bounds $\alpha_\star \|\xi\|_\star^2 \leq Q(\xi) \leq \beta_\star \|\xi\|_\star^2$. Taking the stronger lower and upper estimates proves (88). $\square$

Corollary 4.2 (Linearized deterministic maps). *Let $T_{R,h}$ and $T_{KL,h}$ denote the Riemannian and KL maps of Section 2.5, expressed in whitened coordinates. If $0 < h \leq 2/(4 + \Gamma)$, then*

$$\|DT_{R,h}(a_\star)\xi\|_\star \leq \left(1 - \frac{h}{4 + \Gamma}\right) \|\xi\|_\star, \quad (90)$$

$$\|DT_{KL,h}(a_\star)\xi\|_{\star,h} \leq \left(1 - \frac{h}{4 + 2h + \Gamma}\right) \|\xi\|_{\star,h}, \quad (91)$$

where $\|(u, X)\|_{\star,h}^2 = \|u\|^2 + (1 + h) \|X\|_F^2 / 2$.

4.2 The Gaussian core and a nonlinear local region

Write the whitened Fisher–Rao field as

$$F = F_G + H, \quad F_G(m, C) = (-Cm, C - C^2).$$

The field F_G is the exact natural gradient for the standard Gaussian target. It is not treated as a quadratic Taylor remainder. With $\xi = (m, C - I)$,

$$\langle \xi, F_G(m, C) \rangle_\star = -m^\top Cm - \frac{1}{2} \text{Tr}(C(C - I)^2). \quad (92)$$

Thus its dissipation remains useful well beyond the infinitesimal regime.

To turn an energy level into a covariance band, define

$$\phi_{\alpha_\star}(s) := s - 1 - \log s + \frac{\alpha_\star}{2}(s - 1)^2.$$

If $s_1, \dots, s_d$ are the eigenvalues of $C^{1/2}$, strong convexity and the equilibrium first-order conditions imply

$$\Delta(m, C) \geq \frac{\alpha_\star}{2} \|m\|^2 + \sum_{i=1}^d \phi_{\alpha_\star}(s_i). \quad (93)$$

Let $s_-(\delta) < 1 < s_+(\delta)$ be the two roots of $\phi_{\alpha_\star}(s) = \delta$, and set

$$\ell_\delta = s_-(\delta)^2, \quad U_\delta = s_+(\delta)^2.$$

Then the sublevel $\mathcal{U}_\delta := \{a : \Delta(a) \leq \delta\}$ satisfies $\ell_\delta I \preceq C \preceq U_\delta I$. Define the explicit norm-conversion factors

$$\vartheta(\delta) := \sup_{\substack{s \in [s_-(\delta), s_+(\delta)] \\ s \neq 1}} \frac{(s^2 - 1)^2}{2\phi_{\alpha_\star}(s)} \leq \frac{(1 + s_+(\delta))^2}{\alpha_\star + s_+(\delta)^{-2}}, \quad (94)$$

$$\mathfrak{b}(\delta)^2 := \max \left\{ \frac{2}{\alpha_\star}, \vartheta(\delta) \right\}, \quad \mathfrak{b}_h(\delta)^2 := \max \left\{ \frac{2}{\alpha_\star}, (1 + h)\vartheta(\delta) \right\}. \quad (95)$$

Then

$$\begin{aligned} a \in \mathcal{U}_\delta &\implies \|a - a_\star\|_\star \leq \mathfrak{b}(\delta)\sqrt{\delta}, \\ &\|a - a_\star\|_{\star,h} \leq \mathfrak{b}_h(\delta)\sqrt{\delta}. \end{aligned} \quad (96)$$

Let $\mathcal{H}_\delta$ be the star-shaped hull of $\mathcal{U}_\delta$ about $a_\star$. Every point in this compact hull obeys the same covariance band and the same bound (96): its covariance is a convex combination of I and a covariance in $\mathcal{U}_\delta$, while its displacement from $a_\star$ is shortened by the same interpolation factor. Define the non-Gaussian second-differential modulus

$$K_{\text{ng}}(\delta) := \sup_{\substack{a \in \mathcal{H}_\delta \\ \zeta \neq 0}} \frac{\|D^2 H(a)[\zeta, \zeta]\|_\star}{\|\zeta\|_\star^2}. \quad (97)$$

To make this modulus quantitative, put

$$\mathcal{B}_\delta := \sup_{a \in \mathcal{H}_\delta} \|A(a) - I\|_{\text{op}}, \quad (98)$$

$$\Sigma_\delta := \sup_{a \in \mathcal{H}_\delta} \left(\mathbb{E}_{\rho_a} \left\| \nabla^2 \widehat{V}(X) - A(a) \right\|_{\text{F}}^2 \right)^{1/2}, \quad \bar{\Sigma}_\delta := \min\{\Sigma_\delta, \beta_\star - \alpha_\star\}. \quad (99)$$

The Gaussian score calculation in Section C.1 gives the fully explicit estimate

$$K_{\text{ng}}(\delta) \leq \left\{ K_{\text{ng},m}(\delta)^2 + \frac{1}{2} K_{\text{ng},C}(\delta)^2 \right\}^{1/2}, \quad (100)$$

$$K_{\text{ng},m}(\delta) := \frac{U_\delta^{3/2}}{\ell_\delta^2} \left(\sqrt{3} \bar{\Sigma}_\delta + \Sigma_\delta + \sqrt{\Sigma_\delta^2 + 2\mathcal{B}_\delta^2} \right), \quad (101)$$

$$K_{\text{ng},C}(\delta) := \frac{U_\delta^2}{\ell_\delta^2} \left((4\sqrt{2} + \sqrt{6})\Sigma_\delta + 4\mathcal{B}_\delta \right). \quad (102)$$

Thus $K_{\text{ng}}(\delta)$ measures only departure from the Gaussian core: all three non-Gaussian moduli vanish identically for a Gaussian target. Bounded Hessian and Gaussian score differentiation suffice for finiteness; classical third and fourth derivatives of V are not required.

Theorem 4.3 (Gaussian-core local convergence). *Let $0 < \gamma_0 \leq \gamma_\star$, and define*

$$\rho_{\text{core}}(\delta) := 2\gamma_0 - 2 + 2\ell_\delta - K_{\text{ng}}(\delta)\mathfrak{b}(\delta)\sqrt{\delta}. \quad (103)$$

If $\rho_{\text{core}}(\delta) > 0$ and $\Delta(a_0) \leq \delta$, then the Fisher–Rao flow remains in $\mathcal{U}_\delta$ and

$$\|a_t - a_\star\|_\star^2 \leq e^{-\rho_{\text{core}}(\delta)t} \|a_0 - a_\star\|_\star^2. \quad (104)$$

With $\gamma_0 = \max\{\alpha_\star, (4 + \Gamma)^{-1}\}$, the limiting squared-norm rate as $\delta \downarrow 0$ is at least $2\gamma_0$.

Proof. Energy dissipation makes $\mathcal{U}_\delta$ forward invariant. Equation (92) contributes $-2\ell_\delta \|\xi\|_\star^2$ to the derivative of the squared norm. The linear part of H contributes at most $2(1 - \gamma_0) \|\xi\|_\star^2$. Taylor’s theorem on $\mathcal{H}_\delta$, together with (96), bounds the remaining contribution by $K_{\text{ng}}(\delta)\mathfrak{b}(\delta)\sqrt{\delta} \|\xi\|_\star^2$. The result follows from Grönwall’s inequality. $\square$

Corollary 4.4 (Exact Gaussian region). *For a Gaussian target, $\gamma_\star = 1$, $K_{\text{ng}} \equiv 0$, and $\rho_{\text{core}}(\delta) = 2\ell_\delta$. For every $\rho \in (0, 2)$, the largest energy sublevel on which the flow has squared-norm rate at least ρ is*

$$\Delta \leq \Delta_{\text{G}}^\sharp(\rho) := \frac{1}{2} \left(\frac{\rho}{2} - 1 - \log \frac{\rho}{2} \right). \quad (105)$$

This threshold is independent of dimension.

Proof. For the standard Gaussian, each covariance eigenvalue solves $\dot{c} = c(1 - c)$. The instantaneous squared-norm rate is at least $2\lambda_{\min}(C)$, with equality for a centered covariance perturbation supported on a minimal eigendirection. If $\Delta \leq \Delta_{\text{G}}^\sharp(\rho)$, the Gaussian energy formula forces every covariance eigenvalue to be at least $\rho/2$, so the rate is at least ρ . Conversely, setting one eigenvalue to $c < \rho/2$ gives rate $2c < \rho$ at energy cost $\frac{1}{2}(c - 1 - \log c)$. Letting $c \uparrow \rho/2$ shows that every strictly larger energy sublevel contains such a state, proving sharpness. $\square$

4.3 Discrete local contractions and entry

For the KL map, the Gaussian covariance eigenvalue and mean factors are

$$c^+ = \frac{(1+h)c}{1+hc}, \quad r_C(c) = (1+hc)^{-2}, \quad r_m(c) = (1-hc)^2.$$

Hence, for $q \in (0, 1)$, the exact invariant interval producing contraction factor q is

$$I_h^{\text{KL}}(q) = \left[\frac{q^{-1/2} - 1}{h}, \frac{1 + \sqrt{q}}{h} \right], \quad (106)$$

provided $0 < h < 2$ and $q \geq \max\{(1+h)^{-2}, (1-h)^2\}$. Indeed, the two scalar inequalities $r_C(c), r_m(c) \leq q$ give exactly the endpoints in (106), and the condition on q is precisely what makes that interval contain 1. Since c^+ lies between c and 1, the interval is forward invariant. Hence

$$\text{spec}(C_n) \subseteq I_h^{\text{KL}}(q) \implies \|a_{n+k} - a_\star\|_{\star,h}^2 \leq q^k \|a_n - a_\star\|_{\star,h}^2, \quad k \geq 0.$$

For non-Gaussian targets, let $T_\bullet$ denote either deterministic map and use the norm in Theorem 4.2. Define

$$K_{\text{disc},\bullet}(\delta) := h^{-1} \sup_{\substack{a \in \mathcal{H}_\delta \\ \|\xi\|_{\star,\bullet}=1}} \|D^2 T_\bullet(a)[\xi, \xi]\|_{\star,\bullet}. \quad (107)$$

Here $\|\cdot\|_{\star,\text{R}} = \|\cdot\|_\star$, $\|\cdot\|_{\star,\text{KL}} = \|\cdot\|_{\star,h}$, and

$$\mathbf{b}_\text{R}(\delta) := \mathbf{b}(\delta), \quad \mathbf{b}_\text{KL}(\delta) := \mathbf{b}_h(\delta). \quad (108)$$

The supremum is finite because Gaussian score differentiation makes the two maps twice continuously differentiable on a neighborhood of the compact, spectrally bounded hull; compactness by itself would not be enough. For fixed h , the same continuity gives $K_{\text{disc},\bullet}(\delta) \rightarrow K_{\text{disc},\bullet}(0) < \infty$ as $\delta \downarrow 0$, where the value at zero is the same supremum with $a = a_\star$. Since $\mathbf{b}_\bullet(\delta)\sqrt{\delta} \rightarrow 0$, the radius condition below therefore holds for every sufficiently small positive δ .

Theorem 4.5 (Local deterministic maps). *Suppose*

$$0 < h \leq \min \left\{ \frac{2}{4 + \Gamma}, \frac{1}{\beta_\star \max\{U_\delta, \alpha_\star^{-1}\}} \right\}$$

and

$$K_{\text{disc},\bullet}(\delta) \mathbf{b}_\bullet(\delta) \sqrt{\delta} \leq \gamma_\bullet, \quad \gamma_\text{R} := \frac{1}{4 + \Gamma}, \quad \gamma_\text{KL} := \frac{1}{4 + 2h + \Gamma}.$$

If $\Delta(a_0) \leq \delta$, then the iterates remain in $\mathcal{U}_\delta$ and

$$\|a_n - a_\star\|_{\star,\bullet} \leq \left(1 - \frac{h\gamma_\bullet}{2}\right)^n \|a_0 - a_\star\|_{\star,\bullet}. \quad (109)$$

Proof. Put $\bar{\Lambda}_\delta := \max\{U_\delta, \alpha_\star^{-1}\}$. The stepsize restriction and Theorem 3.6, applied in optimizer-whitened coordinates with curvature bound $\beta_\star$ and covariance upper barrier $\bar{\Lambda}_\delta$, imply $\mathcal{E}(T_\bullet(a)) \leq \mathcal{E}(a)$ for every $a \in \mathcal{U}_\delta$. Thus $\mathcal{U}_\delta$ is invariant by induction.

Write $\xi_n = a_n - a_\star$ and $M_\bullet = DT_\bullet(a_\star)$. Since $T_\bullet(a_\star) = a_\star$, the integral Taylor formula gives

$$a_{n+1} - a_\star = M_\bullet \xi_n + \int_0^1 (1-s) D^2 T_\bullet(a_\star + s\xi_n)[\xi_n, \xi_n] ds.$$

The whole segment lies in $\mathcal{H}_\delta$. By Theorem 4.2, the linear term has norm at most $(1-h\gamma_\bullet) \|\xi_n\|_{\star,\bullet}$, whereas (107), the radius hypothesis, and the energy-to-norm conversion with $\mathbf{b}_\bullet$ bound the remainder by

$$\frac{h}{2} K_{\text{disc},\bullet}(\delta) \|\xi_n\|_{\star,\bullet}^2 \leq \frac{h\gamma_\bullet}{2} \|\xi_n\|_{\star,\bullet}.$$

Adding the two estimates and iterating proves (109). $\square$

Remark 4.6 (What is explicit). The spectral gap, the stepsize restriction, the energy-to-norm conversion, and the exact Gaussian intervals are explicit. For a general target, the certified radius is constructive through (107), but the current theory does not provide a closed-form bound for that supremum in the target parameters. We therefore do not call the general discrete radius fully explicit.

An online switch need not know $\mathcal{E}(a_*)$. In original coordinates, set

$$\mathcal{R}_{\text{BW}}(a)^2 := \|g(a)\|^2 + \text{Tr}[(C^{-1} - A(a))C(C^{-1} - A(a))], \quad (110)$$

where $g(a) = -\mathbb{E}_{\mathcal{N}(m,C)} \nabla V$ and $A(a) = \mathbb{E}_{\mathcal{N}(m,C)} \nabla^2 V$. The Bures–Wasserstein PL inequality gives

$$\mathcal{R}_{\text{BW}}(a)^2 \geq 2\alpha\Delta(a). \quad (111)$$

Thus $\mathcal{R}_{\text{BW}}(a_n)^2 \leq 2\alpha\delta_{\text{loc}}$ certifies entry into any previously validated local level δ_{loc} . This is an exact-expectation gate. A Monte Carlo implementation requires a concentration margin, and the stochastic theory still needs its separate containment argument.

4.4 A sharp growing-dimensional obstruction

The universal lower bound in Theorem 4.1 cannot be replaced by a dimension-free logarithmic rate under a globally Lipschitz Hessian assumption alone.

Theorem 4.7 (Smooth convex-ridge obstruction). *There are numerical constants $c, C > 0$ and a sequence of smooth strongly convex potentials $V_K : \mathbb{R}^{d_K} \rightarrow \mathbb{R}$, $K \rightarrow \infty$, such that the standard Gaussian is the optimizing Gaussian,*

$$\mathbb{E} \nabla V_K(Z) = 0, \quad \mathbb{E} \nabla^2 V_K(Z) = I,$$

and the Hessians have a common global Lipschitz constant. Moreover,

$$d_K = \Theta(K^4), \quad \kappa_* = \Theta(K^2), \quad \|\mathcal{T}\|^2 = \Theta(K), \quad \gamma_* = \Theta(K^{-1}). \quad (112)$$

Equivalently,

$$d_K = \Theta(\kappa_*^2), \quad \|\mathcal{T}\|^2 = \Theta(\kappa_*^{1/2}), \quad \gamma_* = \Theta(\kappa_*^{-1/2}).$$

Proof sketch. The full construction and proof, given in Section C.2, use a convex ridge with one distinguished coordinate and a weak tilt toward a radial block of $d_K - 1$ coordinates. Its averaged Hessian is tuned to be the identity. A distributional Codazzi identity exposes a collective radial fluctuation that gives $\|\mathcal{T}[X]\|^2 \geq K/2 - O(1)$ for an explicit unit Frobenius matrix X . The polarized second-Hermite identity makes the packed generator self-adjoint. The Rayleigh vector

$$Y = \frac{X}{\sqrt{2}}, \quad v = -\frac{\mathcal{T}[X]}{2}$$

has quotient

$$\frac{2 + \langle X, \mathcal{H}[X] \rangle_{\text{F}} - \|\mathcal{T}[X]\|^2}{2 + \|\mathcal{T}[X]\|^2} = O(K^{-1}),$$

which proves $\gamma_* \leq C/K$. On this family $\Gamma = \beta_* - 1 = \Theta(K)$; the lower bound in Theorem 4.1 therefore gives $\gamma_* \geq c/K$. Mollifying the one-dimensional ridge profile and retuning its scalar tilt preserve the exact whitening, the scaling, and the common Hessian-Lipschitz bound. $\square$

Remark 4.8 (Scope of sharpness). The construction saturates the $\beta_* - 1$ branch of Γ , not the $\sqrt{d} \log \kappa_*$ branch. Its dimension grows as $\Theta(\kappa_*^2)$. It rules out a dimension-uniform $(1 + \log \kappa_*)^{-1}$ local rate, but it neither supplies a fixed-dimensional counterexample nor determines the optimal dimension dependence.

4.5 Global-to-local composition

The preceding results now compose without identifying the global energy gap with the local parameter error. Put

$$\underline{\gamma}_\star := \max\left\{\alpha_\star, \frac{1}{4 + \Gamma}\right\}.$$

Theorem 4.9 (Deterministic three-stage complexity). *Assume the global curvature condition (31) and the initial covariance band (49); in particular, Δ_0 , λ_0 , and $\lambda_{\max}$ have the meanings fixed in Section 3, while $\lambda_{0,\star}$ is defined in (35). Let $\delta_c > 0$ satisfy the hypotheses of Theorem 4.3 with $\gamma_0 = \underline{\gamma}_\star$ and $\rho_{\text{core}}(\delta_c) \geq \underline{\gamma}_\star$. Let δ_R and δ_{KL} satisfy the hypotheses of Theorem 4.5 for the indicated schemes. Such levels exist: $\rho_{\text{core}}(\delta) \rightarrow 2\underline{\gamma}_\star$, and, for each fixed admissible h , the discrete radius condition holds at every sufficiently small positive level.*

For the continuous flow, squared local error at most ε is reached by time

$$T_c(\varepsilon) = O\left(\log_+ \frac{1}{\beta_\star \lambda_{0,\star}} + \kappa_\star \log_+ \frac{\Delta_0}{\delta_c} + (4 + \Gamma) \log_+ \frac{\mathbf{b}(\delta_c)^2 \delta_c}{\varepsilon}\right), \quad (113)$$

where the first line may be replaced by the smaller original-coordinate global bound from Theorem 3.2.

For the Riemannian map, suppose

$$0 < h \leq \min\left\{\frac{1}{2\beta\lambda_{\max}}, \frac{2}{4 + \Gamma}, \frac{1}{\beta_\star \max\{U_{\delta_R}, \alpha_\star^{-1}\}}\right\}.$$

Then $\|a_N - a_\star\|_\star^2 \leq \varepsilon$ after

$$N_R(\varepsilon) = O\left(\frac{1}{h} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{h} \log_+ \frac{\Delta_0}{\delta_R} + \frac{4 + \Gamma}{h} \log_+ \frac{\mathbf{b}(\delta_R)^2 \delta_R}{\varepsilon}\right). \quad (114)$$

For the KL map, suppose

$$0 < h \leq \min\left\{\frac{1}{\beta\lambda_{\max}}, \frac{2}{4 + \Gamma}, \frac{1}{\beta_\star \max\{U_{\delta_{\text{KL}}}, \alpha_\star^{-1}\}}\right\}.$$

Then $\|a_N - a_\star\|_{\star,h}^2 \leq \varepsilon$ after

$$N_{\text{KL}}(\varepsilon) = O\left(\frac{1}{h} \log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{h} \log_+ \frac{\Delta_0}{\delta_{\text{KL}}} + \frac{4 + 2h + \Gamma}{h} \log_+ \frac{\mathbf{b}_h(\delta_{\text{KL}})^2 \delta_{\text{KL}}}{\varepsilon}\right). \quad (115)$$

The hidden constants are numerical once the certified entry levels are fixed independently of ε .

Proof. Apply Theorem 3.2 until the continuous trajectory reaches $\mathcal{U}_{\delta_c}$. At entry, (96) bounds the squared local norm by $\mathbf{b}(\delta_c)^2 \delta_c$, and Theorem 4.3 contracts it at rate at least $\underline{\gamma}_\star \geq (4 + \Gamma)^{-1}$. This proves (113).

For either discrete scheme, Theorem 3.7 supplies the covariance-bootstrap and global-localization terms. The energy-to-norm conversion at entry is $\mathbf{b}_\bullet(\delta_\bullet)^2 \delta_\bullet$. Restarting Theorem 4.5 and using $-\log(1 - x) \geq x$ gives the final term in (114) or (115). $\square$

The first two stages above are measured in energy, while the last is measured in an optimizer-frozen parameter norm. Keeping that change of error metric explicit prevents a local spectral statement from being mistaken for a global KL contraction.

5 Price/Hessian stochastic Fisher–Rao algorithms

The deterministic maps of [Section 2.5](#) require Gaussian expectations. We now replace them by minibatch estimates built from a joint gradient–Hessian oracle. The sampled Hessian is not merely a convenient representation of Price’s identity: its samplewise curvature bounds preserve the covariance bands pathwise. This makes the global stochastic theory unconditional. Locally, the score noise has unbounded support, so the appropriate statement is instead a contraction of the killed second moment, paired with a separate finite-horizon containment estimate.

Throughout this section, $\alpha I \preceq \nabla^2 V \preceq \beta I$, $\kappa = \beta/\alpha$, and $a_n = (m_n, C_n)$. We write

$$g(a) := -\mathbb{E}_{\mathcal{N}(m, C)} \nabla V, \quad A(a) := \mathbb{E}_{\mathcal{N}(m, C)} \nabla^2 V, \quad R(a) := C^{-1} - A(a),$$

and recall the Fisher–Rao gradient norm

$$\mathsf{G}(a)^2 = g(a)^\top C g(a) + \frac{1}{2} \left\| C^{1/2} R(a) C^{1/2} \right\|_{\mathsf{F}}^2.$$

5.1 Oracle, estimators, and covariance safety

Definition 5.1 (Price/Hessian oracle). At a query point $x \in \mathbb{R}^d$, the oracle returns the pair $(\nabla V(x), \nabla^2 V(x))$.

Fix a batch size $B \geq 1$. Conditionally on $\mathcal{F}_n := \sigma(a_0, \dots, a_n)$, draw independent $Z_{n,s} \sim \mathcal{N}(0, I)$ and set $X_{n,s} = m_n + C_n^{1/2} Z_{n,s}$. The sticking-the-landing score and Price/Hessian covariance estimates are

$$\bar{b}_n := \frac{1}{B} \sum_{s=1}^B \left[-\nabla V(X_{n,s}) + C_n^{-1/2} Z_{n,s} \right], \quad (116)$$

$$\bar{H}_n := \frac{1}{B} \sum_{s=1}^B \nabla^2 V(X_{n,s}), \quad \bar{R}_n := C_n^{-1} - \bar{H}_n. \quad (117)$$

Gaussian integration by parts and Price’s identity give

$$\mathbb{E}[\bar{b}_n \mid \mathcal{F}_n] = g(a_n), \quad \mathbb{E}[\bar{R}_n \mid \mathcal{F}_n] = R(a_n). \quad (118)$$

The same samples may be used in both averages. None of the arguments below assumes conditional independence between the score and Hessian components. Independence is used only across the B sample pairs.

For a stepsize $h_n > 0$, the two stochastic maps are

$$m_{n+1} = m_n + h_n C_n \bar{b}_n, \quad C_{n+1} = C_n^{1/2} \exp\left(h_n (I - C_n^{1/2} \bar{H}_n C_n^{1/2})\right) C_n^{1/2}, \quad (119)$$

$$m_{n+1} = m_n + h_n C_n \bar{b}_n, \quad C_{n+1} = (1 + h_n)(C_n^{-1} + h_n \bar{H}_n)^{-1}, \quad (120)$$

for the Riemannian and KL variants, respectively. Strong log-concavity holds samplewise:

$$\alpha I \preceq \bar{H}_n \preceq \beta I \quad \text{almost surely.} \quad (121)$$

This property would be lost for the gradient-only Price representation of $A(a)$, whose individual matrices need not have bounded spectrum.

We initialize the stochastic phase by the quadratic rescue from [Section 3](#). One deterministic oracle query at m_{in} gives

$$H_{\text{in}} = \nabla^2 V(m_{\text{in}}), \quad m_0 = m_{\text{in}} - H_{\text{in}}^{-1} \nabla V(m_{\text{in}}), \quad C_0 = H_{\text{in}}^{-1}. \quad (122)$$

Thus $\beta^{-1} I \preceq C_0 \preceq \alpha^{-1} I$. For a quadratic potential, (122) is the exact optimizing Gaussian.

Proposition 5.2 (Pathwise covariance bands). *Initialize by (122).*

(i) *If (119) is run with $0 < h_n \leq 1/\kappa$ for every n , then almost surely*

$$\frac{1}{2\beta}I \preceq C_n \preceq \frac{1}{\alpha}I \quad (n \geq 0).$$

(ii) *If (120) is run with arbitrary $h_n > 0$, then almost surely*

$$\frac{1}{\beta}I \preceq C_n \preceq \frac{1}{\alpha}I \quad (n \geq 0).$$

No stopping, projection, clipping, or local-containment hypothesis is used.

Proof. Condition on an arbitrary realization satisfying (121). For the KL update, the induction hypothesis gives

$$(1 + h_n)\alpha I \preceq C_n^{-1} + h_n \bar{H}_n \preceq (1 + h_n)\beta I.$$

Inversion proves the claimed KL band for every positive stepsize.

For the Riemannian update, put $B_n = C_n^{1/2} \bar{H}_n C_n^{1/2}$. Since $e^M \succeq I + M$ for symmetric M ,

$$C_{n+1}^{-1} = C_n^{-1/2} e^{h_n(B_n - I)} C_n^{-1/2} \succeq (1 - h_n)C_n^{-1} + h_n \bar{H}_n \succeq \alpha I,$$

where $h_n \leq 1/\kappa \leq 1$. This proves the upper covariance bound. It also implies $B_n \preceq \kappa I$, hence $0 \preceq h_n B_n \preceq I$. Convexity of the scalar exponential on $[0, 1]$, transferred spectrally, gives

$$e^{h_n B_n} \preceq I + (e - 1)h_n B_n.$$

If $C_n^{-1} \preceq 2\beta I$, then

$$C_{n+1}^{-1} \preceq e^{-h_n} [C_n^{-1} + (e - 1)h_n \bar{H}_n] \preceq \beta e^{-h_n} [2 + (e - 1)h_n] I \preceq 2\beta I.$$

The last scalar inequality holds for $0 \leq h_n \leq 1$. The rescue lies in the stronger initial band, so induction completes the proof. $\square$

5.2 Intrinsic sticking-the-landing variance

For one draw $X \sim \mathcal{N}(m, C)$, write

$$\hat{b} = -\nabla V(X) + C^{-1}(X - m), \quad \varepsilon_b = \hat{b} - g(a), \quad \varepsilon_H = \nabla^2 V(X) - A(a).$$

The intrinsic quantity controlling the additive error is the fluctuation of the relative Hessian.

Definition 5.3 (Intrinsic Hessian fluctuation). For $a = (m, C)$, define

$$\Psi(a) := \mathbb{E} \left\| C^{1/2} (\nabla^2 V(X) - A(a)) C^{1/2} \right\|_F^2, \quad X \sim \mathcal{N}(m, C). \quad (123)$$

Lemma 5.4 (Intrinsic STL variance). *The single-sample score satisfies*

$$\mathbb{E}[\varepsilon_b^\top C \varepsilon_b] \leq \left\| C^{1/2} R(a) C^{1/2} \right\|_F^2 + \Psi(a). \quad (124)$$

If $\zeta(a; X) = (-C\varepsilon_b, C\varepsilon_H C)$ denotes the single-sample tangent noise, then

$$\mathbb{E} \|\zeta(a; X)\|_{a, \text{FR}}^2 \leq 2\mathbf{G}(a)^2 + \frac{3}{2}\Psi(a). \quad (125)$$

Consequently, the conditional tangent variance of a batch average is at most $B^{-1}(2\mathbf{G}(a_n)^2 + 3\Psi(a_n)/2)$.

Proof. Set $F(x) = -\nabla V(x) + C^{-1}(x - m)$. The Gaussian Poincaré inequality, applied componentwise to $C^{1/2}F(X)$, yields

$$\mathbb{E}[\varepsilon_b^\top C \varepsilon_b] \leq \mathbb{E} \left\| C^{1/2}(C^{-1} - \nabla^2 V(X))C^{1/2} \right\|_{\text{F}}^2.$$

Decompose the matrix on the right as $C^{1/2}R(a)C^{1/2} - C^{1/2}\varepsilon_H C^{1/2}$. The cross term has zero expectation, proving (124). The Fisher–Rao covariance block has weight 1/2, so

$$\mathbb{E} \|\zeta\|_{a, \text{FR}}^2 = \mathbb{E}[\varepsilon_b^\top C \varepsilon_b] + \frac{1}{2}\Psi(a) \leq \left\| C^{1/2}R C^{1/2} \right\|_{\text{F}}^2 + \frac{3}{2}\Psi(a).$$

The covariance residual term is at most $2\mathbf{G}(a)^2$. Centering and independence across the batch divide the variance by B . $\square$

There are two useful ways to control Ψ . The first uses only the curvature bounds and is self-bounding. The second uses a Lipschitz Hessian and retains the exact Gaussian cancellation.

Lemma 5.5 (Two fluctuation regimes). (i) *If $C \preceq \Lambda I$ and $L_\Lambda = \beta\Lambda$, then*

$$\Psi(a) \leq 2L_\Lambda d + L_\Lambda \left\| C^{1/2}R(a)C^{1/2} \right\|_{\text{F}}^2 \leq 2L_\Lambda d + 2L_\Lambda \mathbf{G}(a)^2. \quad (126)$$

In particular, on either rescued upper band, $L_\Lambda = \kappa$, and

$$\mathbb{E} \left\| \widehat{\text{grad}} \mathcal{E}(a) \right\|_{a, \text{FR}}^2 \mid a \leq \rho_B \mathbf{G}(a)^2 + \frac{3\kappa d}{B}, \quad \rho_B := 1 + \frac{2 + 3\kappa}{B}. \quad (127)$$

(ii) *Suppose $\|\nabla^2 V(x) - \nabla^2 V(y)\|_{\text{op}} \leq L_H \|x - y\|$. Then*

$$\Psi(a) \leq L_H^2 \text{Tr}(C^2) \text{Tr} C \leq L_H^2 \|C\|_{\text{op}} (\text{Tr} C)^2. \quad (128)$$

On $C \preceq \alpha^{-1}I$, writing $\bar{L}_H := L_H/\alpha^{3/2}$,

$$\Psi(a) \leq \Psi_H := d^2 \bar{L}_H^2. \quad (129)$$

Proof. For the first part, let $B_X = C^{1/2}\nabla^2 V(X)C^{1/2}$ and $S_A = \mathbb{E}B_X = C^{1/2}A(a)C^{1/2}$. Since $0 \preceq B_X \preceq L_\Lambda I$,

$$\Psi = \mathbb{E} \text{Tr}(B_X^2) - \text{Tr}(S_A^2) \leq L_\Lambda \text{Tr} S_A.$$

The scalar inequality $s \leq 2 + (s - 1)^2$, applied to the eigenvalues of S_A , gives $\text{Tr} S_A \leq 2d + \|S_A - I\|_{\text{F}}^2$. Since $S_A - I = -C^{1/2}RC^{1/2}$, this proves (126). Combining it with (125) gives (127).

For the second part, put $F(Z) = C^{1/2}\nabla^2 V(m + C^{1/2}Z)C^{1/2}$. Componentwise Gaussian Poincaré gives $\Psi \leq \mathbb{E} \|D_Z F(Z)\|_{\text{HS}}^2$. For a unit vector u ,

$$D_Z F(Z)[u] = C^{1/2}M_u C^{1/2}, \quad \|M_u\|_{\text{op}} \leq L_H (u^\top C u)^{1/2}.$$

In an eigenbasis of C , symmetry of M_u and $2c_i c_j \leq c_i^2 + c_j^2$ imply

$$\left\| C^{1/2}M_u C^{1/2} \right\|_{\text{F}}^2 \leq \text{Tr}(C^2 M_u^2) \leq \|M_u\|_{\text{op}}^2 \text{Tr}(C^2).$$

Summing over an orthonormal basis of directions u proves the first inequality in (128). The remaining inequalities follow from $\text{Tr}(C^2) \leq \|C\|_{\text{op}} \text{Tr} C$ and the rescued upper band. $\square$

The anisotropic form in (128) can be much smaller than $d^2 \bar{L}_H^2$ when the covariance spectrum decays. The worst-case d^2 bound is nevertheless the uniform consequence of the band. For a quadratic target, $L_H = 0$ and, more strongly, the integrand in (123) is identically zero.

5.3 Global stochastic convergence

On the Riemannian band of [Theorem 5.2](#), the global gradient-domination estimate from [Section 3](#) reads $\mathsf{G}(a_n)^2 \geq \Delta_n/(2\kappa)$. On the tighter KL band it reads $\mathsf{G}(a_n)^2 \geq \Delta_n/\kappa$, where $\Delta_n = \mathcal{E}(a_n) - \mathcal{E}(a_*)$.

Lemma 5.6 (Intrinsic one-step inequalities). *Conditionally on $\mathcal{F}_n$, the Riemannian update with $0 < h \leq 1/\kappa$ satisfies*

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - h \left[1 - \kappa h \left(1 + \frac{2}{B} \right) \right] \mathsf{G}(a_n)^2 + \frac{3\kappa h^2}{2B} \Psi(a_n). \quad (130)$$

The KL update with $0 < h \leq 1/(8\kappa)$ satisfies

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \frac{h}{2} \mathsf{G}(a_n)^2 + \frac{7\kappa h^2}{4B} \Psi(a_n). \quad (131)$$

Both estimates remain valid when the score and Hessian use the same batch.

Proof. The retraction smoothness estimate from the deterministic analysis gives, pathwise on $C_n \preceq \alpha^{-1}I$,

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - h \left\langle \text{grad } \mathcal{E}(a_n), \widehat{\text{grad}} \mathcal{E}(a_n) \right\rangle_{a_n} + \kappa h^2 \left\| \widehat{\text{grad}} \mathcal{E}(a_n) \right\|_{a_n, \text{FR}}^2.$$

Taking conditional expectation, using unbiasedness and [Theorem 5.4](#), proves (130).

For the KL map, suppress the time index and set

$$S = C^{1/2} R(a) C^{1/2}, \quad D = C^{1/2} (\overline{H} - A(a)) C^{1/2}, \quad P = C^{1/2} \overline{H} C^{1/2}, \quad \overline{S} = S - D.$$

The covariance update has the exponential-retraction representation

$$C^+ = C^{1/2} e^{W(P)} C^{1/2}, \quad W(P) = \log(1 + h)I - \log(I + hP). \quad (132)$$

For every eigenvalue $p \in [1/\kappa, \kappa]$, the integral formula $W(p) = \int_p^1 h(1 + hr)^{-1} dr$ gives

$$\|W(P)\|_F \leq h \|\overline{S}\|_F, \quad \|W(P) - h\overline{S}\|_F \leq \kappa h^2 \|\overline{S}\|_F. \quad (133)$$

Write $M = g^\top C g$, $V = \|S\|_F^2$, and $\sigma^2 = \Psi(a)/B$. Then $\mathbb{E} \|D\|_F^2 = \sigma^2$ and (124) gives $\mathbb{E}[\overline{b}^\top C \overline{b}] \leq M + V/B + \sigma^2$. Applying the same retraction smoothness bound to the logarithmic tangent in (132), and then using (133), yields

$$\mathbb{E}[\Delta^+ \mid \mathcal{F}] \leq \Delta - h(1 - \kappa h)M - h \left[\frac{1}{2} - \kappa h \left(1 + \frac{1}{B} \right) \right] V + \frac{7}{4} \kappa h^2 \sigma^2.$$

This calculation uses only $\mathbb{E}D = 0$ and the two marginal second moments; there is no score–Hessian independence step. If $h \leq 1/(8\kappa)$, the coefficients of M and V are at least $h/2$ and $h/4$, respectively. Since $\mathsf{G}^2 = M + V/2$, (131) follows. $\square$

The next result displays the two fluctuation regimes side by side. Every constant-step conclusion is a geometric approach to the stated residual upper bound. It is not a claim of convergence to arbitrary accuracy, nor a matching lower bound on the algorithm’s stationary error.

Theorem 5.7 (Constant-step global rates and residual floors). *Initialize by (122) and let $B \geq 1$.*

- (i) *Under the Hessian-Lipschitz assumption, the Riemannian method with $h = [2\kappa(1 + 2/B)]^{-1}$ satisfies*

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{1}{8\kappa^2(1 + 2/B)} \right)^N \Delta_0 + \frac{3\kappa}{B + 2} d^2 \overline{L}_H^2. \quad (134)$$

(ii) Under the same assumption, the KL method with $h = 1/(8\kappa)$ satisfies

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{1}{16\kappa^2}\right)^N \Delta_0 + \frac{7\kappa}{16B} d^2 \bar{L}_H^2. \quad (135)$$

(iii) Without Hessian-Lipschitzness, set $\rho_B = 1 + (2 + 3\kappa)/B$. The Riemannian method with $h = (2\kappa\rho_B)^{-1}$ satisfies

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{1}{8\kappa^2\rho_B}\right)^N \Delta_0 + \frac{6\kappa^2 d}{B + 2 + 3\kappa}. \quad (136)$$

(iv) Without Hessian-Lipschitzness, the KL method with $h = \min\{1/(8\kappa), B/(14\kappa^2)\}$ satisfies

$$\mathbb{E}\Delta_N \leq \left(1 - \frac{h}{4\kappa}\right)^N \Delta_0 + \frac{14\kappa^3 h d}{B}. \quad (137)$$

If $B \leq 7\kappa/4$, the last floor is exactly κd and the contraction factor is $1 - B/(56\kappa^3)$.

Proof. For (i), the bracket in (130) equals $1/2$. Use $\Psi \leq \Psi_H = d^2 \bar{L}_H^2$ and $G^2 \geq \Delta/(2\kappa)$, then sum the geometric recursion. Its additive increment is $3\kappa h^2 \Psi_H/(2B)$, and division by the drift $h/(4\kappa)$ gives the floor in (134). Part (ii) follows in the same way from (131) and $G^2 \geq \Delta/\kappa$; the drift is $h/(2\kappa)$.

For (iii), substitute $\Psi \leq 2\kappa d + 2\kappa G^2$ into (130). This gives

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - h(1 - \kappa h \rho_B) G(a_n)^2 + \frac{3\kappa^2 h^2 d}{B}.$$

At $h = (2\kappa\rho_B)^{-1}$, the first bracket is $1/2$. Applying the Riemannian-band gradient domination and unrolling gives (136). For (iv), substitution into (131) gives

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \frac{h}{2} \left(1 - \frac{7\kappa^2 h}{B}\right) G(a_n)^2 + \frac{7\kappa^2 h^2 d}{2B}.$$

The chosen stepsize makes the bracket at least $1/2$. The KL-band gradient domination and a geometric sum prove (137). $\square$

For $B = 1$, the Hessian-Lipschitz bounds require $\tilde{O}(\kappa^2)$ rounds to reach an accuracy above their displayed floors. Under curvature bounds alone, both single-sample bounds require $\tilde{O}(\kappa^3)$ rounds. Taking $B = \Theta(\kappa)$ reduces the latter round complexity to $\tilde{O}(\kappa^2)$, but the number of Price/Hessian oracle pairs remains $\tilde{O}(\kappa^3)$. The one-time rescue adds one oracle pair.

Constant steps are not necessary if convergence all the way to zero is required in the Hessian-Lipschitz regime.

Theorem 5.8 (Floor-free decreasing steps). *Under the Hessian-Lipschitz assumption, initialize by (122) and run either method with*

$$h_n = \frac{8\kappa}{n + n_0}, \quad n_0 \geq 64\kappa^2. \quad (138)$$

Then, for all $N \geq 0$,

$$\mathbb{E}\Delta_N \leq \frac{K}{N + n_0}, \quad K := \max \left\{ \frac{112\kappa^3 d^2 \bar{L}_H^2}{B}, n_0 \Delta_0 \right\}. \quad (139)$$

Thus (139) is a genuine $O(N^{-1})$ decay to zero. The algorithm does not need to know L_H ; that constant appears only in the certificate.

Proof. The schedule satisfies $h_n \leq 1/(8\kappa)$. The KL band persists for all positive steps. The proof of [Theorem 5.2](#) also works step by step for the Riemannian schedule because each $h_n \leq 1/\kappa$. Hence (130) and (131) apply at every iteration. In both cases, using the weaker of the two band-dependent drift constants and $\Psi \leq \Psi_H$,

$$\mathbb{E}\Delta_{n+1} \leq \left(1 - \frac{h_n}{4\kappa}\right) \mathbb{E}\Delta_n + \frac{7\kappa h_n^2}{4B} \Psi_H. \quad (140)$$

With $t = n + n_0$, this is

$$\mathbb{E}\Delta_{n+1} \leq \left(1 - \frac{2}{t}\right) \mathbb{E}\Delta_n + \frac{S}{t^2}, \quad S = \frac{112\kappa^3 \Psi_H}{B}.$$

If $K = \max\{S, n_0\Delta_0\}$, induction gives $\mathbb{E}\Delta_n \leq K/(n + n_0)$: assuming the claim at n ,

$$\mathbb{E}\Delta_{n+1} \leq \frac{K}{t} - \frac{2K - S}{t^2} \leq \frac{K(t-1)}{t^2} \leq \frac{K}{t+1}.$$

This proves (139). $\square$

5.4 Local killed-process contraction and containment

We now work in the optimizer-whitened coordinates of [Section 4](#), so $a_\star = (0, I)$. The two equilibrium norms are

$$\|(m, C) - a_\star\|_{\star, R}^2 := \|m\|^2 + \frac{1}{2} \|C - I\|_F^2, \quad \|(m, C) - a_\star\|_{\star, KL}^2 := \|m\|^2 + \frac{1+h}{2} \|C - I\|_F^2. \quad (141)$$

If the original Hessian has Lipschitz modulus L_H , then the whitened potential has modulus

$$L_{H, \star} \leq L_H \|C_\star\|_{\text{op}}^{3/2} \leq \frac{L_H}{\alpha^{3/2}} = \bar{L}_H. \quad (142)$$

For a fixed method and stepsize, call a local energy level $\delta_\bullet$ stochastically admissible if the deterministic second-derivative criterion of [Section 4.3](#) holds and, on $\mathcal{U}_{\delta_\bullet}$,

$$\frac{1}{2}I \preceq C \preceq \frac{3}{2}I, \quad \|a - a_\star\|_{\star, \bullet} \leq 1. \quad (143)$$

Equivalently, one may require $K_{\text{disc}, \bullet}(\delta_\bullet) \mathbf{b}_\bullet(\delta_\bullet) \sqrt{\delta_\bullet} \leq \gamma_\bullet$, with $\gamma_R = 1/(4 + \Gamma)$ and $\gamma_{KL} = 1/(4 + 2h + \Gamma)$, together with (143). Such levels exist for every fixed admissible h by shrinking $\delta_\bullet$. As in [Theorem 4.6](#), this is a constructive existence statement through $K_{\text{disc}, \bullet}$, not a closed-form radius in the target parameters.

The local noise consists of the Euclidean score error ε_b and the whitened Hessian error ε_H .

Lemma 5.9 (Local variance decomposition). *If $I/2 \preceq C \preceq 3I/2$, then*

$$\mathbb{E} \left[\|\varepsilon_b\|^2 + \frac{1}{2} \|\varepsilon_H\|_F^2 \right] \leq V_0 + V_1 \|a - a_\star\|_{\star, R}^2, \quad V_0 := 3d^2 L_{H, \star}^2, \quad V_1 := 24 + 6d L_{H, \star}^2. \quad (144)$$

For a batch average, the right-hand side is divided by B . At $a_\star$, the sharper bound is

$$\mathbb{E} \left[\|\varepsilon_b\|^2 + \frac{1}{2} \|\varepsilon_H\|_F^2 \right] \leq \frac{3}{2} d^2 L_{H, \star}^2.$$

Proof. Gaussian Poincaré applied to the Hessian entries gives $\mathbb{E} \|\varepsilon_H\|_F^2 \leq (3/2) d^2 L_{H, \star}^2$. Applying it to $F(x) = -\nabla \widehat{V}(x) + C^{-1}(x - m)$ gives

$$\mathbb{E} \|\varepsilon_b\|^2 \leq \frac{3}{2} \left(\|C^{-1} - A(a)\|_F^2 + \mathbb{E} \|\varepsilon_H\|_F^2 \right).$$

On the stated band, $\|C^{-1} - I\|_F \leq 2\|C - I\|_F$. Moreover, coupling $m + C^{1/2}Z$ with Z and using $A(a_\star) = I$ yields

$$\|A(a) - I\|_F^2 \leq dL_{H,\star}^2 \left(\|m\|^2 + \|C^{1/2} - I\|_F^2 \right).$$

Since $\|C^{1/2} - I\|_F \leq \|C - I\|_F$, collecting the terms gives (144). At equilibrium the residual $C^{-1} - A(a)$ vanishes and $\|C\|_{\text{op}} = 1$, which gives the sharper constant. Batch averaging divides centered second moments by B . $\square$

Let $T_{\bullet,h}$ denote the corresponding deterministic local map. The following perturbation estimate records the constants needed below.

Lemma 5.10 (One-step stochastic perturbation). *Suppose (143) holds and $\|T_{\bullet,h}(a) - a_\star\|_{\star,\bullet} \leq 1$, with $0 < h \leq 1$. One stochastic step satisfies*

$$\mathbb{E}[\|a^+ - a_\star\|_{\star,\text{R}}^2 \mid a] \leq \|T_{\text{R},h}(a) - a_\star\|_{\star,\text{R}}^2 + \frac{64h^2}{B} (V_0 + V_1 \|a - a_\star\|_{\star,\text{R}}^2), \quad (145)$$

$$\mathbb{E}[\|a^+ - a_\star\|_{\star,\text{KL}}^2 \mid a] \leq \|T_{\text{KL},h}(a) - a_\star\|_{\star,\text{KL}}^2 + \frac{68h^2}{B} (V_0 + V_1 \|a - a_\star\|_{\star,\text{R}}^2). \quad (146)$$

Proof. For the Riemannian covariance update, put $S = I - C^{1/2}A(a)C^{1/2}$ and replace it in the stochastic map by $\hat{S} = S - C^{1/2}\bar{\varepsilon}_H C^{1/2}$. For $S, \hat{S} \preceq I$ and $h \leq 1$, the Fréchet derivative and its integral remainder give

$$\begin{aligned} \|e^{h\hat{S}} - e^{hS}\|_F &\leq eh \|\hat{S} - S\|_F, \\ \|e^{h\hat{S}} - e^{hS} - D \exp(hS)[h(\hat{S} - S)]\|_F &\leq \frac{e}{2} h^2 \|\hat{S} - S\|_{\text{op}} \|\hat{S} - S\|_F. \end{aligned}$$

The linear term is centered. Expanding the squared norm around the deterministic update, the second-moment and bias coefficients are bounded by $81e^2/16$ and $27e/(4\sqrt{2})$, whose sum is less than 64.

For KL, write $P = C^{-1} + hA(a)$ and $\hat{P} = P + h\bar{\varepsilon}_H$. The identities

$$\hat{P}^{-1} - P^{-1} = -P^{-1}(h\bar{\varepsilon}_H)\hat{P}^{-1}$$

and

$$\hat{P}^{-1} - P^{-1} = -P^{-1}(h\bar{\varepsilon}_H)P^{-1} + P^{-1}(h\bar{\varepsilon}_H)P^{-1}(h\bar{\varepsilon}_H)\hat{P}^{-1}$$

control the second moment and the second-order bias. On the covariance band, their aggregate coefficient is $81/2 + 27 = 67.5 < 68$. The score increment is linear and centered in both maps. Applying Theorem 5.9 completes the proof; in the KL case one also uses $\|a - a_\star\|_{\star,\text{R}} \leq \|a - a_\star\|_{\star,\text{KL}}$. $\square$

Set

$\bullet$	$A_\bullet$	$c_\bullet$	$\eta_\bullet$	$\mu_\bullet := \frac{1}{2A_\bullet}$	(147)
R	$4 + \Gamma$	64	$192h^2d^2L_{H,\star}^2/B$		
KL	$4 + 2h + \Gamma$	68	$204h^2d^2L_{H,\star}^2/B$		

Theorem 5.11 (Local non-exit mean-square contraction). *Fix $\bullet \in \{\text{R}, \text{KL}\}$ and suppose*

$$0 < h \leq \min \left\{ \frac{2}{4 + \Gamma}, \frac{B}{4c_\bullet A_\bullet V_1} \right\}. \quad (148)$$

Let $\delta_\bullet$ be stochastically admissible for this h , start in $\mathcal{U}_{\delta_\bullet}$, and define $\tau_\bullet = \inf\{n \geq 0 : a_n \notin \mathcal{U}_{\delta_\bullet}\}$. Then, for every $N \geq 0$,

$$\mathbb{E} \left[\|a_N - a_\star\|_{\star, \bullet}^2 \mathbf{1}_{\{\tau_\bullet > N\}} \right] \leq (1 - \mu_\bullet h)^N \|a_0 - a_\star\|_{\star, \bullet}^2 + \frac{\eta_\bullet}{\mu_\bullet h}. \quad (149)$$

In particular, the two residual terms are

$$\frac{\eta_R}{\mu_R h} = \frac{384h(4 + \Gamma)d^2 L_{H, \star}^2}{B}, \quad (150)$$

$$\frac{\eta_{KL}}{\mu_{KL} h} = \frac{408h(4 + 2h + \Gamma)d^2 L_{H, \star}^2}{B}. \quad (151)$$

The left-hand side of (149) is a killed, or non-exit, second moment. It is not $\mathbb{E} \|a_{N \wedge \tau_\bullet} - a_\star\|_{\star, \bullet}^2$.

Proof. Write $r_n = \|a_n - a_\star\|_{\star, \bullet}$. On $\{n < \tau_\bullet\}$, the deterministic second-derivative criterion and the linearized bounds of Theorem 4.2 give

$$\|T_{\bullet, h}(a_n) - a_\star\|_{\star, \bullet} \leq (1 - \mu_\bullet h)r_n.$$

Apply Theorem 5.10. Since $\mu_\bullet h \leq 1/2$, $(1 - \mu_\bullet h)^2 \leq 1 - 3\mu_\bullet h/2$. The second restriction in (148) gives $c_\bullet h^2 V_1/B \leq \mu_\bullet h/2$. Therefore

$$\mathbb{E}[r_{n+1}^2 \mid \mathcal{F}_n] \leq (1 - \mu_\bullet h)r_n^2 + \eta_\bullet \quad \text{on } \{n < \tau_\bullet\}. \quad (152)$$

Let $M_n = \mathbb{E}[r_n^2 \mathbf{1}_{\{\tau_\bullet > n\}}]$. Because $\{\tau_\bullet > n+1\} \subseteq \{\tau_\bullet > n\}$, $M_{n+1} \leq (1 - \mu_\bullet h)M_n + \eta_\bullet$. Summing this geometric recursion proves (149); substituting (147) gives (150)–(151). $\square$

The missing mass in (149) must be controlled separately. The next proposition does so without a union bound over all possible exit times.

Proposition 5.12 (Finite-horizon containment). *Choose $r \leq 1/(2\sqrt{2})$ such that the closed $\|\cdot\|_{\star, \bullet}$ -ball of radius r about $a_\star$ is contained in $\mathcal{U}_{\delta_\bullet}$, and set*

$$\tau_r := \inf\{n \geq 0 : \|a_n - a_\star\|_{\star, \bullet} > r\}.$$

Under the assumptions of Theorem 5.11, if $r_0 := \|a_0 - a_\star\|_{\star, \bullet} \leq r$, then

$$\mathbb{P}(\tau_r \leq N) \leq \frac{r_0^2 + N\eta_\bullet}{r^2} \quad (N \geq 1). \quad (153)$$

Consequently, $\mathbb{P}(\tau_r \leq N) \leq \varepsilon_{\text{exit}}$ follows from the two separate conditions

$$r_0^2 \leq \frac{\varepsilon_{\text{exit}}}{2} r^2, \quad B \geq \frac{2c'_\bullet N h^2 d^2 L_{H, \star}^2}{\varepsilon_{\text{exit}} r^2}, \quad c'_R = 192, \quad c'_{KL} = 204. \quad (154)$$

Thus increasing the batch cannot compensate for an initial point near the outer boundary. Deep entry is necessary in this argument, and the batch needed for survival grows linearly with the prescribed horizon N .

Proof. Freeze the radial process at the exit time: $R_n = r_{n \wedge \tau_r}$. On $\{\tau_r > n\}$, (152) and $1 - \mu_\bullet h \leq 1$ give $\mathbb{E}[R_{n+1}^2 \mid \mathcal{F}_n] \leq R_n^2 + \eta_\bullet$. After exit, the process is constant, so the same inequality remains true. Hence $\mathbb{E}R_N^2 \leq r_0^2 + N\eta_\bullet$. On $\{\tau_r \leq N\}$, one has $R_N > r$; Markov's inequality proves (153). Substitution of (147) gives (154). $\square$

The certified local round complexity follows directly. Up to numerical constants, the largest permitted stepsize is

$$h_{\bullet}^* = \Theta\left(\frac{1}{4 + \Gamma} \min\left\{1, \frac{B}{V_1}\right\}\right),$$

where the KL choice may use $4 + \Gamma + 1$ in place of its mildly implicit A_{KL} . Therefore

$$N_{\text{loc}} = O\left((4 + \Gamma)^2 \max\left\{1, \frac{V_1}{B}\right\} \log \frac{r_0^2}{\varepsilon}\right). \quad (155)$$

In the large-batch branch and with the universal spectral certificate $\Gamma = O(1 + \sqrt{d} \log \kappa_*)$, this becomes

$$O\left((1 + d \log^2 \kappa_*) \log \frac{r_0^2}{\varepsilon}\right).$$

The dependence is quadratic in $4 + \Gamma$, not linear. In particular, the present theorem does not prove an unqualified logarithmic dependence on κ_* . Moreover, (155) is a round count for contraction of the killed moment. Enforcing survival over those rounds may increase the batch according to (154), so it is not by itself a high-probability oracle complexity.

The global and local statements can nevertheless be composed without hiding these qualifications.

Corollary 5.13 (Stochastic global-to-local composition). *Let an inner level δ_{in} and an outer radius r satisfy*

$$\mathbf{b}_{\bullet}(\delta_{\text{in}})^2 \delta_{\text{in}} \leq \frac{\varepsilon_{\text{exit}}}{2} r^2,$$

with the radius- r ball contained in a stochastically admissible outer level. Suppose a global phase produces

$$\mathbb{E} \Delta_{N_1} \leq \varepsilon_{\text{ent}} \delta_{\text{in}}. \quad (156)$$

For a constant-step global method, its residual floor in Theorem 5.7 must be below the right-hand side of (156); the decreasing schedule of Theorem 5.8 has no such obstruction. Run the local method for N_2 further steps with the batch required by (154). Then, with $E_{\text{in}} = \{a_{N_1} \in \mathcal{U}_{\delta_{\text{in}}}\}$ and the local exit time shifted to N_1 ,

$$\mathbb{P}(E_{\text{in}} \text{ and no exit through } N_2) \geq 1 - \varepsilon_{\text{ent}} - \varepsilon_{\text{exit}}. \quad (157)$$

The entry-restricted killed second moment satisfies

$$\mathbb{E}\left[\|a_{N_1+N_2} - a_{\star}\|_{\star, \bullet}^2; E_{\text{in}}, \text{ no exit through } N_2\right] \leq (1 - \mu_{\bullet} h)^{N_2} \mathbf{b}_{\bullet}(\delta_{\text{in}})^2 \delta_{\text{in}} + \frac{\eta_{\bullet}}{\mu_{\bullet} h}. \quad (158)$$

Proof. Markov's inequality and (156) give $\mathbb{P}(E_{\text{in}}^c) \leq \varepsilon_{\text{ent}}$. On E_{in} , the norm-conversion estimate of Section 4 gives the deep-entry condition in (154); that proposition bounds the conditional exit mass by $\varepsilon_{\text{exit}}$. A union bound proves (157). Iterating (152) after time N_1 , with the indicator of E_{in} retained, gives (158). $\square$

5.5 Gaussian calibration

The preceding floors measure nonconstant Hessian fluctuations, not Monte Carlo noise in the abstract. Gaussian targets make the distinction exact.

Proposition 5.14 (Gaussian target calibration). *Let $V(x) = \frac{1}{2}(x - \mu)^{\top} H(x - \mu) + c$, $H \succ 0$.*

- (i) *The Hessian is constant, hence $\Psi(a) = 0$ for every a . The additive terms in the intrinsic one-step inequalities (130) and (131) vanish exactly. The nonzero floors displayed by the coarse self-bounding bounds are artifacts of substituting (126), not actual Gaussian floors.*

(ii) In optimizer-whitened coordinates, the single-sample score noise is

$$\varepsilon_b = (C^{-1/2} - C^{1/2})Z. \quad (159)$$

It is nonzero and has unbounded support whenever $C \neq I$. Thus zero additive floor does not imply pathwise deterministic evolution from an arbitrary covariance.

(iii) If $C_0 = C_\star$, then $C_n = C_\star$ for both methods, all sampled noise vanishes, and in whitened coordinates

$$m_{n+1} = (1 - h_n)m_n \quad (160)$$

pathwise. If the quadratic rescue is used, it returns $(m_0, C_0) = (\mu, H^{-1}) = a_\star$ after its single query, so there is no iterative phase.

(iv) Locally, $L_{H,\star} = 0$, hence $V_0 = \eta_\bullet = 0$. The killed contraction has no additive term and the containment bound loses its N -growing term. For a generic unmatched covariance, however, these remain mean-square and probability statements rather than pathwise containment because of (159).

Proof. Constancy of $\nabla^2 V = H$ makes the integrand in (123) zero. After whitening, the target is standard Gaussian and $X = m + C^{1/2}Z$. Since $g(a) = -m$,

$$\hat{b} - g(a) = -C^{1/2}Z + C^{-1/2}Z,$$

which proves (159). When $C = I$, this noise vanishes and $\bar{H} = I$ pathwise. Both covariance maps then fix I , while their common mean update is (160). Finally, direct substitution into (122) gives $m_0 = \mu$, $C_0 = H^{-1}$, and substitution of $L_{H,\star} = 0$ into the local constants proves the last claim. $\square$

6 Beyond log-concavity

Strong log-concavity plays two roles in the preceding theory: it supplies a global functional inequality and it keeps the covariance equation away from negative-curvature amplification. A globally bounded Hessian alone supplies neither. We make the failure explicit on a smooth one-dimensional target and then recover a constrained stationarity statement by clipping the covariance.

6.1 A smooth double well

For $R > 0$, consider

$$V_R(x) := \frac{x^2}{2} - 2R^2 \log \cosh(x/R). \quad (161)$$

Its derivatives are

$$V'_R(x) = x - 2R \tanh(x/R), \quad V''_R(x) = 1 - 2 \operatorname{sech}^2(x/R),$$

so $-1 \leq V''_R \leq 1$ uniformly in R . For $X \sim \mathcal{N}(0, c)$, write

$$A_R(c) := \mathbb{E}[V''_R(X)].$$

For every fixed $c > 0$, dominated convergence gives

$$A_R(c) \longrightarrow -1 \quad \text{as } R \rightarrow \infty. \quad (162)$$

The symmetric mean $m = 0$ is invariant for all the covariance dynamics below.

Proposition 6.1 (Riemannian covariance cascade). *Fix $h > 0$, and let*

$$c_{n+1}^{(R)} = c_n^{(R)} \exp\left\{h(1 - c_n^{(R)} A_R(c_n^{(R)}))\right\}, \quad c_0^{(R)} = 1. \quad (163)$$

For every $M > 0$, there are R and a finite n such that $c_n^{(R)} \geq M$. Consequently, no bound depending only on $\sup_x |V_R''(x)|$, h , and c_0 can control all trajectories of the unprojected Riemannian scheme uniformly over this family.

Proof. The limiting recursion obtained from (162) is

$$\bar{c}_{n+1} = \bar{c}_n e^{h(1+\bar{c}_n)}, \quad \bar{c}_0 = 1,$$

and diverges. Choose a finite N with $\bar{c}_N > 2M$. The map $(R, c) \mapsto A_R(c)$ converges uniformly to -1 on every compact c -interval. An induction over the first N steps therefore gives $c_n^{(R)} \rightarrow \bar{c}_n$ for $0 \leq n \leq N$. For sufficiently large R , $c_N^{(R)} > M$. $\square$

Proposition 6.2 (Pole of the KL covariance resolvent). *For $h = 1$, the unprojected KL covariance map is*

$$c_1 = \frac{2c_0}{1 + c_0 A_R(c_0)}. \quad (164)$$

There are sequences $\varepsilon \downarrow 0$ and $R_\varepsilon \rightarrow \infty$ such that, with $c_0 = 1 - \varepsilon$,

$$c_1 = \Theta(\varepsilon^{-1}), \quad (1 - c_1 A_{R_\varepsilon}(c_1))^2 = \Theta(\varepsilon^{-2}).$$

Thus the pole is caused by approaching loss of positive definiteness in the rational resolvent, despite the uniform Hessian bound.

Proof. For every $c > 0$, writing $X = \sqrt{c}Z$ gives the quantitative bound

$$0 \leq A_R(c) + 1 = 2\mathbb{E} \tanh^2(\sqrt{c}Z/R) \leq \frac{2c}{R^2}. \quad (165)$$

Take $0 < \varepsilon \leq 1/4$ and choose, for example, $R_\varepsilon^2 \geq 2\varepsilon^{-2}$. Then $0 \leq A_{R_\varepsilon}(1 - \varepsilon) + 1 \leq \varepsilon^2$, and the denominator in (164) is then

$$1 - (1 - \varepsilon) + O(\varepsilon^2) = \varepsilon + O(\varepsilon^2),$$

which is positive and gives $c_1 = \Theta(\varepsilon^{-1})$; in particular, $c_1 \leq 2/\varepsilon$. Applying (165) once more,

$$0 \leq A_{R_\varepsilon}(c_1) + 1 \leq \frac{2c_1}{R_\varepsilon^2} \leq 2\varepsilon \leq \frac{1}{2}.$$

This curvature bound yields $|1 - c_1 A_{R_\varepsilon}(c_1)| = \Theta(c_1)$, proving the residual scaling. $\square$

These propositions concern the two discrete maps. They do not say that every bounded-Hessian nonconvex target diverges, nor do they preclude local stability around a favorable stationary point. They show that a global smoothness constant cannot replace covariance control in a trajectory-wise Fisher–Rao theorem.

6.2 A Bures–Wasserstein comparison

The Bures–Wasserstein forward–backward method has a different nonconvex guarantee. Under the assumptions and stepsize of Diao et al. [4, Theorem 5.5], its iterates satisfy a running-minimum bound of the form

$$\min_{0 \leq n < N} \|\text{grad}_{\text{BW}} \mathcal{E}(a_n)\|_{\text{BW}}^2 \leq \frac{150\{\mathcal{E}(a_0) - \inf \mathcal{E}\}}{\eta N}. \quad (166)$$

This is a stationarity statement in the Bures–Wasserstein geometry. It does not control the pointwise Fisher–Rao residual from Theorems 6.1 and 6.2. We use it only as a comparison between geometries, not as the first stage of a general WFR method.

6.3 Covariance-constrained KL stationarity

Fix $0 < \lambda_- \leq \lambda_+ < \infty$ and

$$\mathcal{A}_{[\lambda_-, \lambda_+]} := \{(m, C) : \lambda_- I \preceq C \preceq \lambda_+ I\}.$$

For a symmetric matrix $C = Q \operatorname{diag}(c_i) Q^\top$, let

$$\operatorname{Clip}_{[\lambda_-, \lambda_+]}(C) := Q \operatorname{diag}(\min\{\lambda_+, \max\{\lambda_-, c_i\}\}) Q^\top.$$

Assume only

$$-\beta I \preceq \nabla^2 V(x) \preceq \beta I \quad \text{for all } x. \quad (167)$$

At $a_n = (m_n, C_n)$, set

$$b_n := \mathbb{E}_{\mathcal{N}(m_n, C_n)} \nabla V(X), \quad A_n := \mathbb{E}_{\mathcal{N}(m_n, C_n)} \nabla^2 V(X),$$

and apply the split update

$$m_{n+1} = m_n - h C_n b_n, \quad (168)$$

$$\tilde{C}_{n+1} = (1 + h)(C_n^{-1} + h A_n)^{-1}, \quad C_{n+1} = \operatorname{Clip}_{[\lambda_-, \lambda_+]}(\tilde{C}_{n+1}). \quad (169)$$

The inverse is well defined at the stepsizes below. The natural one-step stationarity measure for this split scheme is

$$S_n := \frac{1}{2} \|m_{n+1} - m_n\|_{C_n^{-1}}^2 + \operatorname{KL}(\mathcal{N}(0, C_n) \parallel \mathcal{N}(0, C_{n+1})). \quad (170)$$

The covariance orientation in the KL term is important.

Theorem 6.3 (Constrained split stationarity). *Suppose (167) holds, $a_0 \in \mathcal{A}_{[\lambda_-, \lambda_+]}$, and*

$$0 < h \leq \frac{1}{2\beta\lambda_+}.$$

Then the iterates (168)–(169) are well defined, remain feasible, and satisfy

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{h} S_n. \quad (171)$$

Consequently, for every $N \geq 1$,

$$\boxed{\min_{0 \leq n < N} S_n \leq \frac{h}{N} \{\mathcal{E}(a_0) - \mathcal{E}(a_N)\}.} \quad (172)$$

Proof. The spectral bounds imply

$$C_n^{-1} + h A_n \succeq (\lambda_+^{-1} - h\beta) I \succeq (2\lambda_+)^{-1} I.$$

The smooth part of the Gaussian objective is relatively smooth with respect to the forward split divergence with constant $2\beta\lambda_+$. The mean optimality equation leaves $\frac{1}{2} \|m_{n+1} - m_n\|_{C_n^{-1}}^2$, while the constrained log-determinant three-point inequality leaves the reverse covariance divergence $\operatorname{KL}(\mathcal{N}(0, C_n) \parallel \mathcal{N}(0, C_{n+1}))$. Spectral clipping is precisely the log-determinant Bregman projection onto the covariance interval. Since $h(2\beta\lambda_+) \leq 1$, the forward remainder is absorbed and the two estimates combine to (171). Summing over n and taking the smallest term gives (172). Full details are recorded in [Section D](#). $\square$

Corollary 6.4 (Full successive-Gaussian KL). *Under the assumptions of [Theorem 6.3](#),*

$$\min_{0 \leq n < N} \text{KL}(\mathcal{N}(m_n, C_n) \parallel \mathcal{N}(m_{n+1}, C_{n+1})) \leq \frac{\lambda_+}{\lambda_-} \frac{h}{N} \{\mathcal{E}(a_0) - \mathcal{E}(a_N)\}. \quad (173)$$

For centered scalar trajectories the mean term vanishes, so the factor-one bound [\(172\)](#) is the quantity plotted in [Section 7.5](#).

Proof. The reverse KL between successive Gaussians splits into the covariance KL in [\(170\)](#) and $\frac{1}{2} \|m_{n+1} - m_n\|_{C_{n+1}^{-1}}^2$. Feasibility gives

$$C_{n+1}^{-1} \preceq \lambda_-^{-1} I \preceq \frac{\lambda_+}{\lambda_-} C_n^{-1}.$$

Therefore the full KL is at most $(\lambda_+/\lambda_-)S_n$, and the conclusion follows from [Theorem 6.3](#). $\square$

Remark 6.5 (Scope). The result certifies stationarity of the projected split map on the covariance-truncated set. If an iterate is pinned at $\lambda_+ I$, S_n can vanish while the unconstrained Fisher–Rao gradient points outward. The theorem therefore does not imply small unconstrained gradient, convergence to the unconstrained Gaussian optimizer, or a stochastic clipped guarantee.

7 Numerical illustrations

The experiments are designed as diagnostics for the mechanisms proved above, not as a search for favorable benchmark instances. Deterministic Gaussian and separable low-dimensional log-concave expectations are evaluated analytically or by fixed Gauss–Hermite quadrature; the nonseparable affine-mode target uses a fixed scrambled Sobol design. The stochastic implementations use float64 arithmetic and explicit seeds; baseline/STL algorithm pairs are driven by the same random-number stream. The selected figures report the five claims that matter for the paper; larger parameter sweeps remain in the accompanying repository.

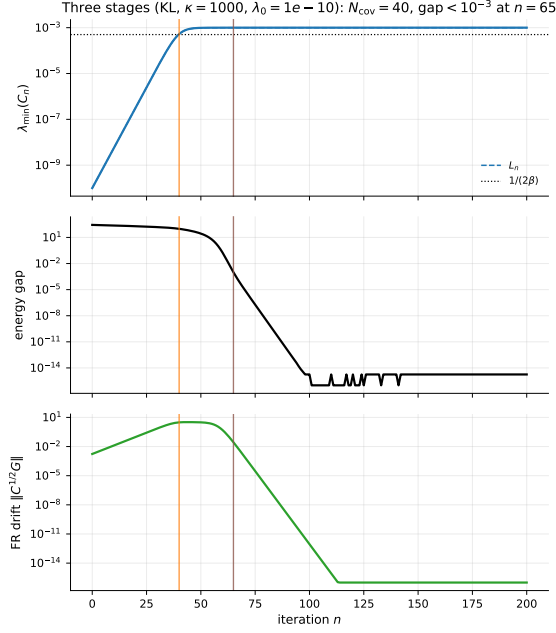
7.1 Covariance bootstrap and transport warm start

The covariance-scaling sweep uses diagonal Gaussian targets with initial covariances as small as $\lambda_{\min}(C_0) = 10^{-10}$; the separate transport experiment also includes a smooth log cosh-perturbed target, with $\lambda_{\min}(C_0) \geq 10^{-8}$. The burn-in endpoint is the first iterate with $\lambda_{\min}(C_n) \geq 1/(2\beta)$. Across the Gaussian scaling sweep, this time is linear in $\log(1/(\beta\lambda_{\min}(C_0)))$, as predicted by the two-sided covariance recursions. [Figure 1](#) shows one hard KL run with $d = 20$ and $\kappa = 1000$: covariance calibration precedes rapid energy decay, which in turn precedes the final decay of the Fisher–Rao mean-block drift $\|C_n^{1/2}g(a_n)\|$.

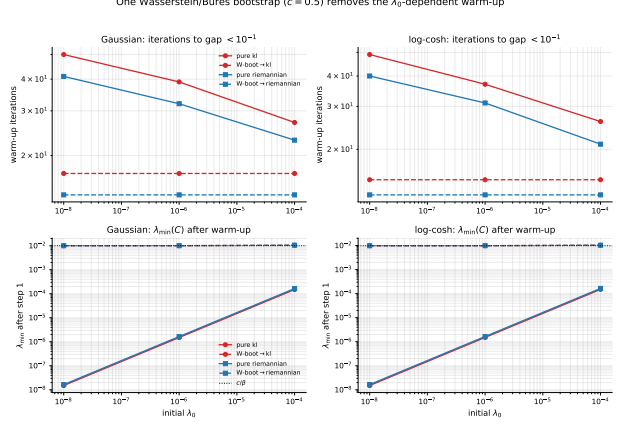
The transport experiment makes a deliberately narrow point. A single Bures–Wasserstein step removes the small-covariance cost and then hands the trajectory back to a Fisher–Rao scheme. It neither changes the κ_* -scale spiral obstruction nor implements a blended WFR flow.

7.2 Riemannian versus KL discretization

The deterministic comparison uses two-dimensional Gaussian and smooth log-concave targets, together with a convex quartic target whose Hessian is unbounded and hence is not globally smooth. Both schemes use the same explicit mean update and evaluate the same population score and Hessian, analytically or by deterministic quadrature. The Riemannian covariance update applies a matrix-exponential retraction; the KL update solves a rational precision resolvent. We report iteration count and elapsed flow time nh separately; the latter is a discretization diagnostic, not wall-clock timing.



(a) A hard KL trajectory.



(b) One transport step, then a Fisher-Rao tail.

Figure 1: **Covariance initialization.** Left: a Gaussian run with $d = 20$, $\kappa = 1000$, and $\lambda_{\min}(C_0) = 10^{-10}$ separates covariance burn-in, global energy localization, and the final local regime. Right: one Bures–Wasserstein bootstrap with $c = 1/2$, hence transport step $\eta = c/\beta$, lifts the minimum covariance eigenvalue to the curvature scale independently of the initial underdispersion; the subsequent Fisher–Rao warm-up is essentially flat in $\lambda_{\min}(C_0)$.

The descriptor *more robust* in this comparison refers specifically to the covariance map. On a log-concave target, $C_n^{-1} + hA_n \succ 0$ for every $h > 0$, so the KL covariance update cannot leave the positive-definite cone. The full algorithm may nevertheless fail at a large step because the mean update contains the factor hC_n .

7.3 STL variance and constant-step floors

The stochastic grid uses Gaussian targets with $d \in \{2, 8, 32\}$, $\kappa \in \{10, 100, 10^4\}$, and smooth logcosh targets with $d \in \{2, 8, 32\}$, $\kappa \in \{10, 100\}$. Each cell uses 20 seeds. Baseline and STL runs share the same samples, so the comparison is paired. The energy shown in the plots is evaluated independently of the update sample.

These plots support the variance decomposition of Section 5. They do not test the high-probability local non-exit theorem, whose horizon-dependent batch requirement is a worst-case containment statement rather than a prediction for the median trajectory.

7.4 Trace and traceless affine modes

The affine-metric experiment varies $\tau \in \{-\omega/(2d), 0, \omega/(2d)\}$ for several values of ω . Five initializations isolate mean, volume-high, volume-low, traceless shape, and mixed perturbations. The Gaussian target uses exact expectations. The non-Gaussian target is strongly log-concave and uses one common fixed, scrambled Sobol quasi-Monte Carlo design across all metric choices.

7.5 The non-log-concave boundary

For (161), all expectations are deterministic Gauss–Hermite quadratures and all trajectories are centered. The first two panels visualize the analytic mechanisms of Theorems 6.1 and 6.2.

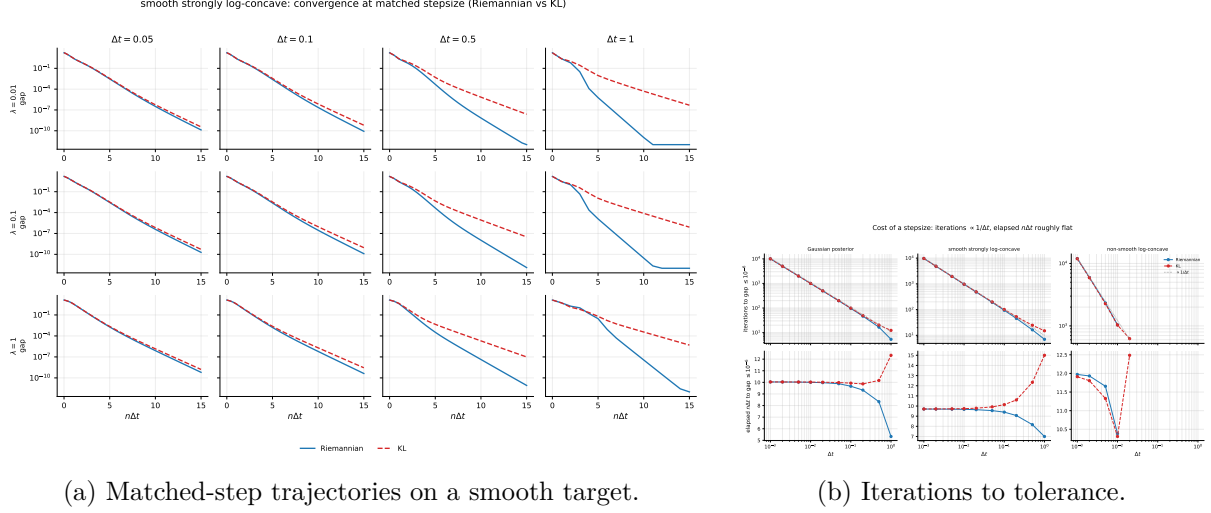


Figure 2: **Deterministic discretizations.** At matched stable stepsizes, the Riemannian and KL trajectories have comparable first-order behavior. The Riemannian map often contracts faster per step in the tested range, whereas the KL covariance resolvent remains well defined for all stepsizes on log-concave targets and avoids a matrix exponential. Both methods can still become unstable through their shared explicit mean update. The common theorem-safe stepsize is sufficient and conservative in these low-dimensional examples.

The last panel evaluates the covariance part of the split stationarity measure in (170); the mean term is zero by symmetry.

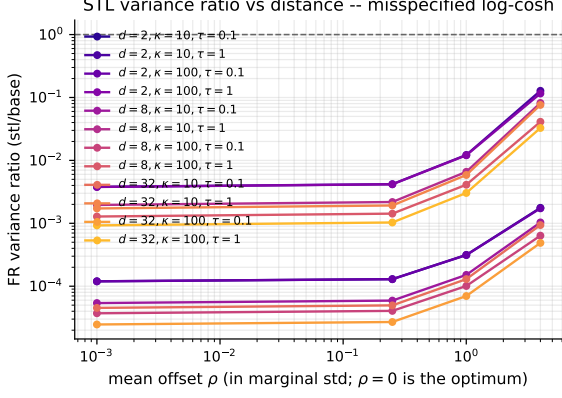
7.6 Reproducibility

The repository contains the experiment scripts, the raw outputs used by the selected panels, and a deterministic asset-building script. The selected submission PDFs are byte-identical to the corresponding report assets. The selected panels can be rebuilt directly from their surviving CSV inputs, but the covariance-bootstrap and STL group-level builder entry points do not run end to end from the current snapshot because some supplementary trajectory and run-metadata files are absent. Section E records the target formulas, grids, seeds, quadrature rules, stopping conditions, exact data-to-figure lineage, and the provenance files that are absent from the current repository snapshot. No general WFR output is used in this paper.

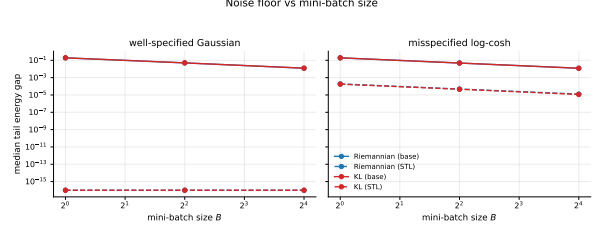
8 Discussion

The Fisher–Rao bounds proved here are not summarized by one condition number. Covariance underdispersion, global nonquadratic localization, and local mean–covariance coupling are three different effects. Their separation explains several facts that otherwise look contradictory. Fisher–Rao is affine invariant, yet its global rate can depend on the initial covariance. A transport warm start can eliminate that dependence, yet a $\kappa_\star$ -scale lower bound remains. Optimizer whitening can remove extrinsic anisotropy, yet the local score generator can still be slow because the whitened target is non-Gaussian.

Three sharpness statements should be read with their scopes attached. The spiral gives a continuous-time $\Omega(\kappa_\star)$ localization time in dimension at least two for a specified anisotropic initialization class. The bump train gives a one-dimensional $\Omega(\kappa/h)$ lower bound for both fixed-step maps, with $h = \gamma/\kappa$ and $0 < \gamma \leq 1/2$, for a constant-factor reduction after covariance burn-in. The convex ridge gives $\gamma_\star = \Theta(\kappa_\star^{-1/2})$ with $d = \Theta(\kappa_\star^2)$. None of these statements



(a) Fisher–Rao variance ratio versus mean displacement.



(b) Tail energy versus batch size.

Figure 3: **Price/Hessian STL diagnostics.** For a non-Gaussian log cosh target, the displayed mean-block Fisher–Rao variance ratio $\text{Var}_{\text{FR}}(b_{\text{STL}})/\text{Var}_{\text{FR}}(b_{\text{base}})$ is positive at the optimizer and increases with the mean displacement in every plotted configuration (left). Mini-batching reduces the median tail energy gap approximately as $1/B$, while STL stays below its paired baseline in the recorded grid (right). For a Gaussian target with matched covariance, the mean estimator and the constant Hessian are deterministic pathwise; this special calibration is not extrapolated to general targets.

alone proves a fixed-dimensional lower bound with the same dependence, and the ridge does not saturate the $\sqrt{d} \log \kappa_*$ branch of the universal estimate.

The stochastic results carry equally important qualifications. A constant-step theorem gives geometric convergence to an explicit residual upper bound; it neither identifies the stationary error nor, when that residual is nonzero, certifies arbitrary accuracy at fixed step and batch size. Under Hessian-Lipschitzness, the decreasing-step result instead certifies decay to zero at an $O(1/N)$ rate. Locally, contraction is first proved for the process killed at the boundary of a certified region. A separate finite-horizon argument turns this into a high-probability non-exit statement, at the price of deeper entry and a batch size that grows with the requested horizon under the present proof. These distinctions are visible in the theorem statements rather than being deferred to caveats.

The classification of affine-invariant metrics explains why Fisher–Rao is a natural default, not why it is fastest on every target. For $d \geq 2$, the parameters (η, ω, τ) set independent time scales for the mean, traceless covariance, and trace covariance modes on a Gaussian target. Fisher–Rao is the unique balanced choice up to scale. A deliberately unbalanced metric can accelerate a known dominant mode, as the trace experiment illustrates. A complete sharp global and stochastic theory for every member of this family would require new arguments; it is not implied by the Gaussian modal calculation.

The non-log-concave example marks a structural boundary. Negative curvature is multiplied by the covariance in the Fisher–Rao dynamics. A uniform Hessian bound therefore does not prevent an unprojected covariance cascade or a pole in the KL resolvent. Spectral clipping changes the problem to constrained optimization and restores a useful $O(1/N)$ displacement guarantee, but the displacement can vanish at an active constraint while the unconstrained gradient remains large.

Several problems remain open. The most immediate is the optimal local spectral rate in fixed dimension. A second is a closed-form general-target radius for the discrete maps, replacing the compact-hull second-derivative supremum. Sharper stochastic local stepsizes and an infinite-horizon stability mechanism that does not require horizon-dependent batching would make

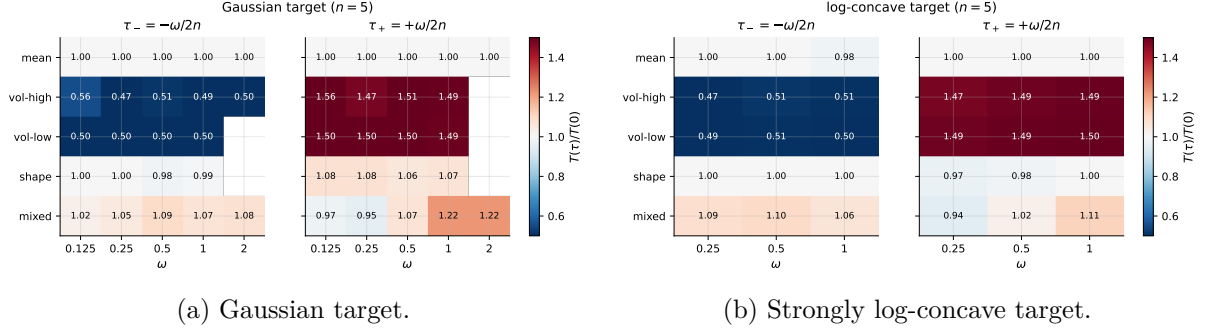


Figure 4: **Affine-invariant mode preconditioning.** The panels show the time-to-tolerance ratio $T(\tau)/T(0)$. For $\tau = -\omega/(2d)$, the trace rate doubles and the volume-dominated runs take about half as long. For $\tau = \omega/(2d)$, their time scale grows by about $3/2$. Mean- and traceless-dominated runs remain near one. The Gaussian panel calibrates the exact modal calculation; the non-Gaussian panel is an illustration, not a general convergence theorem for the full metric family.

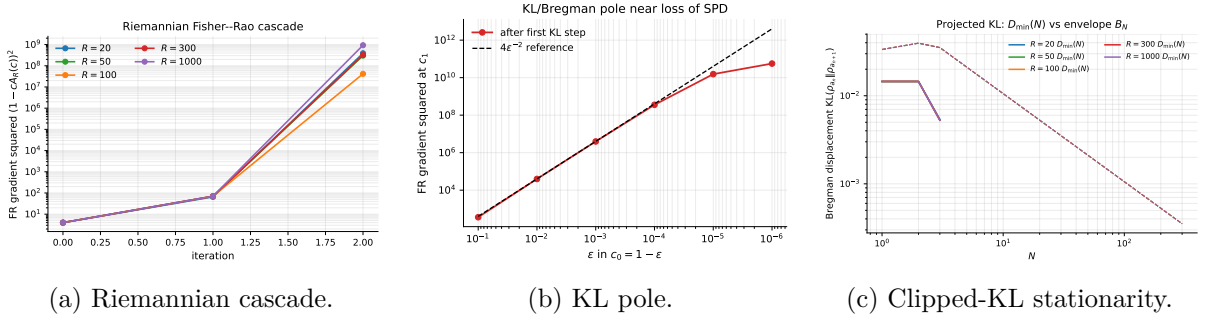


Figure 5: **Beyond log-concavity.** Although $|V_R''| \leq 1$, negative curvature drives a covariance cascade in the Riemannian map and a rational pole in the unprojected KL map. With $C_n \in [\lambda_-, \lambda_+]$ and a theorem-safe step, the running minimum of the centered covariance KL stays below the energy-drop envelope from Theorem 6.3. After the upper constraint has bound, the iterate remains pinned there and the next displacement is zero; this is constrained stationarity, not a small unconstrained Fisher–Rao gradient.

the local theory more practical. Finally, general affine-invariant preconditioning and a full Wasserstein–Fisher–Rao theory are natural continuations, but they should be developed without obscuring the scoped sharp Fisher–Rao bounds established here.

A Affine-invariant geometry: classification details

This appendix records the representation argument behind the metric classification in Section 2. The affine group acts transitively on $\mathbb{R}^d \times \mathbb{S}_{++}(d)$. It is therefore enough to classify inner products on the tangent space at $o = (0, I)$ that are invariant under the isotropy group $O(d)$.

At o , the tangent representation is

$$\mathbb{R}^d \oplus \mathbb{S}(d) = \mathbb{R}^d \oplus \text{span}\{I\} \oplus \mathbb{S}_0(d),$$

where $\mathbb{S}_0(d)$ denotes the traceless symmetric matrices. These three summands are pairwise inequivalent irreducible $O(d)$ -representations (with the familiar low-dimensional interpretation when $d = 1$). Schur’s lemma therefore leaves one scalar weight on each summand. Equivalently, every invariant quadratic form is

$$\eta \|u\|^2 + \omega \text{Tr}(X^2) + \tau (\text{Tr } X)^2.$$

There is no mean–covariance term. One can see this without representation terminology: the isotropy element $-I$ sends u to $-u$ and fixes X , so any cross term changes sign and must vanish.

Transporting this form from o to $a = (m, C)$ by $(u, X) \mapsto (C^{-1/2}u, C^{-1/2}XC^{-1/2})$ gives

$$g_a((u, X), (v, Y)) = \eta u^\top C^{-1}v + \omega \operatorname{Tr}(C^{-1}XC^{-1}Y) + \tau \operatorname{Tr}(C^{-1}X) \operatorname{Tr}(C^{-1}Y).$$

Positivity on the mean and traceless summands requires $\eta > 0$ and $\omega > 0$. On $X = sI$, the covariance part is $d(\omega + d\tau)s^2$, hence $\tau > -\omega/d$. These conditions are also sufficient.

For completeness, let

$$g := -\mathbb{E}_{\mathcal{N}(m, C)} \nabla V, \quad A := \mathbb{E}_{\mathcal{N}(m, C)} \nabla^2 V, \quad R := C^{-1} - A.$$

The Euclidean differentials of the objective are

$$D_m \mathcal{E}[u] = -g^\top u, \quad D_C \mathcal{E}[X] = -\frac{1}{2} \operatorname{Tr}(RX).$$

Solving $g_a(\operatorname{grad} \mathcal{E}, (u, X)) = D\mathcal{E}[(u, X)]$ separately on the mean, traceless, and trace covariance components gives

$$\begin{aligned} \dot{m} &= \frac{1}{\eta} Cg, \\ \dot{C} &= \frac{1}{2\omega} CRC - \frac{\tau}{2\omega(\omega + d\tau)} \operatorname{Tr}(CR) C. \end{aligned}$$

Whitening a Gaussian target at its optimizer yields $\dot{m} = -m/\eta + O(\|a - a_\star\|^2)$. For $C = I + X$, the traceless covariance part obeys $\dot{X}_0 = -X_0/(2\omega) + O(\|X\|^2)$, while the trace part obeys

$$\frac{d}{dt} \frac{\operatorname{Tr} X}{d} = -\frac{1}{2(\omega + d\tau)} \frac{\operatorname{Tr} X}{d} + O(\|X\|^2).$$

Balancing all three linear rates, modulo an overall multiplication of the metric, requires

$$\eta = 1, \quad \omega = \frac{1}{2}, \quad \tau = 0,$$

which is the Fisher–Rao metric.

B Certificates for the deterministic lower bounds

B.1 The logarithmic-spiral construction

This subsection proves [Theorem 3.3](#). The construction is two-dimensional; the extension to higher dimension is recorded at the end.

B.1.1 Potential and global Hessian certificate

Fix $K \geq 16$ and set

$$\alpha_0 := 2\alpha, \quad c := \frac{\sqrt{K-2}}{3}, \quad \delta_K := \frac{10^{-4}}{K}, \quad D_K := \frac{18\alpha_0(K-1)}{K+7}, \quad \beta := \alpha(2K+1). \quad (174)$$

Thus the condition number in the original coordinates is $2K+1$. Define

$$s(u) := 35u^4 - 84u^5 + 70u^6 - 20u^7, \quad 0 \leq u \leq 1,$$

and the even C^3 cutoff

$$\rho(u) := \begin{cases} 1, & |u| \leq \frac{1}{2}, \\ s(2 - 2|u|), & \frac{1}{2} < |u| < 1, \\ 0, & |u| \geq 1. \end{cases} \quad (175)$$

Extend $\rho_{\delta_K}(\phi) := \rho(\phi/\delta_K)$ periodically with period π , and put

$$\bar{\rho}_{\delta_K} := \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \rho_{\delta_K}(\phi) d\phi, \quad g_K(\phi) := \frac{\rho_{\delta_K}(\phi) - \bar{\rho}_{\delta_K}}{1 - \bar{\rho}_{\delta_K}}.$$

Let f_K be the even π -periodic solution of

$$f_K'' = g_K, \quad f_K(0) = f_K'(0) = 0. \quad (176)$$

The normalization gives

$$f_K(\phi) = \frac{1}{2}\phi^2 \quad \text{for } |\phi| \leq \frac{\delta_K}{2}, \quad \|f_K'\|_{\infty} \leq 3\delta_K, \quad \|f_K\|_{\infty} \leq 2\pi\delta_K. \quad (177)$$

For the radial cutoff use

$$\chi(\ell) := \begin{cases} 0, & \ell \leq 0, \\ 35\ell^4 - 84\ell^5 + 70\ell^6 - 20\ell^7, & 0 < \ell < 1, \\ 1, & \ell \geq 1. \end{cases} \quad (178)$$

Fix a scale $r_0 > 0$. In polar coordinates (r, θ) , set

$$\ell := \log \frac{r}{r_0}, \quad \phi := \theta + c\ell,$$

and define

$$V_K(r, \theta) := \frac{1}{2}r^2[\alpha_0 + D_K\chi(\ell)f_K(\phi)]. \quad (179)$$

For $r \leq r_0$, the potential is exactly $V_K(x) = \alpha_0 \|x\|^2/2$. The first three derivatives of ρ and χ match at the endpoints of their pieces, so $V_K \in C^3(\mathbb{R}^2)$. Its evenness follows from the π -periodicity of f_K .

Lemma B.1 (Global Hessian certificate). *The potential in (179) satisfies*

$$0.98\alpha_0 I \preceq \nabla^2 V_K(x) \preceq \alpha_0(K + 0.02)I \quad (x \in \mathbb{R}^2). \quad (180)$$

Consequently, $\alpha I \preceq \nabla^2 V_K \preceq \beta I$, with β from (174). On the flat part of a spiral valley, the two Hessian eigenvalues are exactly α_0 and $\alpha_0 K$.

Proof. Put $q(\ell, \theta) := \chi(\ell)f_K(\theta + c\ell)$. For any profile $A = A(\ell, \theta)$, direct differentiation in the orthonormal polar frame yields

$$\nabla^2 \left(\frac{1}{2}r^2 A \right) = \begin{pmatrix} A + \frac{3}{2}A_{\ell} + \frac{1}{2}A_{\ell\ell} & \frac{1}{2}A_{\theta} + \frac{1}{2}A_{\ell\theta} \\ \frac{1}{2}A_{\theta} + \frac{1}{2}A_{\ell\theta} & A + \frac{1}{2}A_{\ell} + \frac{1}{2}A_{\theta\theta} \end{pmatrix}.$$

It follows that

$$\nabla^2 V_K - \alpha_0 I = \frac{D_K}{2} \chi g_K \begin{pmatrix} c^2 & c \\ c & 1 \end{pmatrix} + D_K \{ \chi B_K + E_K \}, \quad (181)$$

where all scalar functions are evaluated at (ℓ, ϕ) and

$$B_K = \begin{pmatrix} f_K + \frac{3}{2}cf_K' & \frac{1}{2}f_K' \\ \frac{1}{2}f_K' & f_K + \frac{1}{2}cf_K' \end{pmatrix}, \quad E_K = \begin{pmatrix} (\frac{3}{2}\chi' + \frac{1}{2}\chi'')f_K + c\chi'f_K' & \frac{1}{2}\chi'f_K' \\ \frac{1}{2}\chi'f_K' & \frac{1}{2}\chi'f_K \end{pmatrix}.$$

This identity includes the transition annulus $0 < \ell < 1$.

The normalization of the phase cutoff is explicit. Since $\int_{-1}^1 \rho(u) du = 3/2$,

$$\bar{\rho}_{\delta_K} = \frac{3\delta_K}{2\pi}, \quad -\frac{\bar{\rho}_{\delta_K}}{1 - \bar{\rho}_{\delta_K}} \leq g_K \leq 1.$$

For $K \geq 16$,

$$(K-1) \frac{\bar{\rho}_{\delta_K}}{1 - \bar{\rho}_{\delta_K}} < 4.8 \cdot 10^{-5}. \quad (182)$$

The radial cutoff satisfies

$$0 \leq \chi \leq 1, \quad \|\chi'\|_\infty = \frac{35}{16}, \quad \|\chi''\|_\infty < 7.514.$$

Using (177), $D_K/\alpha_0 \leq 18$, and $c \leq \sqrt{K}/3$, the maximum-row-sum bound in (181) gives

$$\begin{aligned} \frac{D_K}{\alpha_0} \|\chi B_K + E_K\|_{\text{op}} &\leq 18 \left[\left(1 + \frac{3}{2} \frac{35}{16} + \frac{7.514}{2} \right) 2\pi\delta_K \right. \\ &\quad \left. + \left(\frac{3}{2}c + \frac{1}{2} + c \frac{35}{16} + \frac{1}{2} \frac{35}{16} \right) 3\delta_K \right] < 0.0078. \end{aligned}$$

Every term in the last bound decreases with K , so evaluation at $K = 16$ suffices.

The rank-one matrix in the first term of (181) has nonzero eigenvalue $1 + c^2 = (K+7)/9$, and

$$\frac{D_K}{2}(1 + c^2) = \alpha_0(K-1).$$

Its positive part is at most $\alpha_0(K-1)I$, while (182) bounds its negative part by $4.8 \cdot 10^{-5}\alpha_0 I$. Thus the calculation in fact proves

$$0.992\alpha_0 I \preceq \nabla^2 V_K \preceq \alpha_0(K + 0.0078)I,$$

which implies (180). On an uncut valley core, $g_K = 1$ and $f_K(0) = f'_K(0) = 0$, so the two eigenvalues are α_0 and $\alpha_0 K$. Finally, $0.98\alpha_0 = 1.96\alpha > \alpha$ and $\alpha_0(K+0.02) < \alpha(2K+1) = \beta$. $\square$

B.1.2 Reference orbit

Let R_θ be rotation through angle θ , put $s_K := \sqrt{K-2}$, and define

$$Q_K := \frac{1}{K+7} \begin{pmatrix} (K+1)^2 & 2s_K(K+1) \\ 2s_K(K+1) & 6(K+1) \end{pmatrix}. \quad (183)$$

Its trace and determinant are

$$\text{Tr } Q_K = K+1, \quad \det Q_K = \frac{2(K+1)^2}{K+7},$$

and

$$\det \left(Q_K - \frac{3}{2}I \right) = \frac{2K^2 - 23K + 29}{4(K+7)} > 0 \quad (K \geq 16).$$

The principal minors of $Q_K - 3I/2$ and $KI - Q_K$ therefore give

$$\frac{3}{2}I \prec Q_K \prec KI, \quad Q_K^{-1}e_1 = \frac{1}{K+1} \begin{pmatrix} 3 \\ -\sqrt{K-2} \end{pmatrix}. \quad (184)$$

Choose $\|m_0\| = e^2 r_0$ and an angle θ_0 such that

$$\theta_0 + c \log \frac{\|m_0\|}{r_0} \equiv 0 \pmod{\pi}, \quad (185)$$

and initialize the precision by

$$P_0 := C_0^{-1} := \alpha_0 R_{\theta_0} Q_K R_{\theta_0}^\top. \quad (186)$$

By (184),

$$\frac{1}{\beta} I \preceq C_0 \preceq \frac{1}{\alpha} I. \quad (187)$$

The point-system reference orbit is

$$r_t^{\text{ref}} := e^2 r_0 \exp\left(-\frac{3t}{K+1}\right), \quad \theta_t^{\text{ref}} := \theta_0 + \frac{\sqrt{K-2}}{K+1} t, \quad (188)$$

with

$$P_t^{\text{ref}} := \alpha_0 R_{\theta_t^{\text{ref}}} Q_K R_{\theta_t^{\text{ref}}}^\top.$$

Indeed, on the valley $\nabla V_K(m) = hm$, and (184) gives

$$\dot{r} = -\frac{3}{K+1} r, \quad r\dot{\theta} = \frac{\sqrt{K-2}}{K+1} r.$$

A direct substitution also verifies $\dot{P}^{\text{ref}} = \nabla^2 V_K(m^{\text{ref}}) - P^{\text{ref}}$.

B.1.3 Gaussian shadowing and the first-exit argument

Lemma B.2 (Gaussian shadowing certificate). *Assume*

$$r_0 \sqrt{\alpha} \geq 10^{25} K^{3/2} \sqrt{\log(eK)}. \quad (189)$$

Let (m_t, C_t) be the exact Gaussian Fisher–Rao flow for V_K , initialized by (185)–(186). Then, for $0 \leq t \leq K/100$,

$$-\frac{4}{K} \leq \frac{d}{dt} \log \|m_t\| \leq -\frac{2}{K}. \quad (190)$$

In particular,

$$\|m_t\| \geq e^{-0.04} \|m_0\| \quad (0 \leq t \leq K/100). \quad (191)$$

Proof. Write $P_t = C_t^{-1}$, choose the continuous lift of the polar angle starting at θ_0 , and introduce

$$z_t := \theta_t + c \log \frac{r_t}{r_0}, \quad S_t := \frac{1}{\alpha_0} R_{\theta_t}^\top P_t R_{\theta_t}, \quad E_t := S_t - Q_K. \quad (192)$$

The phase z_t is represented near zero modulo π . Put

$$w := \begin{pmatrix} c \\ 1 \end{pmatrix}, \quad J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad d_K := \frac{D_K}{\alpha_0} = \frac{18(K-1)}{K+7}.$$

On the uncut flat phase core $f_K(z) = z^2/2$. The normalized point gradient and Hessian there are

$$b_K(z) := \frac{1}{\alpha_0 r} R_\theta^\top \nabla V_K(m) = \begin{pmatrix} 1 + \frac{d_K}{2} z^2 + \frac{d_K c}{2} z \\ \frac{d_K}{2} z \end{pmatrix}, \quad (193)$$

$$\begin{aligned} H_K(z) &:= \frac{1}{\alpha_0} R_\theta^\top \nabla^2 V_K(m) R_\theta \\ &= I + \frac{d_K}{2} w w^\top + d_K \begin{pmatrix} \frac{1}{2} z^2 + \frac{3}{2} c z & \frac{1}{2} z \\ \frac{1}{2} z & \frac{1}{2} z^2 + \frac{1}{2} c z \end{pmatrix}. \end{aligned} \quad (194)$$

For the exact Gaussian flow define

$$\bar{b}_t := \frac{1}{\alpha_0 r_t} R_{\theta_t}^\top \mathbb{E}[\nabla V_K(X_t)], \quad \bar{H}_t := \frac{1}{\alpha_0} R_{\theta_t}^\top \mathbb{E}[\nabla^2 V_K(X_t)] R_{\theta_t}, \quad q_t := S_t^{-1} \bar{b}_t.$$

The mean and precision equations become

$$\frac{\dot{r}_t}{r_t} = -(q_t)_1, \quad \dot{\theta}_t = -(q_t)_2, \quad \dot{z}_t = -w^\top q_t, \quad \dot{S}_t = \bar{H}_t - S_t - (q_t)_2(S_t J - J S_t). \quad (195)$$

If the Gaussian averages are replaced by (193)–(194), then $(z, S) = (0, Q_K)$ is an equilibrium. With $\omega_K = s_K/(K+1)$,

$$Q_K^{-1} e_1 = \begin{pmatrix} 3/(K+1) \\ -\omega_K \end{pmatrix}, \quad w^\top Q_K^{-1} e_1 = 0, \quad \mathbf{H}_K(0) - Q_K + \omega_K(Q_K J - J Q_K) = 0.$$

We next certify the linearization and nonlinear remainder. Let

$$\mathbf{T}_K := \text{diag}(K, \sqrt{K}, 1, \sqrt{K}), \quad y := \mathbf{T}_K(z, E_{11}, E_{12}, E_{22})^\top. \quad (196)$$

For $S = Q_K + E$, let $\mathcal{F}_K(y)$ be the scaled point vector field obtained from (195) with $\bar{b} = b_K(z)$ and $\bar{H} = \mathbf{H}_K(z)$. Set

$$A_K := D\mathcal{F}_K(0), \quad \mathcal{N}_K(y) := \mathcal{F}_K(y) - A_K y.$$

Writing $s = \sqrt{K-2}$, direct differentiation gives

$$A_K = \mathbf{T}_K \mathbf{B}_K \mathbf{T}_K^{-1}, \quad (197)$$

where, in the order $(z, E_{11}, E_{12}, E_{22})$,

$$\mathbf{B}_K = \begin{pmatrix} -\frac{3(K-1)}{2(K+1)} & 0 & \frac{K+7}{2(K+1)^2} & -\frac{s(K+7)}{6(K+1)^2} \\ \frac{3s(K-1)}{K+7} & -\frac{K^2+20K-17}{(K+1)(K+7)} & \frac{12s}{K+7} & -\frac{2(K-2)}{K+7} \\ \frac{3(K-1)(K+1)}{2(K+7)} & \frac{2(K-11)s}{(K+1)(K+7)} & -\frac{7K^2-10K+19}{2(K+1)(K+7)} & \frac{s(K^2-2K+9)}{2(K+1)(K+7)} \\ \frac{9s(K-1)}{K+7} & \frac{12(K-2)}{(K+1)(K+7)} & -\frac{12s}{K+7} & \frac{K-11}{K+7} \end{pmatrix}.$$

Its characteristic polynomial factors as

$$\det(\lambda I - A_K) = \frac{\lambda+1}{2(K+1)^3} p_K(\lambda+1), \quad (198)$$

where

$$p_K(\mu) = 2(K+1)^3 \mu^3 + 2(K-5)(K+1)^2 \mu^2 + 4(K-2)(K+1)^2 \mu + (K-2)(K^2 - 10K - 23).$$

For $K \geq 16$, all coefficients are positive and the cubic Routh–Hurwitz determinant is

$$6(K-2)(K-1)^2(K+1)^3 > 0.$$

Thus every eigenvalue of A_K has real part at most -1 . The entries of the scaled matrix satisfy

$$|(A_K)_{ij}| \preceq \begin{pmatrix} 3/2 & 0 & 2/3 & 1/4 \\ 3 & 3/2 & 12 & 2 \\ 3/2 & 1/8 & 7/2 & 1/2 \\ 9 & 3/4 & 12 & 1 \end{pmatrix},$$

so $\|A_K\|_{\text{op}} < 21$. In a complex Schur decomposition $A_K = U(D + N)U^*$, the matrix N is strictly upper triangular, $\|N\|_{\text{op}} \leq 42$, and $\|e^{Dt}\|_{\text{op}} \leq e^{-t}$. Iterated Duhamel expansion terminates after three factors of N and yields

$$\|e^{A_K t}\|_{\text{op}} \leq e^{-t} \sum_{j=0}^3 \frac{(42t)^j}{j!} < 2 \cdot 10^6 e^{-t/2}. \quad (199)$$

The nonlinear estimate is a finite-dimensional calculation. Insert the explicit inverse of S into (193)–(194). For $\|y\| \leq 10^{-12}$ and $K \geq 16$,

$$\sup_{j,\ell=1}^4 \left| \partial_{j\ell}^2(\mathcal{F}_K)_i \right| \leq (100, 2000, 500, 2000)_i. \quad (200)$$

Indeed, these bounds follow term by term from $d_K \leq 18$, $c/\sqrt{K} \leq 1/3$, $Q_K \succeq 3I/2$, and

$$E = \begin{pmatrix} y_2/\sqrt{K} & y_3 \\ y_3 & y_4/\sqrt{K} \end{pmatrix};$$

on this ball, $\|(Q_K + E)^{-1}\|_{\text{op}} \leq 1$. Taylor's theorem then gives

$$\|\mathcal{N}_K(y)\| \leq \frac{1}{2} \sqrt{100^2 + 2000^2 + 500^2 + 2000^2} \|y\|^2 < 4000 \|y\|^2. \quad (201)$$

It remains to compare Gaussian averages with the point fields at the actual state. Define the first-exit time

$$\tau := \frac{K}{100} \wedge \inf \left\{ t \geq 0 : \|y_t\| = 10^{-12} \text{ or } r_t \notin [e^{3/2}r_0, e^{5/2}r_0] \right\}. \quad (202)$$

On $[0, \tau]$, (184) and (196) imply $S_t \succeq I$, hence $C_t \preceq \alpha_0^{-1}I$. Put $Z_t = X_t - m_t$ and

$$\mathcal{G}_t := \left\{ \|Z_t\| \leq \frac{\delta_K r_t}{8\sqrt{K}} \right\}, \quad p_K := \sup_{t \leq \tau} \mathbb{P}(\mathcal{G}_t^c), \quad M_K := \sup_{t \leq \tau} \mathbb{E} \left[\sqrt{\alpha_0} \|Z_t\| \mathbf{1}_{\mathcal{G}_t^c} \right].$$

On $\mathcal{G}_t$, every point of the segment $m_t + uZ_t$, $0 \leq u \leq 1$, satisfies

$$|\Delta\theta| + c|\Delta\ell| \leq 2(1+c) \frac{\|Z_t\|}{r_t} < \frac{\delta_K}{4}.$$

Because $|z_t| \leq 10^{-12}/K < \delta_K/4$ and $r_t \geq e^{3/2}r_0$, the whole segment remains in the uncut flat phase core. Under a common Gaussian representation, $\sqrt{\alpha_0} \|Z_t\| \leq \|G\|$ for $G \sim \mathcal{N}(0, I_2)$. The exact two-dimensional radial tail gives

$$p_K \leq \exp \left(-\frac{\delta_K^2 \alpha_0 r_0^2}{128K} \right) \leq 10^{-50} K^{-30}, \quad M_K \leq \sqrt{2p_K} \leq \sqrt{2} 10^{-25} K^{-15}. \quad (203)$$

On the flat core, differentiation of (194) in Cartesian coordinates gives

$$\text{Lip}_{\text{op}}(\nabla^2 V_K) \leq \frac{15\alpha_0 K}{r_t} \quad (204)$$

along these segments. For example, $\|\partial_z \mathbf{H}_K\|_{\text{op}} \leq 12\sqrt{K}$, $\|\nabla z\| \leq 2\sqrt{K}/(5r)$, and rotation of the polar frame contributes at most $2\alpha_0(K + 0.02)/r$; allowing a factor two for the radius along the segment gives the displayed constant.

Define the normalized Gaussian errors

$$\begin{aligned}\varepsilon_b(t) &:= \left\| \frac{1}{\alpha_0 r_t} R_{\theta_t}^\top (\mathbb{E}[\nabla V_K(X_t)] - \nabla V_K(m_t)) \right\|, \\ \varepsilon_H(t) &:= \left\| \frac{1}{\alpha_0} R_{\theta_t}^\top (\mathbb{E}[\nabla^2 V_K(X_t)] - \nabla^2 V_K(m_t)) R_{\theta_t} \right\|_{\text{op}}.\end{aligned}$$

Let $A_0 := \sqrt{\alpha} r_0$. Split the expectations at $\mathcal{G}_t$, use (204) for the Taylor remainder on $\mathcal{G}_t$, the global Hessian bound on its complement, $\mathbb{E}Z_t = 0$, and $\mathbb{E}\|Z_t\|^2 \leq 2/\alpha_0$. This gives

$$\varepsilon_b(t) \leq 8 \frac{K}{A_0^2} + 3 \frac{K}{A_0} M_K, \quad \varepsilon_H(t) \leq 15 \frac{K}{A_0} + 3K p_K. \quad (205)$$

Let $\xi_K(t)$ be the contribution of these actual-state errors to the scaled transverse system. Since $\|S_t^{-1}\|_{\text{op}} \leq 1$, $\|S_t\|_{\text{op}} \leq 2K$, and $\|w\| \leq 2\sqrt{K}/5$, equations (195) and (196) imply

$$\begin{aligned}\|\xi_K(t)\| &\leq \frac{2}{5} K^{3/2} \varepsilon_b(t) + \sqrt{2K} (\varepsilon_H(t) + 4K \varepsilon_b(t)) \\ &\leq 23 \frac{K^{3/2}}{A_0} + 5K^{3/2} p_K + 52 \frac{K^{5/2}}{A_0^2} + 20 \frac{K^{5/2}}{A_0} M_K.\end{aligned} \quad (206)$$

The radius condition (189) and (203) make the last line smaller than $3 \cdot 10^{-24}$, hence smaller than 10^{-22} , throughout $[0, \tau]$.

The exact transverse equation is therefore

$$\dot{y}_t = A_K y_t + \mathcal{N}_K(y_t) + \xi_K(t), \quad y_0 = 0. \quad (207)$$

Variation of constants together with (199), (201), and (206) yields, for $t \leq \tau$,

$$\|y_t\| \leq 2 \cdot 10^6 \int_0^t e^{-(t-s)/2} (4000 \cdot 10^{-24} + 10^{-22}) ds < 2 \cdot 10^{-14}.$$

Thus the transverse component cannot trigger the exit in (202).

Finally, let $\mathcal{R}_K(y) := -e_1^\top S^{-1} b_K(z)$ be the point radial field. Differentiating the same explicit 2×2 inverse gives

$$\sup_{\|y\| \leq 10^{-12}} \|\nabla \mathcal{R}_K(y)\| \leq \frac{20}{K}, \quad \mathcal{R}_K(0) = -\frac{3}{K+1}.$$

The Gaussian correction is at most $\varepsilon_b(t)$, so

$$\left| \frac{d}{dt} \log r_t + \frac{3}{K+1} \right| \leq \frac{20}{K} \|y_t\| + \varepsilon_b(t) < \frac{1}{4K}.$$

This proves $-4/K \leq \dot{r}_t/r_t \leq -2/K$. The radius can neither reach the upper barrier nor, before time $K/100$, fall below $e^{2-0.04} r_0 > e^{3/2} r_0$. Hence $\tau = K/100$, which proves (190); integration gives (191). $\square$

B.1.4 Optimizer comparison and completion of the lower bound

Lemma B.3 (Optimizer covariance and intrinsic condition number). *Let $C_\star$ be the optimizing covariance for V_K , and define*

$$\eta_K := K \exp\left(-\frac{\alpha r_0^2}{2}\right).$$

Under (189), $\eta_K \leq 10^{-3}$ and

$$\alpha_0(1 - \eta_K)I \preceq C_\star^{-1} \preceq \alpha_0(1 + \eta_K)I. \quad (208)$$

Moreover,

$$K \leq \kappa_*(V_K) \leq 1.03K, \quad (209)$$

and

$$\beta_* \lambda_{0,*} \geq \frac{1 - \eta_K}{1 + \eta_K}. \quad (210)$$

Proof. The potential is exactly quadratic with Hessian $\alpha_0 I$ on $\{\|x\| \leq r_0\}$. Since $C_* \preceq \alpha^{-1} I$, write $X_* = C_*^{1/2} Z$, $Z \sim \mathcal{N}(0, I_2)$. Then $\|X_*\| > r_0$ implies $\|Z\| > \sqrt{\alpha} r_0$, and $\mathbb{P}(\|Z\| > s) = e^{-s^2/2}$ in two dimensions. Combining this tail identity with $\|\nabla^2 V_K - \alpha_0 I\|_{\text{op}} \leq \alpha_0 K$ and

$$C_*^{-1} = \mathbb{E}[\nabla^2 V_K(X_*)]$$

proves (208).

The global Hessian certificate and (208) give

$$\alpha_* \geq \frac{0.98}{1 + \eta_K}, \quad \beta_* \leq \frac{K + 0.02}{1 - \eta_K},$$

and hence $\kappa_* \leq 1.03K$. At the origin the Hessian is $\alpha_0 I$, while at a valley point it has eigenvalue $\alpha_0 K$. Therefore

$$\alpha_* \leq \alpha_0 \lambda_{\min}(C_*), \quad \beta_* \geq \alpha_0 K \lambda_{\min}(C_*),$$

which proves $\kappa_* \geq K$.

Finally, (187) gives $C_0 \succeq (\alpha_0 K)^{-1} I$, whereas $C_*^{-1} \succeq \alpha_0(1 - \eta_K) I$. Thus $\lambda_{0,*} \geq (1 - \eta_K)/K$. The valley eigenvalue and (208) also give $\beta_* \geq K/(1 + \eta_K)$, proving (210). $\square$

Proof of Theorem 3.3. Theorem B.1 proves the global C^3 regularity and curvature bounds, while (187) proves admissibility of the anisotropic initialization. The potential is even, so its optimizing mean is $m_* = 0$. Strong convexity in the mean and global optimality of $(0, C_*)$ imply

$$\Delta(m, C) \geq \frac{\alpha}{2} \|m\|^2 \quad \text{for every } (m, C).$$

By Theorem B.2, for $0 \leq t \leq K/100$,

$$\Delta(a_t) \geq \frac{\alpha}{2} e^{-0.08} \|m_0\|^2.$$

Therefore no level $\delta_{\text{loc}} < (\alpha/2) e^{-0.08} \|m_0\|^2$ can be reached before $K/100$. The intrinsic comparison $\kappa_* \leq 1.03K$ in Theorem B.3 gives

$$T_{\text{loc}} \geq \frac{K}{100} \geq \frac{\kappa_*(V_K)}{103}.$$

$\square$

For $d > 2$, define

$$V_K^{(d)}(x) := V_K(x_1, x_2) + \frac{\alpha_0}{2} \sum_{j=3}^d x_j^2$$

and initialize the added coordinates at equilibrium. The block-diagonal subsystem is invariant. Hadamard's determinant inequality shows that a cross-covariance block can only decrease the Gaussian determinant while leaving the separated potential expectation unchanged, so the optimizing covariance is block diagonal as well. The two-dimensional slow subsystem and its intrinsic condition comparison are unchanged.

The construction can also meet any prescribed positive Hessian-Lipschitz constant. Differentiating (181), using $\|\chi'''\|_\infty \leq 52.5$ and $\|g'_K\|_\infty \leq 4.376/\delta_K$, gives

$$\text{Lip}_{\text{op}}(\nabla^2 V_K) \leq 10^8 \frac{\alpha_0 K^{5/2}}{r_0}. \quad (211)$$

Thus, for $L_H > 0$, it is enough to choose

$$r_0 \geq \max \left\{ \frac{10^8 \alpha_0 K^{5/2}}{L_H}, \frac{10^{25} K^{3/2} \sqrt{\log(eK)}}{\sqrt{\alpha}} \right\}.$$

The initial radius is allowed to grow with K , and with L_H^{-1} in this version. At $L_H = 0$ the target is quadratic, and this construction does not assert a linear lower bound under a radius budget fixed independently of L_H .

B.2 The one-dimensional bump train

This subsection proves Theorem 3.8. Normalize $\alpha = 1$, $\beta = \kappa$, and write a scalar Gaussian parameter as $a = (m, c)$. For $X \sim \mathcal{N}(m, c)$, set

$$b(m, c) := \mathbb{E}[V'(X)], \quad A(m, c) := \mathbb{E}[V''(X)]. \quad (212)$$

The two deterministic maps reduce respectively to

$$m^+ = m - hc b(m, c), \quad c^+ = c \exp(h[1 - cA(m, c)]), \quad (213)$$

$$m^+ = m - hc b(m, c), \quad c^+ = \frac{(1 + h)c}{1 + hcA(m, c)}. \quad (214)$$

Fix

$$h = \frac{\gamma}{\kappa}, \quad 0 < \gamma \leq \frac{1}{2}, \quad (215)$$

with γ independent of κ .

B.2.1 Bump profile, placement, and regularity

Choose an even $\varphi \in C_c^\infty(\mathbb{R})$ such that

$$0 \leq \varphi \leq 1, \quad \varphi(u) = 1 \quad (|u| \leq \tfrac{1}{2}), \quad \varphi(u) = 0 \quad (|u| \geq 1). \quad (216)$$

Let

$$I_\varphi := \int_{\mathbb{R}} \varphi(u) du, \quad M_\varphi := \|\varphi'\|_\infty.$$

For a prescribed $L_0 > 0$, define

$$w_\kappa := \frac{(\kappa - 1)M_\varphi}{L_0}, \quad B_\kappa := w_\kappa I_\varphi, \quad H_\kappa := (\kappa - 1)B_\kappa. \quad (217)$$

Then $w_\kappa = \Theta(\kappa)$, $H_\kappa = \Theta(\kappa^2)$, and

$$\frac{\kappa - 1}{w_\kappa} \|\varphi'\|_\infty = L_0. \quad (218)$$

Choose $T > 0$ sufficiently small and $Y > 0$ sufficiently large, depending only on φ and L_0 , and set

$$N_\kappa := \left\lfloor \frac{T\kappa^2}{\gamma} \right\rfloor, \quad x_0 := \frac{Y\kappa^4}{\gamma}. \quad (219)$$

Define the positive centers recursively by

$$x_{j+1} := x_j - \frac{\gamma}{\kappa^2} \left[x_j + H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) \right], \quad 0 \leq j < N_\kappa. \quad (220)$$

Lemma B.4 (Geometry of the centers). *For the preceding choices and all sufficiently large κ ,*

$$x_{N_\kappa} \geq \frac{3}{4}x_0 \quad (221)$$

and

$$x_j - x_{j+1} \geq \frac{3Y}{4}\kappa^2, \quad 0 \leq j < N_\kappa. \quad (222)$$

Hence the intervals $[x_j - w_\kappa, x_j + w_\kappa]$ are pairwise disjoint and disjoint from their reflections about the origin.

Proof. Summing (220) and using $x_j \leq x_0$ gives

$$\begin{aligned} x_0 - x_{N_\kappa} &\leq \frac{\gamma N_\kappa}{\kappa^2} x_0 + \frac{\gamma H_\kappa}{\kappa^2} \sum_{j=0}^{N_\kappa-1} \left(N_\kappa - j + \frac{1}{2} \right) \\ &\leq T x_0 + C_\varphi \frac{T^2 \kappa^4}{\gamma}, \end{aligned}$$

where C_φ depends only on φ, L_0 . Since $x_0 = Y\kappa^4/\gamma$, choosing T small and then Y large makes the right side at most $x_0/4$, proving (221). The recursion and this lower bound give

$$x_j - x_{j+1} \geq \frac{\gamma}{\kappa^2} x_{N_\kappa} \geq \frac{3Y}{4}\kappa^2.$$

Because $w_\kappa = O(\kappa)$, the positive supports are disjoint for large κ . Also $x_{N_\kappa} \asymp \kappa^4/\gamma \gg w_\kappa$, so they are disjoint from the reflected supports. $\square$

Define V_κ by

$$V_\kappa''(x) := 1 + (\kappa - 1) \sum_{j=0}^{N_\kappa} \left[\varphi\left(\frac{x - x_j}{w_\kappa}\right) + \varphi\left(\frac{x + x_j}{w_\kappa}\right) \right], \quad (223)$$

with $V_\kappa'(0) = V_\kappa(0) = 0$.

Lemma B.5 (Regularity and exact condition number). *For all sufficiently large κ , the potential V_κ is even and belongs to $C^\infty(\mathbb{R})$. Moreover,*

$$1 \leq V_\kappa''(x) \leq \kappa \quad (x \in \mathbb{R}), \quad (224)$$

both endpoints are attained, and

$$\|V_\kappa'''\|_\infty \leq L_0. \quad (225)$$

The density e^{-V_κ} is integrable, and the original and optimizer-whitened condition numbers are both κ .

Proof. The reflected construction makes V_κ'' even, so the normalizations at zero make V_κ' odd and V_κ even. By Theorem B.4, at most one summand in (223) is nonzero at any point. Hence $1 \leq V_\kappa'' \leq 1 + (\kappa - 1) = \kappa$. The lower value is attained outside the supports, and the upper value on each flat top. Likewise,

$$|V_\kappa'''(x)| \leq \frac{\kappa - 1}{w_\kappa} \|\varphi'\|_\infty = L_0.$$

Since $V_\kappa'' \geq 1$ and $V_\kappa(0) = V_\kappa'(0) = 0$, one has $V_\kappa(x) \geq x^2/2$, proving integrability. In one dimension, optimizer whitening multiplies every Hessian value by the same positive scalar $c_\star$, so it does not change the ratio of the largest and smallest Hessian values. Thus $\kappa_\star = \kappa$. $\square$

B.2.2 Gaussian averages on the train

Set

$$c_\kappa := \frac{1}{\kappa} \quad (226)$$

and define ideal states $a_j^\circ = (x_j, c_\kappa)$.

Lemma B.6 (Averaged score and curvature). *There exist constants $C, c > 0$ and an integer $p \geq 1$, independent of κ , such that, for $0 \leq j \leq N_\kappa$,*

$$A(x_j, c_\kappa) = \kappa + r_{j,\kappa}^A, \quad |r_{j,\kappa}^A| \leq C\kappa^p e^{-c\kappa^3}, \quad (227)$$

$$b(x_j, c_\kappa) = x_j + H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) + r_{j,\kappa}^b, \quad |r_{j,\kappa}^b| \leq C\kappa^p e^{-c\kappa^3}. \quad (228)$$

The same estimates are uniform on

$$\mathcal{T}_{j,\kappa} := \left\{ (m, c) : |m - x_j| \leq \frac{w_\kappa}{8}, \quad |\kappa c - 1| \leq \frac{1}{8} \right\}. \quad (229)$$

More precisely, for $(m, c) \in \mathcal{T}_{j,\kappa}$,

$$|A(m, c) - \kappa| \leq \varepsilon_\kappa, \quad (230)$$

$$\left| b(m, c) - x_j - H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) - \kappa(m - x_j) \right| \leq \varepsilon_\kappa, \quad (231)$$

where

$$\varepsilon_\kappa := C\kappa^p e^{-c\kappa^3}. \quad (232)$$

Proof. At the ideal state write $X = x_j + Z/\sqrt{\kappa}$, with $Z \sim \mathcal{N}(0, 1)$. The current bump equals one whenever $|X - x_j| \leq w_\kappa/2$, and $w_\kappa \asymp \kappa$, so

$$\mathbb{P} \left(|X - x_j| > \frac{w_\kappa}{2} \right) \leq 2e^{-c\kappa^3}.$$

Every other bump is at distance at least $c\kappa^2$, or $c\kappa^{5/2}$ Gaussian standard deviations, from x_j . Summing over $N_\kappa + 1 = O(\kappa^2)$ bumps changes only the polynomial prefactor. This proves (227).

Let

$$\Phi(s) := \int_{-\infty}^s \varphi(u) du.$$

For $x > 0$, except on an exponentially unlikely event on which the Gaussian sample crosses the origin, integration of (223) gives

$$V'_\kappa(x) = x + (\kappa - 1)w_\kappa \sum_{k=0}^{N_\kappa} \Phi \left(\frac{x - x_k}{w_\kappa} \right).$$

The bumps with $k > j$ lie between the origin and x_j and contribute their full mass I_φ ; those with $k < j$ contribute zero, up to the same Gaussian-tail error. For the current bump, $\Phi(s) + \Phi(-s) = I_\varphi$, and symmetry gives

$$\mathbb{E} \Phi \left(\frac{X - x_j}{w_\kappa} \right) = \frac{I_\varphi}{2}.$$

Multiplication by $(\kappa - 1)w_\kappa$ proves (228).

Now take $(m, c) \in \mathcal{T}_{j,\kappa}$. Its mean stays at least $3w_\kappa/8$ from the edge of the current flat top, and its standard deviation is at most $\sqrt{9/(8\kappa)}$. The probability of reaching a transition layer remains bounded by $Ce^{-c\kappa^3}$, uniformly on the tube. This proves (230). Since $\partial_m b(m, c) = A(m, c)$, integration in m gives the term $\kappa(m - x_j)$ with error ε_κ . Price's identity also gives

$$\partial_c b(m, c) = \frac{1}{2} \mathbb{E}[V_\kappa'''(X)].$$

The third derivative is supported on transition layers, which are reached with probability $O(e^{-c\kappa^3})$ throughout the tube. Integration in c proves (231), after enlarging the polynomial prefactor in (232). $\square$

B.2.3 Uniform shadowing for both covariance maps

Let (m_j, c_j) be the iterates of either (213) or (214), initialized by

$$m_0 = x_0, \quad c_0 = c_\kappa = \frac{1}{\kappa}. \quad (233)$$

Define

$$e_j := m_j - x_j, \quad q_j := \kappa c_j - 1. \quad (234)$$

Lemma B.7 (Uniform discrete shadowing). *For either covariance map, there exist constants $C, c > 0$ and an integer $p \geq 1$ such that, for all sufficiently large κ and every $0 \leq j \leq N_\kappa$,*

$$|e_j| + |q_j| \leq C\kappa^p e^{-c\kappa^3}. \quad (235)$$

In particular, every iterate lies in $\mathcal{T}_{j,\kappa}$ and

$$m_j = x_j + o(1), \quad c_j = \frac{1}{\kappa}(1 + o(1))$$

uniformly for $0 \leq j \leq N_\kappa$.

Proof. Write

$$B_j := x_j + H_\kappa \left(N_\kappa - j + \frac{1}{2} \right).$$

The center recursion says $x_{j+1} = x_j - \gamma B_j / \kappa^2$. On the tube, [Theorem B.6](#) gives

$$b(m_j, c_j) = B_j + \kappa e_j + r_j, \quad |r_j| \leq \varepsilon_\kappa.$$

Since $c_j = (1 + q_j)/\kappa$, the mean error obeys the exact recursion

$$e_{j+1} = \left(1 - \frac{\gamma(1 + q_j)}{\kappa} \right) e_j - \frac{\gamma q_j}{\kappa^2} B_j - \frac{\gamma(1 + q_j)}{\kappa^2} r_j. \quad (236)$$

The center geometry gives $|B_j| \leq C\kappa^4/\gamma$. As long as $|q_j| \leq 1/8$,

$$|e_{j+1}| \leq \left(1 - \frac{7\gamma}{8\kappa} \right) |e_j| + C\kappa^2 |q_j| + C \frac{\varepsilon_\kappa}{\kappa^2}. \quad (237)$$

For the Riemannian covariance map write $A(m_j, c_j) = \kappa + \delta_j$, with $|\delta_j| \leq \varepsilon_\kappa$. Then

$$1 + q_{j+1} = (1 + q_j) \exp \left[\frac{\gamma}{\kappa} \left(1 - (1 + q_j) \left(1 + \frac{\delta_j}{\kappa} \right) \right) \right]. \quad (238)$$

For $|q_j| \leq 1/8$ and large κ , the derivative of the right side with respect to q_j lies between zero and $1 - \gamma/(2\kappa)$, while its variation with respect to δ_j is at most $C\gamma|\delta_j|/\kappa^2$. Since the map fixes $q = 0$ when $\delta = 0$, the mean-value theorem yields

$$|q_{j+1}| \leq \left(1 - \frac{\gamma}{2\kappa} \right) |q_j| + C \frac{\varepsilon_\kappa}{\kappa^2}. \quad (239)$$

For the KL/Bregman covariance map,

$$1 + q_{j+1} = \frac{(1 + \gamma/\kappa)(1 + q_j)}{1 + (\gamma/\kappa)(1 + q_j)(1 + \delta_j/\kappa)}. \quad (240)$$

The same mean-value argument gives

$$|q_{j+1}| \leq \left(1 - \frac{\gamma}{2\kappa} \right) |q_j| + C \frac{\varepsilon_\kappa}{\kappa^2}. \quad (241)$$

Since $q_0 = 0$, either covariance recursion implies

$$|q_j| \leq C \frac{\varepsilon_\kappa}{\gamma \kappa} \quad (0 \leq j \leq N_\kappa). \quad (242)$$

Substitute this into (237), use $e_0 = 0$, and sum the geometric recursion to obtain

$$|e_j| \leq C_\gamma \kappa^p e^{-c\kappa^3} \quad (243)$$

after increasing p if necessary. Exponential decay dominates every polynomial, so (242)–(243) are eventually smaller than $1/8$ and $w_\kappa/8$. The tube hypotheses close by induction, proving (235). $\square$

B.2.4 Objective comparison and completion

For $c > 0$, define

$$F_c(m) := \mathbb{E}_{X \sim \mathcal{N}(m, c)}[V_\kappa(X)]. \quad (244)$$

Up to a target-dependent additive constant,

$$\mathcal{E}_\kappa(m, c) = F_c(m) - \frac{1}{2} \log c. \quad (245)$$

Lemma B.8 (Constant-factor objective lower bound). *There is a numerical $q \in (0, 1)$, independent of κ , such that the iterates of either scheme satisfy*

$$\Delta(m_j, c_j) \geq q \Delta(m_0, c_0), \quad 0 \leq j \leq N_\kappa, \quad (246)$$

for all sufficiently large κ .

Proof. The potential is even, so F_c is even and $F'_c(0) = 0$. Moreover, $F''_c(m) = A(m, c) \geq 1$, whence

$$F_c(m) - F_c(0) \geq \frac{1}{2} m^2. \quad (247)$$

At the matched covariance, Theorem B.4 gives

$$F_{c_\kappa}(x_j) - F_{c_\kappa}(0) \geq \frac{1}{2} x_j^2 \geq \frac{1}{2} x_{N_\kappa}^2 \geq \frac{9}{32} x_0^2. \quad (248)$$

Integration of (223) gives the global estimate

$$|V'_\kappa(x)| \leq |x| + H_\kappa(N_\kappa + 1). \quad (249)$$

Therefore, for $m \geq 0$,

$$F'_{c_\kappa}(m) \leq m + \sqrt{\frac{2}{\pi \kappa}} + H_\kappa(N_\kappa + 1).$$

Integrating from zero to x_0 gives

$$F_{c_\kappa}(x_0) - F_{c_\kappa}(0) \leq \frac{1}{2} x_0^2 + \left[H_\kappa(N_\kappa + 1) + \sqrt{\frac{2}{\pi \kappa}} \right] x_0 \leq C_0 x_0^2, \quad (250)$$

where C_0 is independent of κ ; here $H_\kappa(N_\kappa + 1) = O(\kappa^4/\gamma)$ and $x_0 = Y \kappa^4/\gamma$. Combining (248) and (250) yields

$$F_{c_\kappa}(x_j) - F_{c_\kappa}(0) \geq q_0 [F_{c_\kappa}(x_0) - F_{c_\kappa}(0)], \quad q_0 := \frac{9}{32 C_0} > 0. \quad (251)$$

For every fixed covariance, $m \mapsto \mathcal{E}_\kappa(m, c)$ is strictly convex and even, so its unique minimizing mean is zero. Let $c_\star$ be the optimizing covariance and set

$$D_\kappa := \mathcal{E}_\kappa(0, c_\kappa) - \mathcal{E}_\kappa(0, c_\star) \geq 0.$$

At covariance c_κ ,

$$\Delta(x_j, c_\kappa) = F_{c_\kappa}(x_j) - F_{c_\kappa}(0) + D_\kappa.$$

After replacing q_0 by $\min\{q_0, 1\}$, (251) implies

$$\Delta(x_j, c_\kappa) \geq q_0 \Delta(x_0, c_\kappa). \quad (252)$$

It remains to replace the ideal states by the actual iterates. On the tube,

$$|\partial_m \mathcal{E}_\kappa| = |b(m, c)| \leq C\kappa^4/\gamma,$$

while

$$|\partial_c \mathcal{E}_\kappa(m, c)| = \frac{1}{2} \left| A(m, c) - \frac{1}{c} \right| \leq C(\varepsilon_\kappa + \kappa|q|).$$

Together with Theorem B.7, the mean-value theorem gives

$$\sup_{0 \leq j \leq N_\kappa} |\mathcal{E}_\kappa(m_j, c_j) - \mathcal{E}_\kappa(x_j, c_\kappa)| = o(x_0^2). \quad (253)$$

By (247), $\Delta(x_0, c_\kappa) \geq x_0^2/2$, so the error in (253) is $o(\Delta(x_0, c_\kappa))$. Absorbing it in (252), after reducing q_0 by a fixed factor, proves (246). $\square$

Proof of Theorem 3.8. Theorem B.5 proves $1 \leq V_\kappa'' \leq \kappa$, $\|V_\kappa'''\|_\infty \leq L_0$, and $\kappa_\star = \kappa$. The initialization (233) lies at the lower edge of the curvature-scale band $[1/\kappa, 1]$, so the construction begins after covariance burn-in. By Theorem B.8,

$$\Delta_n \geq q\Delta_0 \quad \left(0 \leq n \leq N_\kappa = \left\lfloor \frac{T\kappa^2}{\gamma} \right\rfloor \right)$$

for either exact-expectation map. Since $h = \gamma/\kappa$,

$$\frac{\kappa^2}{\gamma} = \frac{\kappa}{h},$$

which proves the claimed $\Omega(\kappa/h)$ constant-factor lower bound for both discretizations. $\square$

The construction is tied to the fixed value of γ and does not cover adaptive relative-curvature steps, line searches, or implicit updates. Indeed, along the shadowed path,

$$c_j A(m_j, c_j) = 1 + o(1), \quad c_j \simeq \frac{1}{\kappa}, \quad \frac{b(m_j, c_j)}{m_j} = \Theta(1).$$

Thus the relative mean motion is

$$h \frac{c_j b(m_j, c_j)}{m_j} = \Theta\left(\frac{\gamma}{\kappa^2}\right),$$

which is the precise source of the fixed-step slowdown.

C Local score calculus and the convex-ridge obstruction

C.1 Second-score identities and the non-Gaussian modulus

This subsection proves the score identities and the explicit bounds (100)–(102). We work in the optimizer-whitened coordinates of Section 4.2. Assume throughout that $\widehat{V} \in C^2(\mathbb{R}^d)$ and

$$\alpha_\star I \preceq \nabla^2 \widehat{V}(x) \preceq \beta_\star I, \quad x \in \mathbb{R}^d.$$

Boundedness of the Hessian implies at most quadratic growth of $\widehat{V}$ and at most linear growth of its gradient. It therefore justifies all Gaussian density differentiations below, locally uniformly on sets whose covariance eigenvalues are bounded away from zero and infinity. No classical third or fourth derivative is used. The formulas also apply entrywise to vector- and matrix-valued integrands.

Fix $a = (m, C)$, a tangent vector $\zeta = (u, U)$, and set

$$v := C^{-1/2}u, \quad S := C^{-1/2}UC^{-1/2}, \quad t^2 := \|v\|^2 + \frac{1}{2}\|S\|_{\mathbb{F}}^2. \quad (254)$$

If $Z \sim \mathcal{N}(0, I)$ and $X = m + C^{1/2}Z$, define

$$q_\zeta(Z) := v^\top Z + \frac{1}{2}(Z^\top SZ - \text{Tr } S), \quad (255)$$

$$p_\zeta(Z) := \frac{1}{2}\text{Tr}(S^2) - \|v\|^2 - 2v^\top SZ - Z^\top S^2 Z, \quad \chi_\zeta := q_\zeta^2 + p_\zeta. \quad (256)$$

Lemma C.1 (First and second Gaussian score identities). *Let $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ be measurable and of polynomial growth, and put $\Phi(m, C) := \mathbb{E}_{\mathcal{N}(m, C)}[\phi]$. Then*

$$D\Phi(a)[\zeta] = \mathbb{E}[\phi(X)q_\zeta(Z)], \quad D^2\Phi(a)[\zeta, \zeta] = \mathbb{E}[\phi(X)\chi_\zeta(Z)]. \quad (257)$$

Moreover, q_ζ belongs to the first two Hermite chaoses, whereas χ_ζ belongs to chaoses two, three, and four. In particular,

$$\mathbb{E}q_\zeta = \mathbb{E}\chi_\zeta = 0, \quad \mathbb{E}[Z\chi_\zeta] = 0, \quad \mathbb{E}q_\zeta^2 = t^2, \quad \|\chi_\zeta\|_{L^2} \leq \sqrt{6}t^2. \quad (258)$$

The constant $\sqrt{6}$ is sharp.

Proof. Let $m_s = m + su$ and $C_s = C + sU$, which remains positive definite for small $|s|$. Differentiating the log-density of $\mathcal{N}(m_s, C_s)$ at the fixed point $x = m + C^{1/2}Z$ gives

$$\partial_s \log \rho_{m_s, C_s}(x)|_{s=0} = q_\zeta(Z), \quad \partial_s^2 \log \rho_{m_s, C_s}(x)|_{s=0} = p_\zeta(Z).$$

The identity $\partial_s^2 \rho_s = \rho_s[(\partial_s \log \rho_s)^2 + \partial_s^2 \log \rho_s]$ proves (257).

For the chaos calculation, write

$$\ell := v^\top Z, \quad Q := Z^\top SZ - \text{Tr } S, \quad R_2 := Z^\top S^2 Z - \text{Tr } S^2.$$

Then $q_\zeta = \ell + Q/2$, and orthogonality of the first and second chaoses gives $\mathbb{E}q_\zeta^2 = \|v\|^2 + \|S\|_{\mathbb{F}}^2/2 = t^2$. The Gaussian product formula yields

$$(Q^2)_0 = 2\|S\|_{\mathbb{F}}^2, \quad (Q^2)_2 = 4R_2, \quad \ell Q = 2v^\top SZ + G_3,$$

where G_3 is pure chaos three. Substitution in $\chi_\zeta = q_\zeta^2 + p_\zeta$ cancels its chaos-zero and chaos-one parts and leaves the mutually orthogonal components

$$(\chi_\zeta)_2 = \ell^2 - \|v\|^2, \quad (\chi_\zeta)_3 = G_3, \quad (\chi_\zeta)_4 = \frac{1}{4}(Q^2)_4.$$

This proves the first three assertions in (258).

For the L^2 norm, abbreviate

$$a_0 := \|v\|^2, \quad b_0 := \|S\|_F^2, \quad c_0 := v^\top S^2 v, \quad d_4 := \text{Tr}(S^4).$$

Wick's formula and the preceding orthogonal decomposition give

$$\|\chi_\zeta\|_{L^2}^2 = 2a_0^2 + 2a_0b_0 + 4c_0 + \frac{1}{2}b_0^2 + d_4. \quad (259)$$

Since $c_0 \leq a_0b_0$ and $d_4 \leq b_0^2$,

$$\|\chi_\zeta\|_{L^2}^2 \leq 2a_0^2 + 6a_0b_0 + \frac{3}{2}b_0^2 \leq 6 \left(a_0 + \frac{b_0}{2}\right)^2 = 6t^4.$$

Equality holds when $v = 0$ and S has rank one, proving sharpness. $\square$

The Ornstein–Uhlenbeck generator $\mathcal{L}_{\text{OU}} = \Delta_z - z^\top \nabla_z$ acts as multiplication by $-k$ on the k -th Hermite chaos. Thus, for a centered $\psi = \sum_{k \geq 1} \psi_k$,

$$\mathbb{E} \left\| \nabla(-\mathcal{L}_{\text{OU}})^{-1} \psi \right\|^2 = \sum_{k \geq 1} \frac{1}{k} \|\psi_k\|_{L^2}^2. \quad (260)$$

In particular, the right-hand side is at most $\|\psi\|_{L^2}^2 / 2$ when ψ has no chaos-one component. This factor $1/2$ is applicable to χ_ζ , but not to a general centered score.

Lemma C.2 (Explicit non-Gaussian second-differential modulus). *Let $H = F - F_G$ be the non-Gaussian part of the whitened Fisher–Rao field. On the star hull $\mathcal{H}_\delta$,*

$$H_m(a) = C(g(a) + m), \quad H_C(a) = -C(A(a) - I)C.$$

Then

$$K_{\text{ng}}(\delta) := \sup_{\substack{a \in \mathcal{H}_\delta \\ \zeta \neq 0}} \frac{\|D^2 H(a)[\zeta, \zeta]\|_\star}{\|\zeta\|_\star^2} \leq \left(K_{\text{ng},m}(\delta)^2 + \frac{1}{2} K_{\text{ng},C}(\delta)^2 \right)^{1/2}, \quad (261)$$

Here the two block constants are

$$K_{\text{ng},m}(\delta) := \frac{U_\delta^{3/2}}{\ell_\delta^2} \left(\sqrt{3} \bar{\Sigma}_\delta + \Sigma_\delta + \sqrt{\Sigma_\delta^2 + 2\mathcal{B}_\delta^2} \right), \quad (262)$$

$$K_{\text{ng},C}(\delta) := \frac{U_\delta^2}{\ell_\delta^2} \left((4\sqrt{2} + \sqrt{6}) \Sigma_\delta + 4\mathcal{B}_\delta \right). \quad (263)$$

These are the quantities stated in (101)–(102).

Proof. Fix $a \in \mathcal{H}_\delta$. The covariance band gives $\ell_\delta I \preceq C \preceq U_\delta I$. Put

$$x := \|v\|, \quad y := \frac{\|S\|_F}{\sqrt{2}}, \quad t^2 = x^2 + y^2.$$

Then

$$\|U\|_{\text{op}} \leq \sqrt{2} U_\delta y, \quad \|U\|_F \leq \sqrt{2} U_\delta y, \quad t \leq \ell_\delta^{-1} \|\zeta\|_\star. \quad (264)$$

The final inequality uses $\ell_\delta < 1$.

For the mean block, Gaussian integration by parts through $\psi_1 = (-\mathcal{L}_{\text{OU}})^{-1} q_\zeta$, for which $\nabla \psi_1 = v + SZ/2$, gives

$$D(g + m)[\zeta] = -\mathbb{E} \left[(\nabla^2 \hat{V}(X) - I) C^{1/2} \left(v + \frac{1}{2} SZ \right) \right]. \quad (265)$$

Split $\nabla^2 \widehat{V} - I = (\nabla^2 \widehat{V} - A) + (A - I)$. The centered Hessian annihilates the constant v , while the deterministic part annihilates SZ . By the definitions of $\mathcal{B}_\delta$ and Σ_δ ,

$$\|D(g+m)[\zeta]\| \leq \sqrt{U_\delta} \left(\mathcal{B}_\delta x + \frac{\Sigma_\delta}{\sqrt{2}} y \right). \quad (266)$$

For the second derivative, let $\psi_2 = (-\mathcal{L}_{\text{OU}})^{-1} \chi_\zeta$. By (260) and (258),

$$\mathbb{E} \|\nabla \psi_2\|^2 \leq 3t^4, \quad \mathbb{E} \nabla \psi_2 = 0.$$

Another Gaussian integration by parts therefore gives

$$D^2 g[\zeta, \zeta] = -\mathbb{E} \left[(\nabla^2 \widehat{V}(X) - A(a)) C^{1/2} \nabla \psi_2 \right].$$

This expectation can be bounded either with the Hessian dispersion Σ_δ or with the pointwise operator bound $\|\nabla^2 \widehat{V} - A\|_{\text{op}} \leq \beta_\star - \alpha_\star$. Consequently,

$$\|D^2 g[\zeta, \zeta]\| \leq \sqrt{3U_\delta} \bar{\Sigma}_\delta t^2. \quad (267)$$

This is an operator-norm alternative, not the false pointwise assertion $\|\nabla^2 \widehat{V} - A\|_{\text{F}} \leq \beta_\star - \alpha_\star$.

Since C is affine in the covariance coordinate,

$$D^2 H_m[\zeta, \zeta] = 2U D(g+m)[\zeta] + C D^2 g[\zeta, \zeta].$$

Combining (266) and (267) with (264) yields

$$\|D^2 H_m[\zeta, \zeta]\| \leq U_\delta^{3/2} \left(2\sqrt{2} \mathcal{B}_\delta x y + 2\Sigma_\delta y^2 + \sqrt{3} \bar{\Sigma}_\delta t^2 \right).$$

The largest eigenvalue of the first two terms as a quadratic form in (x, y) is $\Sigma_\delta + \sqrt{\Sigma_\delta^2 + 2\mathcal{B}_\delta^2}$. Using $t \leq \ell_\delta^{-1} \|\zeta\|_\star$ proves

$$\|D^2 H_m[\zeta, \zeta]\| \leq K_{\text{ng},m}(\delta) \|\zeta\|_\star^2. \quad (268)$$

For the covariance block, write $W = A - I$. The score identities imply

$$\|DW[\zeta]\|_{\text{F}} \leq \Sigma_\delta t, \quad \|D^2 W[\zeta, \zeta]\|_{\text{F}} \leq \sqrt{6} \Sigma_\delta t^2. \quad (269)$$

Indeed, $DA[\zeta] = \mathbb{E}[(\nabla^2 \widehat{V} - A)q_\zeta]$ and $D^2 A[\zeta, \zeta] = \mathbb{E}[(\nabla^2 \widehat{V} - A)\chi_\zeta]$. The product rule gives

$$D^2(CWC)[\zeta, \zeta] = 2U(DW)C + 2C(DW)U + 2UWU + C(D^2 W)C.$$

Using $\|W\|_{\text{op}} \leq \mathcal{B}_\delta$ and (264)–(269),

$$\begin{aligned} \|D^2 H_C[\zeta, \zeta]\|_{\text{F}} &\leq U_\delta^2 \left(4\sqrt{2} \Sigma_\delta t y + 4\mathcal{B}_\delta y^2 + \sqrt{6} \Sigma_\delta t^2 \right) \\ &\leq K_{\text{ng},C}(\delta) \|\zeta\|_\star^2. \end{aligned}$$

Finally, $\|D^2 H\|_\star^2 = \|D^2 H_m\|^2 + \|D^2 H_C\|_{\text{F}}^2 / 2$, which proves (261). Each constant is proportional to $\mathcal{B}_\delta$, Σ_δ , or $\bar{\Sigma}_\delta$, so the modulus vanishes for a Gaussian target. $\square$

C.2 The convex-ridge construction

All constants implicit in $O(\cdot)$ below are numerical and independent of K and the dimension. Let $p \in \mathbb{N}$ tend to infinity and set

$$K := (8p)^{1/4}, \quad d_K := p + 1, \quad p = \frac{K^4}{8}. \quad (270)$$

Write $z = (t, y) \in \mathbb{R} \times \mathbb{R}^p$, $r = \|y\|$, and use the regularized radius

$$\rho(y) := \sqrt{p + r^2}, \quad \rho_0 := \sqrt{2p} = \frac{K^2}{2}, \quad c := \rho_0 + \frac{1}{2}. \quad (271)$$

The additive p in ρ is essential: it keeps $D^2\rho$ and $D^3\rho$ uniformly controlled at $y = 0$.

Define a convex $C^{2,1}$ profile $F_K : \mathbb{R} \rightarrow \mathbb{R}$, up to an irrelevant additive constant, by

$$F_K''(x) := (K - |x - c|)_+, \quad F_K'(-\infty) = 0. \quad (272)$$

Its total curvature mass is K^2 , and

$$0 \leq F_K'' \leq K, \quad 0 \leq F_K' \leq K^2, \quad \text{Lip}(F_K'') \leq 1. \quad (273)$$

For $s \geq 0$, set

$$U_s(t, y) := F_K(\rho(y) + st), \quad x := \rho(y) + st, \quad q := \nabla x = \left(s, \frac{y}{\rho}\right). \quad (274)$$

Since

$$D^2\rho(y) = \frac{I_p}{\rho} - \frac{yy^\top}{\rho^3} \succeq 0,$$

the ridge Hessian is

$$\mathcal{A}_s := D^2U_s = F_K''(x)qq^\top + F_K'(x) \begin{pmatrix} 0 & 0 \\ 0 & D^2\rho(y) \end{pmatrix} \succeq 0. \quad (275)$$

Let $Z = (\Theta, Y) \sim \mathcal{N}(0, I_{d_K})$, put $R = \|Y\|$, and continue to write $\rho = \sqrt{p + R^2}$. Define $\xi := \rho - \rho_0$.

Lemma C.3 (Dimension-free radial fluctuations). *For every fixed $q < \infty$, there is $C_q < \infty$ such that*

$$\|\xi\|_{L^q} \leq C_q, \quad \mathbb{E}\xi = -\frac{\mathbb{E}\xi^2}{2\rho_0} = O(K^{-2}). \quad (276)$$

Uniformly for $0 \leq s \leq 2K^{-1/2}$, the random variable

$$D_s := \xi + s\Theta - \frac{1}{2}$$

has dimension-free sub-Gaussian tails. In particular, for every fixed $q < \infty$, there are $c_q, C_q > 0$ such that

$$\sup_{0 \leq s \leq 2K^{-1/2}} \mathbb{E} \left[(1 + |D_s|)^q \mathbf{1}_{\{|D_s| > K\}} \right] \leq C_q e^{-c_q K^2}. \quad (277)$$

Proof. The identity

$$\xi = \frac{R^2 - p}{\rho + \rho_0}$$

and $\rho + \rho_0 \geq \rho_0 \asymp \sqrt{p}$ give $\|\xi\|_{L^q} \leq \|R^2 - p\|_{L^q} / \rho_0 \leq C_q$; the last inequality is the standard fixed-moment bound for a centered χ_p^2 variable. Moreover, $\rho^2 - \rho_0^2 = 2\rho_0\xi + \xi^2$ has mean zero, which proves the exact mean identity in (276).

The map $y \mapsto \sqrt{p + \|y\|^2}$ is 1-Lipschitz. Gaussian concentration therefore makes $\rho - \mathbb{E}\rho$ sub-Gaussian with a numerical variance proxy. Since $\mathbb{E}\xi = O(K^{-2})$ and $s\Theta$ has variance at most $4/K$, the same is true uniformly for D_s . Integrating the sub-Gaussian tail, or applying Cauchy–Schwarz against its fixed polynomial moments, proves (277). $\square$

The tilt is chosen to make the averaged ridge Hessian isotropic. Define

$$\bar{K}(s) := \mathbb{E}F_K''(\rho + s\Theta), \quad (278)$$

$$G(s) := \frac{1}{p} \mathbb{E} \left[F_K''(\rho + s\Theta) \frac{R^2}{\rho^2} + F_K'(\rho + s\Theta) \left(\frac{p}{\rho} - \frac{R^2}{\rho^3} \right) \right]. \quad (279)$$

Lemma C.4 (Averaged profile and exact isotropic tuning). *For every $s \geq 0$,*

$$\mathbb{E}\mathcal{A}_s = \text{diag}(s^2\bar{K}(s), G(s)I_p). \quad (280)$$

Uniformly for $0 \leq s \leq 2K^{-1/2}$,

$$\bar{K}(s) = K + O(1), \quad \mathbb{E}F_K'(\rho + s\Theta) = \rho_0 - \frac{K}{2} + O(1), \quad G(s) = 1 - \frac{1}{K} + O(K^{-2}). \quad (281)$$

For all sufficiently large K , there is $s_K \in (0, 2K^{-1/2})$ such that

$$s_K^2 \bar{K}(s_K) = G(s_K) =: a_K. \quad (282)$$

For this choice,

$$s_K^2 = K^{-1} + O(K^{-2}), \quad a_K = 1 - K^{-1} + O(K^{-2}), \quad \alpha_K := 1 - a_K = K^{-1} + O(K^{-2}), \quad (283)$$

and $\mathbb{E}\mathcal{A}_{s_K} = a_K I_{d_K}$.

Proof. The (t, t) entry of $\mathbb{E}\mathcal{A}_s$ is $s^2\bar{K}(s)$. Every mixed entry is odd under one of the reflections $Y_i \mapsto -Y_i$. Rotational symmetry makes the yy block a scalar multiple of I_p , and averaging its diagonal entries using (275) gives exactly $G(s)$. This proves (280).

Since $\rho + s\Theta = c + D_s$, on $\{|D_s| \leq K\}$ the profile has the exact form

$$F_K''(c + D_s) = K - |D_s|, \quad F_K'(c + D_s) = \frac{K^2}{2} + KD_s - \frac{1}{2}D_s|D_s|. \quad (284)$$

The tail contribution outside this event is exponentially small by (277). Since $\mathbb{E}|D_s| = O(1)$, $\mathbb{E}[D_s | D_s|] = O(1)$, and $\mathbb{E}D_s = -1/2 + O(K^{-2})$, formula (284) gives the first two estimates in (281).

For the last estimate, decompose $G = \text{(I)} + \text{(II)} - \text{(III)}$, where

$$\text{(I)} = \frac{1}{p} \mathbb{E} \left[F_K'' \frac{R^2}{\rho^2} \right], \quad \text{(II)} = \mathbb{E} \left[\frac{F_K'}{\rho} \right], \quad \text{(III)} = \frac{1}{p} \mathbb{E} \left[F_K' \frac{R^2}{\rho^3} \right].$$

The profile bounds imply $\text{(I)} = O(K/p) = O(K^{-3})$. Also,

$$\left\| \frac{1}{\rho} - \frac{1}{\rho_0} \right\|_{L^2} = \left\| \frac{\xi}{\rho\rho_0} \right\|_{L^2} \leq \frac{\|\xi\|_{L^2}}{\sqrt{2}p} = O(K^{-4}),$$

and $F_K' \leq K^2$, so

$$\text{(II)} = \frac{\mathbb{E}F_K'}{\rho_0} + O(K^{-2}).$$

Finally, $\sup_{r \geq 0} r^2(p + r^2)^{-3/2} \leq p^{-1/2}$, whence $\text{(III)} \leq K^2 p^{-3/2} = O(K^{-4})$. Using $\rho_0 = K^2/2$ proves $G(s) = 1 - K^{-1} + O(K^{-2})$.

The function $\Phi_K(s) := s^2\bar{K}(s) - G(s)$ is continuous by dominated convergence. The preceding estimates give

$$\Phi_K(0) = -1 + O(K^{-1}) < 0, \quad \Phi_K(2K^{-1/2}) = 3 + O(K^{-1}) > 0.$$

The intermediate value theorem produces s_K . At a zero, $a_K = G(s_K) = 1 - K^{-1} + O(K^{-2})$, and division by $\bar{K}(s_K) = K + O(1)$ gives the remaining estimates in (283). $\square$

Fix this tuned $s = s_K$, let

$$b_K := \mathbb{E}[\nabla U_s(Z)], \quad V_K(z) := \frac{\alpha_K}{2} \|z\|^2 + U_s(z) - b_K^\top z, \quad B_K(z) := \nabla^2 V_K(z) = \alpha_K I + \mathcal{A}_s(z). \quad (285)$$

The expectation defining b_K is finite because $\|\nabla U_s\| \leq \sqrt{2}K^2$.

Lemma C.5 (Exact optimizer whitening and ellipticity). *The standard Gaussian is the unique optimizing Gaussian for V_K , and*

$$\mathbb{E}[\nabla V_K(Z)] = 0, \quad \mathbb{E}[B_K(Z)] = I_{d_K}. \quad (286)$$

If

$$L_K := \sup_{z \in \mathbb{R}^{d_K}} \|\mathcal{A}_s(z)\|_{\text{op}}, \quad \beta_K := \alpha_K + L_K,$$

then, for all sufficiently large K ,

$$\frac{K}{2} \leq L_K \leq 3K, \quad \alpha_K I \preceq B_K(z) \preceq \beta_K I, \quad \frac{\beta_K}{\alpha_K} = \Theta(K^2). \quad (287)$$

Proof. The definition of b_K and $\mathbb{E}Z = 0$ give the first equality in (286). The second follows from $\mathbb{E}\mathcal{A}_s = \alpha_K I$ and $\alpha_K + \alpha_K = 1$.

These first-order conditions identify the optimizer, not merely a stationary point. Parameterize a Gaussian by $\mathcal{N}(\mu, P^2)$ with $P \in \mathbb{S}_{++}(d_K)$. Up to a constant, its reverse-KL objective is

$$(\mu, P) \mapsto \mathbb{E}[V_K(\mu + PZ)] - \log \det P.$$

The first term is α_K -strongly convex in (μ, P) , because $\mathbb{E}\|\delta\mu + \delta PZ\|^2 = \|\delta\mu\|^2 + \|\delta P\|_{\text{F}}^2$, and $-\log \det P$ is convex. At $(0, I)$, the derivative in μ vanishes by (286); the derivative in P is $\mathbb{E}[\nabla V_K(Z)Z^\top] - I = \mathbb{E}[B_K(Z)] - I = 0$, by Gaussian integration by parts. Strong convexity therefore makes $(0, I)$ the unique minimizer.

For the upper Hessian bound, $\rho \geq \sqrt{p} = K^2/\sqrt{8}$ and $s^2 \leq 4/K$ imply, for large K ,

$$\|q\|^2 = s^2 + \frac{r^2}{\rho^2} \leq 2, \quad \|D^2\rho\|_{\text{op}} \leq \frac{1}{\rho} \leq \frac{\sqrt{8}}{K^2}.$$

Equations (273) and (275) yield

$$\|\mathcal{A}_s\|_{\text{op}} \leq F_K'' \|q\|^2 + F_K' \|D^2\rho\|_{\text{op}} \leq 2K + \sqrt{8} \leq 3K.$$

For the lower bound, take $r^2 = p$, so $\rho = \rho_0$, and $t = 1/(2s)$, so $x = c$. Then $F_K''(x) = K$ and $\|q\|^2 = s^2 + 1/2 \geq 1/2$; the rank-one first term in (275) has an eigenvalue at least $K/2$. This proves the bounds on L_K . Positive semidefiniteness of $\mathcal{A}_s$ gives the lower curvature bound, and it is attained by taking t sufficiently negative that $F_K' = F_K'' = 0$. Finally, $\alpha_K = \Theta(K^{-1})$ and $L_K = \Theta(K)$ prove the condition-number estimate. $\square$

Lemma C.6 (A dimension-free Lipschitz Hessian). *The tent-profile potential V_K belongs to $C^{2,1}(\mathbb{R}^{d_K})$ and satisfies*

$$\left\| \nabla^2 V_K(z) - \nabla^2 V_K(z') \right\|_{\text{op}} \leq 64 \|z - z'\|, \quad z, z' \in \mathbb{R}^{d_K}. \quad (288)$$

Proof. At points where the Lipschitz function F_K'' is differentiable, $|F_K'''| \leq 1$. For unit vectors u, v, w , differentiating (275) gives

$$D^3 U_s[u, v, w] = F_K'''(x) q[u] q[v] q[w]$$

$$\begin{aligned}
& + F_K''(x)(q[u]D^2\rho[v, w] + q[v]D^2\rho[u, w] + q[w]D^2\rho[u, v]) \\
& + F_K'(x)D^3\rho[u, v, w].
\end{aligned} \tag{289}$$

Here derivatives of ρ ignore the distinguished coordinate. Direct differentiation in the y variables gives

$$\begin{aligned}
D^3\rho[u, v, w] = & - \frac{(u^\top v)(y^\top w) + (u^\top w)(y^\top v) + (v^\top w)(y^\top u)}{\rho^3} \\
& + \frac{3(y^\top u)(y^\top v)(y^\top w)}{\rho^5}.
\end{aligned}$$

Consequently,

$$\sup_{\|u\|=\|v\|=\|w\|=1} |D^3\rho[u, v, w]| \leq \frac{3r}{\rho^3} + \frac{3r^3}{\rho^5} \leq \frac{6}{\rho^2} \leq \frac{6}{p} = \frac{48}{K^4}.$$

Together with $\|q\| \leq \sqrt{2}$, $\|D^2\rho\|_{\text{op}} \leq \sqrt{8}/K^2$, and (273), this gives almost everywhere

$$\sup_{\|u\|=\|v\|=\|w\|=1} |D^3U_s[u, v, w]| \leq 2\sqrt{2} + \frac{12}{K} + \frac{48}{K^2} \leq 64.$$

To pass from the almost-everywhere derivative bound to (288), fix unit u, v and consider $f_{u,v}(z) = u^\top \mathcal{A}_s(z)v$. This function is locally Lipschitz, and its weak gradient has norm at most 64 almost everywhere. Convolving $f_{u,v}$ with a standard mollifier gives smooth 64-Lipschitz functions; letting the mollifier radius tend to zero preserves the bound. Taking the supremum over unit u, v proves the operator-norm estimate. The quadratic and linear terms in (285) do not change the Hessian Lipschitz seminorm. $\square$

Define the first- and second-Hermite operators of the constructed potential by

$$\mathcal{T}_K[X] := \mathbb{E}[\text{Tr}(XB_K(Z))Z], \tag{290}$$

$$\mathcal{H}_K[X] := \mathbb{E}[B_K(Z)(Z^\top XZ - \text{Tr } X)]. \tag{291}$$

Use the unit-Frobenius radial direction

$$X_K := \text{diag}\left(0, \frac{I_p}{\sqrt{p}}\right), \quad \|X_K\|_F = 1. \tag{292}$$

Lemma C.7 (Codazzi conversion and the large Hermite coupling). *The vector $\mathcal{T}_K[X_K]$ points in the distinguished coordinate direction and satisfies the exact identity*

$$\|\mathcal{T}_K[X_K]\| = \frac{s}{\sqrt{p}} \mathbb{E}\left[F_K''(\rho + s\Theta) \frac{R^2}{\rho}\right]. \tag{293}$$

Moreover,

$$\|\mathcal{T}_K[X_K]\|^2 = \frac{K}{2} + O(1). \tag{294}$$

Proof. The constant matrix $\alpha_K I$ has zero first-Hermite coefficient. Furthermore,

$$\text{Tr}(X_K \mathcal{A}_s) = \frac{1}{\sqrt{p}} \sum_{i=1}^p \partial_{ii} U_s$$

is a function only of (Θ, R) . Reflection in Y_i shows that every radial component of $\mathcal{T}_K[X_K]$ vanishes. Its distinguished component is therefore

$$\frac{1}{\sqrt{p}} \sum_{i=1}^p \mathbb{E}[\Theta \partial_{ii} U_s(\Theta, Y)].$$

The bounded Hessian $\mathcal{A}_s = D^2 U_s$ defines a tempered distribution. Commutation of weak derivatives, followed by Gaussian integration by parts in the distinguished and i -th radial coordinates, gives the Codazzi identity

$$\mathbb{E}[\Theta(\mathcal{A}_s)_{ii}] = \mathbb{E}[Y_i(\mathcal{A}_s)_{0i}].$$

Indeed, writing φ for the standard Gaussian density,

$$-\int (\mathcal{A}_s)_{ii} \partial_0 \varphi = \langle \partial_0(\mathcal{A}_s)_{ii}, \varphi \rangle = \langle \partial_i(\mathcal{A}_s)_{0i}, \varphi \rangle = -\int (\mathcal{A}_s)_{0i} \partial_i \varphi.$$

Compactly supported cutoffs justify the pairings if desired. From (275), $(\mathcal{A}_s)_{0i} = sF_K''(\rho + s\Theta)Y_i/\rho$. Summation over i proves (293).

To estimate the expectation, put $q_0(\rho) = R^2/\rho = \rho - p/\rho$. It is 2-Lipschitz on $[\sqrt{p}, \infty)$, and

$$q_0(\rho_0) = \frac{\rho_0}{2} = \frac{K^2}{4}.$$

Using $0 \leq F_K'' \leq K$ and $\mathbb{E}|\rho - \rho_0| = O(1)$,

$$\mathbb{E}[F_K'' q_0(\rho)] = \overline{K}(s) \frac{K^2}{4} + O(K). \quad (295)$$

Since $p = K^4/8$, $s^2 \overline{K}(s) = a_K$, $\overline{K}(s) = K + O(1)$, and $a_K = 1 + O(K^{-1})$,

$$\begin{aligned} \|\mathcal{T}_K[X_K]\|^2 &= \frac{s^2}{p} \left(\overline{K}(s) \frac{K^2}{4} + O(K) \right)^2 \\ &= \frac{s^2 \overline{K}(s)^2}{2} + O(K^{-1}) = \frac{a_K \overline{K}(s)}{2} + O(K^{-1}) = \frac{K}{2} + O(1). \end{aligned}$$

□

The next identity is the polarized Hessian compatibility relation needed for self-adjointness of the local generator.

Lemma C.8 (Second-Hermite–Bochner identity). *Let $V \in C^2(\mathbb{R}^d)$, suppose $B = \nabla^2 V$ is bounded, and assume $\mathbb{E}B(Z) = I$. For*

$$\mathcal{H}_B[X] := \mathbb{E}[B(Z)(Z^\top X Z - \text{Tr } X)],$$

one has, for every $X, Y \in \mathbb{S}(d)$,

$$\langle Y, \mathcal{H}_B[X] \rangle_F = \mathbb{E}[(YZ)^\top (B(Z) - I)(XZ)]. \quad (296)$$

In particular, $\mathcal{H}_B$ is self-adjoint.

Proof. Let φ_d be the standard Gaussian density. In the sense of distributions,

$$\partial_{cd} \varphi_d = (z_c z_d - \delta_{cd}) \varphi_d, \quad \partial_{cd} B_{ab} = \partial_{bc} B_{ad},$$

where the second equality follows from $B = D^2 V$. Hence

$$\mathbb{E}[(Z_c Z_d - \delta_{cd}) B_{ab}(Z)] = \mathbb{E}[(Z_b Z_c - \delta_{bc}) B_{ad}(Z)]. \quad (297)$$

Boundedness of B makes the distributional pairings legitimate; one may again use compactly supported cutoffs of φ_d .

Expanding the left side of (296) in indices and applying (297) gives

$$\begin{aligned} \langle Y, \mathcal{H}_B[X] \rangle_F &= \sum_{a,b,c,d} Y_{ab} X_{cd} \mathbb{E}[(Z_c Z_d - \delta_{cd}) B_{ab}(Z)] \\ &= \mathbb{E}[(YZ)^\top B(Z)(XZ)] - \text{Tr}(YX). \end{aligned}$$

The whitening condition gives the trace term in the second line, while $\text{Tr}(YX) = \mathbb{E}[(YZ)^\top (XZ)]$. This proves the identity. Its right side is symmetric in X, Y , because $B - I$ is symmetric. □

Set

$$\tau_K^2 := \|\mathcal{T}_K[X_K]\|^2, \quad h_K := \langle X_K, \mathcal{H}_K[X_K] \rangle_{\mathbb{F}}. \quad (298)$$

Lemma C.9 (Second-Hermite asymptotics and the Rayleigh direction). *The test direction satisfies*

$$h_K = \frac{K}{2} + O(1), \quad h_K - \tau_K^2 = O(1). \quad (299)$$

Let L_K^{pack} denote the operator

$$L_K^{\text{pack}} = \begin{pmatrix} I & \mathcal{T}_K/\sqrt{2} \\ \mathcal{T}_K^*/\sqrt{2} & I + \mathcal{H}_K/2 \end{pmatrix} \quad \text{on } \mathbb{R}^{d_K} \oplus \mathbb{S}(d_K). \quad (300)$$

It is isometrically equivalent to the local generator (86). With

$$Q_K := \frac{X_K}{\sqrt{2}}, \quad v_K := -\frac{\mathcal{T}_K[X_K]}{2}, \quad (301)$$

one has the exact quotient

$$\frac{\langle (v_K, Q_K), L_K^{\text{pack}}(v_K, Q_K) \rangle_{\text{pack}}}{\|v_K\|^2 + \|Q_K\|_{\mathbb{F}}^2} = \frac{2 + h_K - \tau_K^2}{2 + \tau_K^2} = O(K^{-1}). \quad (302)$$

Proof. Since

$$X_K Z = \left(0, \frac{Y}{\sqrt{p}}\right), \quad B_K - I = \mathcal{A}_s - a_K I,$$

Theorem C.8 gives

$$h_K = \frac{1}{p} \mathbb{E} \left[F_K'' \frac{R^4}{\rho^2} \right] + \mathbb{E} \left[F_K' \frac{R^2}{\rho^3} \right] - a_K. \quad (303)$$

For $u = R^2/p$, the function $u \mapsto u^2/(1+u)$ is 1-Lipschitz on $[0, \infty)$, equals $1/2$ at $u = 1$, and $\|u - 1\|_{L^2} = \sqrt{2/p} = O(K^{-2})$. Therefore

$$\frac{1}{p} \mathbb{E} \left[F_K'' \frac{R^4}{\rho^2} \right] = \frac{\bar{K}(s)}{2} + O(K^{-1}) = \frac{K}{2} + O(1).$$

The remaining positive expectation in (303) is bounded by

$$K^2 \sup_{r \geq 0} \frac{r^2}{(p + r^2)^{3/2}} \leq \frac{K^2}{\sqrt{p}} = 2\sqrt{2},$$

and $a_K = O(1)$. This proves $h_K = K/2 + O(1)$. On the other hand, the last calculation in Theorem C.7 gives

$$\tau_K^2 = \frac{a_K \bar{K}(s)}{2} + O(K^{-1}).$$

Thus

$$h_K - \tau_K^2 = \frac{(1 - a_K) \bar{K}(s)}{2} + O(1) = \frac{\alpha_K K}{2} + O(1) = O(1).$$

The map $(v, Q) \mapsto (v, \sqrt{2}Q)$ is an isometry from the packed norm to $\|(v, X)\|_{\star}^2 = \|v\|^2 + \|X\|_{\mathbb{F}}^2/2$, and conjugates (86) to (300). The quadratic form of L_K^{pack} at (301) is

$$\begin{aligned} \|v_K\|^2 + 2 \langle v_K, (\mathcal{T}_K/\sqrt{2}) Q_K \rangle + \|Q_K\|_{\mathbb{F}}^2 + \frac{1}{2} \langle Q_K, \mathcal{H}_K[Q_K] \rangle_{\mathbb{F}} \\ = \frac{\tau_K^2}{4} - \frac{\tau_K^2}{2} + \frac{1}{2} + \frac{h_K}{4} = \frac{2 + h_K - \tau_K^2}{4}. \end{aligned}$$

The squared norm of the vector is $(2 + \tau_K^2)/4$, proving the exact quotient. Its final order follows from $\tau_K^2 = K/2 + O(1)$ and (299). $\square$

Lemma C.10 (Smooth realization). *After increasing the lower threshold on K , the profiles can be chosen so that $V_K \in C^\infty(\mathbb{R}^{d_K})$, while preserving exact whitening, all asymptotic estimates above, and the common Hessian-Lipschitz constant 64.*

Proof. Let $f_K(x) = (K - |x - c|)_+$, fix an even nonnegative $\eta \in C_c^\infty((-1, 1))$ with $\int \eta = 1$, and choose any fixed $\varepsilon \in (0, 1]$. Put

$$\eta_\varepsilon(u) = \varepsilon^{-1} \eta(u/\varepsilon), \quad f_{K,\varepsilon} = f_K * \eta_\varepsilon, \quad F''_{K,\varepsilon} = f_{K,\varepsilon}, \quad F'_{K,\varepsilon}(-\infty) = 0.$$

Convolution preserves

$$0 \leq f_{K,\varepsilon} \leq K, \quad \text{Lip}(f_{K,\varepsilon}) \leq 1, \quad \int_{\mathbb{R}} f_{K,\varepsilon} = K^2,$$

and

$$\|f_{K,\varepsilon} - f_K\|_{L^\infty} \leq \varepsilon.$$

Since F''_K is 1-Lipschitz and the even mollifier has zero first moment,

$$F'_{K,\varepsilon} = F'_K * \eta_\varepsilon, \quad \|F'_{K,\varepsilon} - F'_K\|_{L^\infty} \leq \frac{\varepsilon^2}{2}.$$

It follows uniformly for $0 \leq s \leq 2K^{-1/2}$ that

$$|\bar{K}_\varepsilon(s) - \bar{K}(s)| \leq \varepsilon,$$

and

$$|G_\varepsilon(s) - G(s)| \leq \frac{\varepsilon}{p} + \frac{\varepsilon^2}{2}(p^{-1/2} + p^{-3/2}).$$

Thus (281) is unchanged. The same intermediate-value argument retunes $s = s_{K,\varepsilon}$ and restores exact isotropy. Since $f_{K,\varepsilon}(c) \geq K - \varepsilon$, the ellipticity lower bound remains of order K . The proof of Theorem C.6 uses only $\|f'_{K,\varepsilon}\|_\infty \leq 1$, and hence retains the same constant 64. The coupling, second-Hermite, and Rayleigh calculations change by at most $O(1)$, so all their asymptotic conclusions persist. $\square$

Proof of Theorem 4.7. Use the smooth, retuned potential supplied by Theorem C.10. The optimizer and whitening conditions follow by applying the argument of Theorem C.5 to the mollified profile, and

$$\begin{aligned} d_K &= \frac{K^4}{8} + 1, & \alpha_\star &= \alpha_K = \Theta(K^{-1}), \\ \beta_\star &= \beta_K = \Theta(K), & \kappa_\star &= \Theta(K^2). \end{aligned} \tag{304}$$

The common global Hessian-Lipschitz bound is (288).

Theorem C.7 gives $\|\mathcal{T}_K\|^2 \geq K/2 - O(1)$. For the converse bound, diagonalize an arbitrary unit-Frobenius X and write its eigenvalue vector as λ . In that basis set

$$\tilde{T}_{ji} := \mathbb{E}[(B_K)_{ii}(Z)Z_j], \quad D_{ij} := \mathbb{E}[(B_K)_{ii}(Z)(Z_j^2 - 1)].$$

For every u , Gaussian integration by parts gives

$$\mathbb{E}[(u + \text{diag}(\lambda)Z)^\top B_K(Z)(u + \text{diag}(\lambda)Z)] = \|u\|^2 + 2u^\top \tilde{T}\lambda + \|\lambda\|^2 + \lambda^\top D\lambda.$$

Positivity of B_K and the Schur complement yield $\tilde{T}^\top \tilde{T} \preceq I + D$, while $B_K \preceq \beta_K I$ gives $I + D \preceq \beta_K I$. Hence $\|\mathcal{T}_K[X]\|^2 \leq \beta_K \|X\|_{\mathbb{F}}^2 = O(K)$. Together with the explicit test direction,

$$\|\mathcal{T}_K\|^2 = \Theta(K). \tag{305}$$

By Rayleigh–Ritz and [Theorem C.9](#), $\gamma_\star \leq C/K$. Conversely, the second branch in [\(87\)](#) is of order at least K^2 , whereas $\beta_\star - 1 = \Theta(K)$. Thus $\Gamma = \beta_\star - 1 = \Theta(K)$, and [Theorem 4.1](#) gives

$$\gamma_\star \geq \max \left\{ \alpha_\star, \frac{1}{4 + \Gamma} \right\} \geq \frac{c}{K}.$$

This proves $\gamma_\star = \Theta(K^{-1})$. Substituting $\kappa_\star = \Theta(K^2)$ and $d_K = K^4/8 + 1$ yields every equivalence in [\(112\)](#). $\square$

Remark C.11 (Scope and minimum regularity). The construction has $d_K = \Theta(\kappa_\star^2)$; it does not settle the fixed-dimensional question or the sharpness of the $\sqrt{d} \log \kappa_\star$ branch of Γ . The Codazzi and second-Hermite identities use only commutation of distributional derivatives for a bounded Hessian. Thus the tent profile, which is $C^{2,1}$, already proves the analytic estimates; mollification is used solely to realize the theorem with C^∞ potentials.

D Clipped-KL descent in split coordinates

We prove the one-step estimate in [Theorem 6.3](#). The point that requires care is the orientation of the Gaussian divergences. The forward divergence controls the smoothness remainder and appears in the proximal model, whereas the covariance part of the reverse divergence is what remains in the descent certificate.

Write

$$\mathcal{E}(m, C) = \Phi(C) + \mathcal{E}_2(m, C) + \text{const}, \quad \Phi(C) := -\frac{1}{2} \log \det C, \quad \mathcal{E}_2(m, C) := \mathbb{E}_{\mathcal{N}(m, C)} V(X).$$

For two feasible states $a = (m, C)$ and $a' = (m', C')$, put

$$\begin{aligned} D_m(a', a) &:= \frac{1}{2} \|m' - m\|_{C^{-1}}^2, \\ D_C^+(C', C) &:= \text{KL}(\mathcal{N}(0, C') \parallel \mathcal{N}(0, C)), \\ D_C^-(C, C') &:= \text{KL}(\mathcal{N}(0, C) \parallel \mathcal{N}(0, C')). \end{aligned}$$

Lemma D.1 (Relative smoothness on the covariance box). *Suppose $-\beta I \preceq \nabla^2 V \preceq \beta I$ and both C, C' lie in $[\lambda_- I, \lambda_+ I]$. If $b = \mathbb{E}_{\mathcal{N}(m, C)} \nabla V$ and $A = \mathbb{E}_{\mathcal{N}(m, C)} \nabla^2 V$, then*

$$\mathcal{E}_2(m', C') \leq \mathcal{E}_2(m, C) + b^\top (m' - m) + \frac{1}{2} \text{Tr}(A(C' - C)) + 2\beta\lambda_+ \left\{ D_m(a', a) + D_C^+(C', C) \right\}. \quad (306)$$

Proof. Set $u = m' - m$, $M = C^{-1/2} C' C^{-1/2}$, $B_0 = C^{1/2}$, $B' = C^{1/2} M^{1/2}$, and $H = B' - B_0$. Then $B' B'^\top = C'$. Couple

$$X = m + B_0 Z, \quad Y = m' + B' Z, \quad Z \sim \mathcal{N}(0, I).$$

The upper Hessian bound in absolute value gives

$$\mathbb{E} V(Y) \leq \mathbb{E} V(X) + \mathbb{E} \left[\nabla V(X)^\top (u + HZ) \right] + \frac{\beta}{2} \left(\|u\|^2 + \|H\|_F^2 \right).$$

Gaussian integration by parts and

$$C' - C = B_0 H^\top + H B_0^\top + H H^\top$$

show that

$$\mathbb{E} \left[\nabla V(X)^\top H Z \right] = \text{Tr}(H^\top A B_0)$$

$$= \frac{1}{2} \text{Tr}(A(C' - C)) - \frac{1}{2} \text{Tr}(AHH^\top).$$

Since $\|A\|_{\text{op}} \leq \beta$, the last term and the Taylor remainder together are bounded above by

$$\frac{\beta}{2} \|u\|^2 + \beta \|H\|_{\text{F}}^2.$$

The covariance box yields

$$\frac{\beta}{2} \|u\|^2 \leq \beta \lambda_+ D_m(a', a).$$

Moreover, $\|H\|_{\text{F}}^2 \leq \lambda_+ \|M^{1/2} - I\|_{\text{F}}^2$, while the scalar inequality

$$x - 1 - \log x \geq (\sqrt{x} - 1)^2, \quad x > 0,$$

gives

$$\|M^{1/2} - I\|_{\text{F}}^2 \leq \sum_i \{\lambda_i(M) - 1 - \log \lambda_i(M)\} = 2D_C^+(C', C).$$

Substitution proves (306). $\square$

We next identify the clipped update as a constrained proximal step. At $a_n = (m_n, C_n)$, define

$$\mathcal{Q}_n(m, C) := \Phi(C) + b_n^\top (m - m_n) + \frac{1}{2} \text{Tr}(A_n(C - C_n)) + \frac{1}{h} \left\{ D_m((m, C), a_n) + D_C^+(C, C_n) \right\}. \quad (307)$$

With the trace pairing on symmetric matrices,

$$D_C^+(C, C_n) = \Phi(C) - \Phi(C_n) - \text{Tr}(\nabla \Phi(C_n)(C - C_n)), \quad \nabla \Phi(C) = -\frac{1}{2}C^{-1}. \quad (308)$$

Thus the covariance penalty in $\mathcal{Q}_n$ is the log-determinant Bregman divergence in its first argument.

The mean and covariance variables separate. The mean minimizer is $m_n - hC_nb_n$. The unconstrained covariance minimizer is

$$\tilde{C}_{n+1} = (1 + h)(C_n^{-1} + hA_n)^{-1}.$$

Under $h\beta\lambda_+ \leq 1/2$, its inverse is positive definite because

$$C_n^{-1} + hA_n \succeq (\lambda_+^{-1} - h\beta)I \succeq (2\lambda_+)^{-1}I.$$

More precisely, if $\alpha = 1 + h^{-1}$, the covariance-dependent part of $\mathcal{Q}_n$ is, up to a constant,

$$\alpha D_\Phi(C, \tilde{C}_{n+1}), \quad D_\Phi(C, \tilde{C}) := \Phi(C) - \Phi(\tilde{C}) - \text{Tr}(\nabla \Phi(\tilde{C})(C - \tilde{C})). \quad (309)$$

Indeed, $\alpha \tilde{C}_{n+1}^{-1} = h^{-1}C_n^{-1} + A_n$, which matches the linear coefficient in (307).

For completeness, the Bregman projection in (309) is exactly spectral clipping. Write $\tilde{C}_{n+1} = Q \text{diag}(\tilde{c}_i)Q^\top$ and $C^+ = Q \text{diag}(c_i^+)Q^\top$, where $c_i^+ = \min\{\lambda_+, \max\{\lambda_-, \tilde{c}_i\}\}$. Then

$$G := \nabla_C D_\Phi(C^+, \tilde{C}_{n+1}) = \frac{1}{2}(\tilde{C}_{n+1}^{-1} - (C^+)^{-1}) = Q \text{diag}(g_i)Q^\top.$$

Here $g_i > 0$ only when $c_i^+ = \lambda_-$, $g_i < 0$ only when $c_i^+ = \lambda_+$, and $g_i = 0$ otherwise. Hence every feasible C satisfies

$$\text{Tr}(G(C - C^+)) = \sum_i g_i \{q_i^\top C q_i - c_i^+\} \geq 0.$$

This is the variational inequality for minimizing $D_\Phi(\cdot, \tilde{C}_{n+1})$ over the covariance box; strict convexity in the first argument gives uniqueness. Consequently, $C^+ = \text{Clip}_{[\lambda_-, \lambda_+]}(\tilde{C}_{n+1}) = C_{n+1}$, and (m_{n+1}, C_{n+1}) is the unique minimizer of (307) over the covariance box.

For the mean minimizer, with $u_n = m_{n+1} - m_n$,

$$b_n^\top u_n = -\frac{1}{h} \|u_n\|_{C_n^{-1}}^2 = -\frac{2}{h} D_m(a_{n+1}, a_n). \quad (310)$$

For the covariance minimizer, constrained first-order optimality gives, for every feasible C ,

$$\text{Tr}\left(\left\{\nabla\Phi(C_{n+1}) + \frac{1}{2}A_n + \frac{1}{h}(\nabla\Phi(C_{n+1}) - \nabla\Phi(C_n))\right\}(C - C_{n+1})\right) \geq 0. \quad (311)$$

Set $C = C_n$ and $\Delta C_n = C_{n+1} - C_n$. The Bregman identities

$$\begin{aligned} \text{Tr}(\nabla\Phi(C_{n+1})\Delta C_n) &= \Phi(C_{n+1}) - \Phi(C_n) + D_C^-(C_n, C_{n+1}), \\ \text{Tr}(\{\nabla\Phi(C_{n+1}) - \nabla\Phi(C_n)\}\Delta C_n) &= D_C^-(C_n, C_{n+1}) + D_C^+(C_{n+1}, C_n) \end{aligned}$$

therefore give the stronger estimate

$$\begin{aligned} &\Phi(C_{n+1}) - \Phi(C_n) + \frac{1}{2} \text{Tr}(A_n(C_{n+1} - C_n)) \\ &\leq -\left(1 + \frac{1}{h}\right) D_C^-(C_n, C_{n+1}) - \frac{1}{h} D_C^+(C_{n+1}, C_n) \\ &\leq -\frac{1}{h} \left\{ D_C^-(C_n, C_{n+1}) + D_C^+(C_{n+1}, C_n) \right\}. \end{aligned} \quad (312)$$

Apply Theorem D.1 with $a = a_n$, $a' = a_{n+1}$, and combine (310)–(312). With $L = 2\beta\lambda_+$,

$$\begin{aligned} \mathcal{E}(a_{n+1}) - \mathcal{E}(a_n) &\leq -\left(\frac{2}{h} - L\right) D_m(a_{n+1}, a_n) - \frac{1}{h} D_C^-(C_n, C_{n+1}) \\ &\quad - \left(\frac{1}{h} - L\right) D_C^+(C_{n+1}, C_n). \end{aligned}$$

Because $hL \leq 1$, the final term is nonpositive and $2/h - L \geq 1/h$. Hence

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{h} \left\{ \frac{1}{2} \|m_{n+1} - m_n\|_{C_n^{-1}}^2 + D_C^-(C_n, C_{n+1}) \right\},$$

which is exactly (171). Summation proves the running-minimum estimate.

Finally, $S_n = 0$ if and only if $(m_{n+1}, C_{n+1}) = (m_n, C_n)$. At such a fixed point, mean optimality gives $b_n = 0$, while (311) reduces to

$$\text{Tr}\left(\frac{1}{2}(A_n - C_n^{-1})(C - C_n)\right) \geq 0 \quad \text{for every feasible } C.$$

These are exactly the first-order conditions for $\mathcal{E}$ on the covariance box. Thus S_n is a constrained fixed-point residual, not an unconstrained Fisher–Rao gradient norm. Moreover, the full successive reverse KL is

$$\text{KL}(\mathcal{N}(m_n, C_n) \parallel \mathcal{N}(m_{n+1}, C_{n+1})) = D_C^-(C_n, C_{n+1}) + \frac{1}{2} \|m_{n+1} - m_n\|_{C_{n+1}^{-1}}^2.$$

Its mean block uses C_{n+1}^{-1} , rather than C_n^{-1} , which is why the comparison in Theorem 6.4 incurs the condition-number factor λ_+/λ_- .

E Experimental protocols and data lineage

All paths below are relative to the repository root. The numerical kernels use float64 arrays. The objective reported throughout is the Gaussian variational objective, up to an additive target-dependent constant,

$$\mathcal{E}(m, C) = -\frac{1}{2} \log \det C + \mathbb{E}_{X \sim \mathcal{N}(m, C)} V(X).$$

When deterministic quadrature is used, NumPy’s Gauss–Hermite nodes (x_j, w_j) are rescaled according to

$$\mathbb{E}f(Z) = \frac{1}{\sqrt{\pi}} \sum_{j=1}^K w_j f(\sqrt{2}x_j), \quad Z \sim \mathcal{N}(0, 1).$$

Unless stated otherwise, a run is evolved for its fixed iteration budget; a “time to tolerance” is a diagnostic computed from the saved trajectory, not an early-stopping rule. The PDF files used in [Section 7](#) are byte-for-byte copies of the corresponding files in `reports/assets/figs`. The script `reports/make_report_assets.py` performs only post-processing and never reruns a dynamics experiment.

E.1 Covariance bootstrap

Let

$$H = \text{diag}(\nu_1, \dots, \nu_d), \quad (\nu_i)_{i=1}^d = \text{logspace}(1, \kappa, d).$$

The Gaussian target is

$$V(x) = \frac{1}{2} x^\top H x, \quad (m_\star, C_\star) = (0, H^{-1}),$$

and the non-Gaussian target used only in the transport experiment is

$$V(x) = \frac{1}{2} x^\top H x + \gamma \sum_{i=1}^d \log \cosh(x_i), \quad \gamma = 1.$$

For the latter, $m_\star = 0$, and each diagonal entry of $C_\star$ solves

$$c_i^{-1} = \nu_i + \gamma \mathbb{E}_{Z \sim \mathcal{N}(0, c_i)} \text{sech}^2 Z.$$

The fixed-point tolerance is 10^{-13} , with at most 500 iterations and 80-point Gauss–Hermite quadrature. Thus $\alpha = 1$, while $\beta = \kappa$ for the Gaussian target and $\beta = \kappa + 1$ for the log cosh target.

Write $G = \mathbb{E} \nabla V(X)$, $A = \mathbb{E} \nabla^2 V(X)$. The two Fisher–Rao steps share

$$m_{n+1} = m_n - h C_n G_n$$

and use, respectively,

$$\begin{aligned} C_{n+1}^R &= C_n^{1/2} \exp \left[h \{ I - C_n^{1/2} A_n C_n^{1/2} \} \right] C_n^{1/2}, \\ C_{n+1}^{\text{KL}} &= (1 + h)(C_n^{-1} + h A_n)^{-1}. \end{aligned}$$

Every run starts from

$$m_0 = m_\star + 3 \sqrt{\text{diag}(C_\star)}, \quad C_0 = \lambda_0 I.$$

The Gaussian scaling grid is

$$d \in \{10, 20\}, \quad \kappa \in \{10, 100, 1000\}, \quad \lambda_0 \in \{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\},$$

for both covariance maps, with $h = 0.5$ and 200 steps. The burn-in time is the first n such that $\lambda_{\min}(C_n) \geq (2\beta)^{-1}$. The left panel of [Figure 1](#) selects KL, $d = 20$, $\kappa = 1000$, and $\lambda_0 = 10^{-10}$. Its three ordinates are $\lambda_{\min}(C_n)$, the exact Gaussian energy gap, and $\|C_n^{1/2}G_n\|$. Its vertical markers are the covariance threshold and the first saved iterate with energy gap at most 10^{-3} .

For the transport grid, the Gaussian and log cosh targets both have $d = 20$ and $\kappa = 50$. It uses

$$\lambda_0 \in \{10^{-8}, 10^{-6}, 10^{-4}\}, \quad c \in \{0.25, 0.5, 1\}, \quad h = 0.5,$$

both Fisher–Rao tails, and a 6000-step budget. When $\lambda_0 < c/\beta$, one Bures step of size $\eta = c/\beta$ is applied:

$$\begin{aligned} m^+ &= m - \eta G, & \tilde{C} &= (I - \eta A)C(I - \eta A), \\ C^+ &= \frac{1}{2} \left\{ \tilde{C} + 2\eta I + [\tilde{C}(\tilde{C} + 4\eta I)]^{1/2} \right\}. \end{aligned}$$

The right panel of [Figure 1](#) selects $c = 0.5$ and plots, for every target, tail, and λ_0 , the first iteration whose gap is at most 10^{-1} , together with $\lambda_{\min}(C)$ after the first step.

Data and figure lineage. The configuration is `configs/natural_gradient_covariance_bootstrap/covariance_bootstrap.yaml`; the runner is `scripts/natural_gradient_covariance_bootstrap/run_experiments.py`; and the kernels and figure selectors are in `src/natural_gradient_covariance_bootstrap`. The left panel reads `outputs/natural_gradient_covariance_bootstrap/results_long.csv` through `plotting.py::fig_three_stage`. The right panel reads `outputs/natural_gradient_covariance_bootstrap/wasserstein_bootstrap_summary.csv` through `plotting.py::fig_wasserstein_bootstrap`. The final assets are `reports/assets/figs/fig_covboot_three_stage.pdf` and `reports/assets/figs/fig_covboot_wasserstein_bootstrap.pdf`.

E.2 Deterministic discretization grid

All three main targets have dimension two and use $\lambda \in \{0.01, 0.1, 1\}$. They are

$$\begin{aligned} V_G(x, y) &= \frac{1}{2}(x, y) \operatorname{diag}(1, \lambda)(x, y)^\top, \\ V_q(x, y) &= \frac{(\sqrt{\lambda}x - y)^2}{20} + \frac{y^4}{20}, \\ V_s(x, y) &= \frac{(\sqrt{\lambda}x - y)^2}{20} + \frac{\delta}{2}(x^2 + y^2) + \gamma \log \cosh y, \quad (\delta, \gamma) = (0.05, 1). \end{aligned}$$

The Gaussian initialization is $m_0 = (10, 10)$, $C_0 = \operatorname{diag}(0.5, 2)$; the other two use $m_0 = (10, 10)$, $C_0 = 4I$. Gaussian and quartic expectations are analytic, using exact Gaussian moments through order four. The smooth target uses 80-point one-dimensional Gauss–Hermite quadrature for the y -marginal. The Gaussian optimum is analytic. The other optima are computed in Cholesky coordinates by four deterministic L-BFGS-B starts, with a 5000-iteration budget, gradient tolerance 10^{-10} , and function tolerance 10^{-16} .

The common grid is

$$h \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10\}, \quad T = 15, \quad N = \lceil T/h \rceil.$$

The steps are the two formulas in the preceding subsection, with exact population G_n, A_n . A Radau solution of the continuous flow, with relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and 400 output times, is used only for the terminal discretization-accuracy diagnostic. Simulation stops early only on a nonfinite update or $\lambda_{\min}(C_n) \leq 10^{-12}$. A completed trajectory is called

stable when it is SPD-feasible, ends below its initial energy, and never exceeds 10^3 times its initial energy gap. Monotonicity allows an absolute per-step tolerance of 10^{-10} ; accuracy means

$$\|m_N - m(T)\| + \frac{\|C_N - C(T)\|_F}{\|C(T)\|_F} \leq 10^{-2}.$$

The left panel of [Figure 2](#) uses only V_s , all three λ 's, and the matched stable steps

$$h \in \{0.05, 0.1, 0.5, 1\}.$$

It plots energy gap against nh . The right panel retains every stable run that reaches gap 10^{-6} . For each target, method, and h , it averages the hitting iteration and hitting time nh over the eligible λ 's. Its top row is iteration count and its bottom row is physical flow time, not measured wall time.

Data and figure lineage. The configuration and runner are `configs/natural_gradient_discretization_stepsize/stepsize_grid.yaml` and `scripts/natural_gradient_discretization_stepsize/run_stepsize_grid.py`. Target, method, optimizer, ODE, and classification code is in `src/natural_gradient_discretization_stepsize`. Both selected panels read `outputs/natural_gradient_discretization_stepsize/results_long.csv` and/or `outputs/natural_gradient_discretization_stepsize/summary.csv`; their builders are `scripts/natural_gradient_discretization_stepsize/plot_results.py::fig_convergence_speed_grid` and `plot_results.py::fig_iterations_vs_stepsize`. The assets are `reports/assets/figs/fig_convergence_speed_smooth_logconcave.pdf` and `reports/assets/figs/fig_iterations_vs_stepsize.pdf`.

E.3 STL estimator variance and constant-step floors

Here $A = \text{diag}(a_1, \dots, a_d)$, with (a_i) log-spaced in $[1, \kappa]$. The targets are

$$\begin{aligned} V_G(x) &= \frac{1}{2} x^\top A x, \\ V_{lc}(x) &= \frac{1}{2} x^\top A x + \tau \sum_{i=1}^d \log \cosh(x_i). \end{aligned}$$

The grid is

$$\begin{aligned} V_G : \quad & d \in \{2, 8, 32\}, \quad \kappa \in \{10, 100, 10^4\}, \\ V_{lc} : \quad & d \in \{2, 8, 32\}, \quad \kappa \in \{10, 100\}, \quad \tau \in \{0.1, 1\}. \end{aligned}$$

For V_{lc} , $m_\star = 0$ and

$$(C_\star)_{ii}^{-1} = a_i + \tau \mathbb{E}_{Z \sim \mathcal{N}(0, (C_\star)_{ii})} \text{sech}^2 Z.$$

This fixed point uses 80-point Gauss–Hermite quadrature, tolerance 10^{-13} , and at most 500 iterations.

For the estimator-level experiment,

$$b_{\text{base}}(X) = \nabla \log \pi(X), \quad b_{\text{STL}}(X) = \nabla \log \pi(X) + C^{-1}(X - m).$$

At each target configuration the four states plotted on the left of [Figure 3](#) have $C = C_\star$ and

$$m = m_\star + \rho \sqrt{\text{diag}(C_\star)}, \quad \rho \in \{0, 0.25, 1, 4\}.$$

The full estimator grid additionally includes $(m_\star, 0.25C_\star)$ and $(m_\star, 4C_\star)$. For every state and each seed $s \in \{0, \dots, 19\}$, $M = 100000$ Gaussian samples are drawn in one chunk (the configured chunk cap is 200000). The reported mean-block Fisher–Rao variance is

$$\text{Var}_{\text{FR}}(b) = \mathbb{E}[(b - \mathbb{E}b)^\top C (b - \mathbb{E}b)].$$

The left panel plots the median over the 20 seeds of $\text{Var}_{\text{FR}}(b_{\text{STL}})/\text{Var}_{\text{FR}}(b_{\text{base}})$ against ρ , separately for all 12 non-Gaussian configurations. For display on a logarithmic axis, the plotting code adds 10^{-3} to ρ and clips ratios below 10^{-16} .

The algorithm-level experiment compares $\{\text{R}, \text{R-STL}, \text{KL}, \text{KL-STL}\}$. At every step it draws B samples from the current Gaussian, averages the corresponding mean estimator, and averages the diagonal pointwise log-target Hessian for the covariance step. The two mean estimators within a covariance scheme use the same standard-normal stream. The settings are

$$h = 0.1, \quad B \in \{1, 4, 16\}, \quad m_0 = m_\star + 3\sqrt{\text{diag}(C_\star)}, \quad C_0 = 0.5C_\star,$$

with 2000 steps for V_G and 3000 for V_{lc} . Gaussian energy gaps are analytic; non-Gaussian gaps use 40-point Gauss–Hermite quadrature and are therefore independent of the mini-batch used for the update. An emergency eigenvalue guard clips values below 10^{-12} . The runner counts such clips in a cell diagnostic printed during execution, but the surviving summary CSV does not preserve that count.

The replicate labels are $0, \dots, 19$. For the algorithm grid, one batched random stream advances these 20 leading-array entries. Its cell seed is the first 15 hexadecimal digits of

$$\text{SHA256}(20260623|\text{kind}|\text{d}|\text{kappa}|\text{tau}|\text{B}),$$

interpreted as an integer; the method name is deliberately absent, so all four paired methods receive the same standard-normal sequence. For the estimator grid, the generator seed is the replicate label itself.

For each run the empirical floor is the median gap over the final 20 percent of the full, undecimated trajectory. The right panel of [Figure 3](#) then takes the median of these per-run medians over every target configuration and seed, separately by target kind, method, and B . Nonpositive display values are clipped to 10^{-16} .

Data and figure lineage. The configuration is `configs/natural_gradient_stl_variance/stl_variance.yaml`. The two runners are `scripts/natural_gradient_stl_variance/run_estimator_variance.py` and `scripts/natural_gradient_stl_variance/run_algorithm_grid.py`; kernels and plot selectors are in `src/natural_gradient_stl_variance`. The left panel reads `outputs/natural_gradient_stl_variance/estimator_variance.csv` through `plotting.py::fig_variance_by_distance` with `kind=log_cosh`. The right panel reads `outputs/natural_gradient_stl_variance/tail_noise_floor.csv` through `plotting.py::fig_batchsize_effect`. The assets are `reports/assets/figs/fig_stl_variance_by_distance_to_optimum.pdf` and `reports/assets/figs/fig_stl_batchsize_effect.pdf`.

E.4 Trace and traceless affine modes

For the Gaussian calibration target, $\pi = \mathcal{N}(0, I_d)$, expectations are exact. The covariance step is diagonalized as

$$\lambda_i^+ = \lambda_i \exp\left[\frac{h}{2\omega}\{-\lambda_i + q_\tau(C)\}\right], \quad q_\tau(C) = \frac{\omega + \tau \text{Tr } C}{\omega + d\tau},$$

and $m^+ = m - hCm$. The grid is

$$d \in \{2, 5, 10\}, \quad \omega \in \{0.125, 0.25, 0.5, 1, 2\}, \quad \tau \in \{-\omega/(2d), 0, \omega/(2d)\},$$

with $h = 0.02$, $T = 20$, and metrics saved every five steps.

The non-Gaussian target is

$$V_\rho(x) = \frac{1}{2} \|x\|^2 + \frac{\rho}{M} \sum_{\ell=1}^M \log \cosh(a_\ell^\top x), \quad d = 5, \quad \rho = 5, \quad M = 4d = 20.$$

The unit vectors a_ℓ are normalized Gaussian draws from NumPy seed 123. At a state (m, C) , set $A_C = \mathbb{E} \nabla^2 V_\rho(X)$ and $B = C^{1/2} A_C C^{1/2}$. The covariance update is

$$C^+ = C^{1/2} \exp \left[\frac{h}{2\omega} \left\{ -B + \frac{\omega + \tau \operatorname{Tr} B}{\omega + d\tau} I \right\} \right] C^{1/2}, \quad m^+ = m - hC \mathbb{E} \nabla V_\rho(X).$$

The dynamics expectations use one fixed scrambled Sobol design, transformed coordinatewise by the Gaussian inverse CDF, with $K = 4096$ and seed 2026. The same points are reused at every state and for every (ω, τ) and initialization. The reference optimum uses an independent scrambled Sobol design with $K_{\text{ref}} = 8192$ and seed 2027, followed by Cholesky-coordinate L-BFGS-B with a 2000-iteration budget, gradient tolerance 10^{-6} , and function tolerance 10^{-15} . The dynamics grid is

$$\omega \in \{0.25, 0.5, 1\}, \quad \tau \in \{-\omega/(2d), 0, \omega/(2d)\}, \quad h = 0.005, \quad T = 40,$$

with metrics saved every ten steps. There are no random replicates: QMC is a single common deterministic design once the seeds are fixed.

The five Gaussian initializations are

$$\begin{aligned} \text{mean:} \quad m_0 &= 3\mathbf{1}/\sqrt{d}, \quad C_0 = I, \\ \text{volume-high/low:} \quad m_0 &= 0, \quad C_0 = 4I \text{ or } I/4, \\ \text{shape:} \quad m_0 &= 0, \quad C_0 = \operatorname{diag}(e^2, e^{-2}, 1, \dots, 1), \\ \text{mixed:} \quad m_0 &= 2\mathbf{1}/\sqrt{d}, \quad C_0 = 2 \operatorname{diag}(e^{1.5}, e^{-1.5}, 1, \dots, 1). \end{aligned}$$

For the non-Gaussian target the same formulas are applied in the $(m_\star, C_\star)$ -whitened coordinates. Thus, for example, the shape covariance is $C_\star^{1/2} \operatorname{diag}(e^2, e^{-2}, 1, \dots, 1) C_\star^{1/2}$.

Both panels of [Figure 4](#) show $T(\tau)/T(0)$ for every ω and initialization. The Gaussian panel selects $d = 5$ and the first saved time at which the exact Gaussian KL, normalized by its initial value, is at most 10^{-4} . The non-Gaussian panel uses the fixed-QMC objective gap, normalized by its initial gap, at tolerance 10^{-2} . A cell is left blank if either hitting time is not finite within the horizon.

Data and figure lineage. The configurations are `configs/omega_tau_modes/gaussian_target.yaml` and `configs/omega_tau_modes/logconcave_target.yaml`; runners and kernels are under `scripts/omega_tau_modes` and `src/omega_tau_modes`. The report builder reads the legacy committed locations `outputs/gaussian_grid/summary.csv` and `outputs/logconcave_grid/summary.csv`, through `reports/make_report_assets.py::fig_tau_speedup`. The resulting assets are `reports/assets/figs/fig_ot_gaussian_tau_speedup.pdf` and `reports/assets/figs/fig_ot_logconcave_tau_speedup.pdf`. To reproduce these exact locations with the current runners, their `-outdir` arguments must override the nested `output_dir` values in the YAML files.

E.5 Non-log-concave boundary

All selected runs are centered, scalar, deterministic, and use 160-point Gauss–Hermite quadrature for

$$V_R(x) = \frac{1}{2}x^2 - 2R^2 \log \cosh(x/R), \quad V_R''(x) = -1 + 2 \tanh^2(x/R) \in [-1, 1].$$

For $X \sim \mathcal{N}(0, c)$, write $A_R(c) = \mathbb{E} V_R''(X)$. The plotted Fisher–Rao gradient squared is $(1 - cA_R(c))^2$.

The Riemannian cascade uses

$$c_{n+1} = c_n \exp\{h(1 - c_n A_R(c_n))\}, \quad R \in \{20, 50, 100, 300, 1000\}, \quad c_0 = 1, \quad h = 1.$$

It has a 20-step cap and stops before the next update if $c_n > R^2/4$, if $\log c$ would exceed 700, or on nonfinite arithmetic. The first panel of [Figure 5](#) plots every recorded iterate for all five R 's.

The pole experiment uses the unprojected update

$$c_{n+1} = \frac{(1+h)c_n}{1+hc_n A_R(c_n)}, \quad R = 1000, \quad h = 1, \quad c_0 = 1 - \varepsilon,$$

with $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and a two-step cap. The middle panel plots the gradient squared after the first step against ε , together with the reference $4\varepsilon^{-2}$.

The clipped experiment uses

$$c_{n+1} = \text{clip}_{[0.5, 2]} \frac{(1+h)c_n}{1+hc_n A_R(c_n)}, \quad c_0 = 1, \quad \beta = 1.$$

The report panel selects the primary safety fraction 0.5, hence

$$L_{\text{clip}} = 2\beta\lambda_+ = 4, \quad h = 0.5/L_{\text{clip}} = 0.125,$$

and runs 300 steps for each of the five R 's. The full output also contains safety fractions 0.25 and 0.9, plus the deliberately outside-theorem step $h = 1/(\beta\lambda_+) = 0.5$; the selected panel excludes those rows. At step n , the code records

$$\begin{aligned} D_n &= \text{KL}(\mathcal{N}(0, c_n) \parallel \mathcal{N}(0, c_{n+1})), \\ D_{\min}(N) &= \min_{0 \leq n < N} D_n, \\ B_N &= \frac{h}{N} \{\mathcal{E}(0, c_0) - \mathcal{E}(0, c_N)\}. \end{aligned}$$

The right panel plots $D_{\min}(N)$ and B_N . In the recorded primary runs, the upper clip first activates at step 2 for every R ; the step-2 displacement is nonzero, while the next and subsequent pinned steps have $D_n = 0$.

Data and figure lineage. The configuration is

`configs/natural_gradient_nonconvex_instability/nonconvex_instability.yaml`; the runner and kernels are in `scripts/natural_gradient_nonconvex_instability` and `src/natural_gradient_nonconvex_instability`. The cascade panel reads `outputs/natural_gradient_nonconvex_instability/results_long.csv` through `plotting.py::fig_riemannian_cascade`; the pole panel reads `outputs/natural_gradient_nonconvex_instability/kl_pole_summary.csv` through `plotting.py::fig_kl_pole`; and the clipped panel reads `outputs/natural_gradient_nonconvex_instability/clipped_kl_stationarity.csv` through `plotting.py::fig_clipped_kl_stationarity`, restricted to `dt_rule=theorem_safe`. The assets are, respectively, `reports/assets/figs/fig_nonconvex_riemannian_cascade.pdf`, `reports/assets/figs/fig_nonconvex_kl_pole.pdf`, and `reports/assets/figs/fig_nonconvex_clipped_kl_stationarity.pdf`.

E.6 Provenance limits of the committed snapshot

The scientific parameters above are recoverable from the configurations, source, and raw columns used by the selected figures. Some run-level provenance files that the runners are designed to write are not committed:

- (i) `outputs/natural_gradient_covariance_bootstrap` lacks `target_metadata.json` and `run_metadata.json`;
- (ii) `outputs/natural_gradient_stl_variance` lacks `algorithm_results_long.csv`, `target_metadata.json`, `run_metadata.json`, `estimator_target_metadata.json`, and `estimator_run_metadata.json`; and

(iii) the Gaussian affine-mode runner does not write a run-metadata file.

Consequently, the exact Python, NumPy, Torch, CUDA, and hardware versions of those original runs cannot be reconstructed from this snapshot. The estimator-level STL CSV itself records the Torch/CUDA backend, but not the GPU model; the algorithm-level backend is not recorded in the surviving summary files. The repository’s `requirements.txt` and `requirements-gpu.txt` give lower dependency bounds rather than a locked environment.

These omissions do not change the selected data-to-figure mappings: the two STL panels need only the surviving estimator and tail CSVs, and the covariance panels need only the surviving long and transport-summary CSVs. They do mean that `reports/make_report_assets.py`, invoked with either `-only stl_variance` or `-only covariance_bootstrap`, does not run to completion from the current snapshot, because the group-level builder also loads the missing metadata or supplementary trajectory files before producing all of that group’s tables and figures. Rerunning the corresponding experiment scripts restores those files; no missing value has been guessed here.

General WFR experiments under `outputs/wfr_gradient_flow` are not used in any figure or claim in this manuscript.

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